SAMHSA
Substance Abuse and Mental Health Services Administration

2024 National Survey on Drug Use and Health (NSDUH)

Methodological Summary and Definitions

2024 National Survey on Drug Use and Health (NSDUH): Methodological Summary and Definitions

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1. Introduction

This report summarizes methods and other supporting information relevant to estimates of substance use and mental health conditions from the 2024 National Survey on Drug Use and Health (NSDUH), an annual survey of the civilian, noninstitutionalized population of the United States aged 12 years or older. NSDUH is the primary source of statistical information on the use of tobacco, alcohol, prescription psychotherapeutic drugs (pain relievers, tranquilizers, stimulants, and sedatives), and other substances (e.g., marijuana, cocaine) by people aged 12 or older in that population. The survey also includes extensive information on substance use disorders, substance use treatment, mental health conditions, and mental health treatment.

1.1 Background on the 2024 NSDUH

Conducted by the federal government since 1971, the 2024 survey collected data with a representative sample of the population. The survey is sponsored by the Substance Abuse and Mental Health Services Administration (SAMHSA), U.S. Department of Health and Human Services, and is planned and managed by SAMHSA's Center for Behavioral Health Statistics and Quality (CBHSQ). Data collection and analysis are conducted under contract with RTI International.[1]

NSDUH targets the collection of information from residents of households (e.g., individuals living in houses or townhouses, apartments, and condominiums; civilians living in housing on military bases) and individuals in noninstitutional group quarters (e.g., shelters, rooming or boarding houses, college dormitories, migratory workers' camps, halfway houses). Not included are individuals with no fixed household address (e.g., people experiencing homelessness and not in shelters), military personnel on active duty, and residents of institutional group quarters, such as jails or hospitals. People also are excluded during data collection if they cannot complete the survey in either English or Spanish or they are not physically or mentally capable of completing the interview.

The 2024 NSDUH used multimode data collection, in which respondents completed the survey in person or via the web. Methodological investigations led to the conclusion that estimates based on multimode data collection since 2021 are not comparable with estimates from 2020 or prior years. Chapter 6 in the *2021 National Survey on Drug Use and Health (NSDUH): Methodological Summary and Definitions* (CBHSQ, 2022) discusses these methodological investigations in greater detail. However, the 2024 estimates are comparable with the 2021-2023 estimates that are calculated with an adjusted weight (see Section 2.3.4.2).

1.2 Organization of the Report

This report is organized into four chapters, including this introductory chapter. Chapter 2 describes the survey, including information about the sample design, data collection procedures, and key aspects of data processing (e.g., development of analysis weights).

[1] RTI International is a trade name of Research Triangle Institute. RTI and the RTI logo are U.S. registered trademarks of Research Triangle Institute.

Chapter 3 presents technical details on the statistical methods and measurement, such as suppression criteria for unreliable estimates, statistical testing procedures, response rates, and issues for selected measures for substance use, mental health status, and the receipt of substance use treatment or mental health treatment. Chapter 4 discusses special topics related to prescription psychotherapeutic drugs.

1.3 Summary of Other Relevant Sources of NSDUH Information

Data and findings for the 2024 NSDUH are presented in five key products:

1. The *Key Substance Use and Mental Health Indicators in the United States: Results from the 2024 National Survey on Drug Use and Health* report (CBHSQ, 2025g) is a national-level report focusing on estimates among people aged 12 years or older among the civilian, noninstitutionalized population in the United States. This report presents trends in estimates from 2021 to 2024.
2. The *2024 Companion Infographic Report: Results from the 2021 to 2024 National Surveys on Drug Use and Health* (CBHSQ, 2025f) shows selected estimates from 2021 to 2024 among the population aged 12 or older.
3. SAMHSA will produce a series of reports using pooled data from the 2022-2024 NSDUHs that will examine associations in greater depth between selected demographic characteristics and substance use and mental health indicators. Reports in this series will be available on the NSDUH Data Collection web page.
4. The *Results from the 2024 National Survey on Drug Use and Health: Detailed Tables* (CBHSQ, 2025h) are a comprehensive set of tables on substance use and mental health issues that include estimated numbers of people with a characteristic of interest (e.g., numbers of substance users, numbers of adults with mental illness), corresponding percentages, and standard errors of estimates. These tables show estimates for 2023 and 2024 and whether there was a statistically significant change between 2023 and 2024 when applicable. Multiyear tables are also included that show selected estimates for 2021-2024 and whether there are significant changes between estimates in 2024 and corresponding estimates in 2021-2023.

Additional 2024 national reports and products can be found on the NSDUH Data Collection web page. State-level estimates and substate estimates for substance use and mental health outcomes[2] are also available on SAMHSA's website.

As in previous years, CBHSQ will construct a public use data file for the 2024 NSDUH that will be available in late 2025 on the NSDUH Data Collection web page. Users of NSDUH data files, including the public use file, will see important questionnaire updates for a given survey year being reflected by changes to the variable names, labels, and codebook documentation.

[2] The most recent state-level estimates are for 2022-2023. The most recent substate estimates are for 2016-2018.

2. Description of the Survey

2.1 Sample Design

The target population for the National Survey on Drug Use and Health (NSDUH) is the civilian, noninstitutionalized population aged 12 years or older residing within the United States.

The survey includes the following:

- residents of households (e.g., individuals living in houses or townhouses, apartments, and condominiums; civilians living in housing on military bases); and
- individuals in noninstitutional group quarters (e.g., shelters, rooming or boarding houses, college dormitories, migratory workers' camps, halfway houses).

The survey excludes the following:

- individuals with no fixed household address (e.g., people experiencing homelessness and not in shelters);
- active-duty military personnel; and
- residents of institutional group quarters, such as jails or hospitals.

People who are unable to take the survey in either English or Spanish are part of the eligible population but are also excluded.

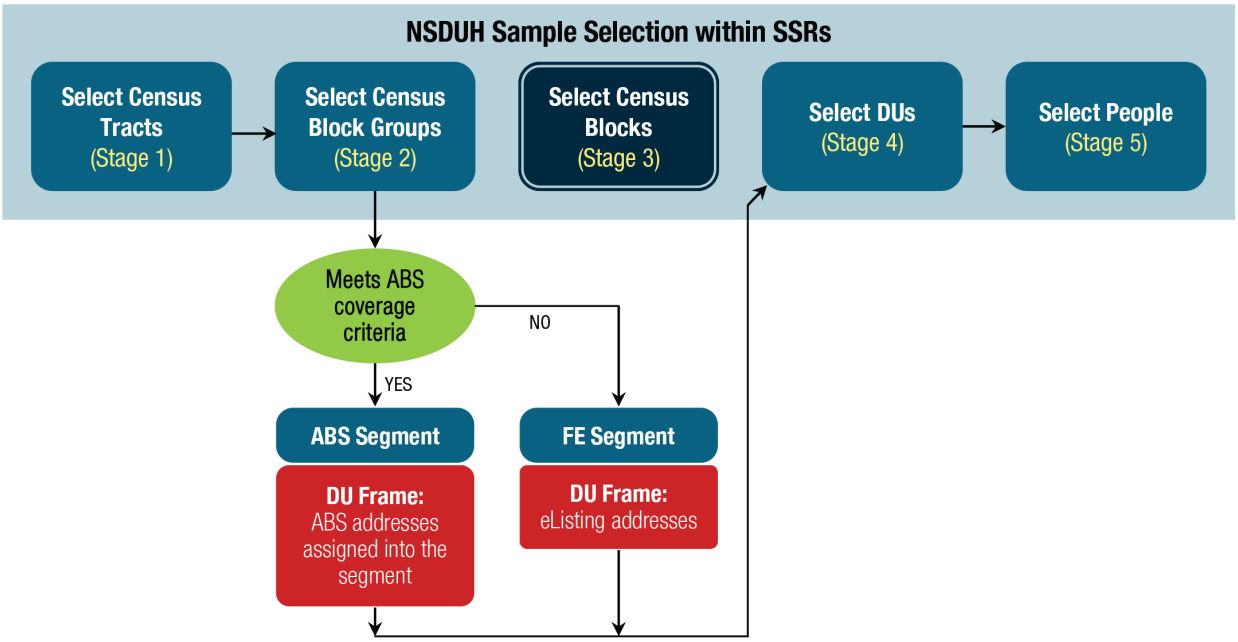
2.1.1 Coordinated Sample Design for 2014-2024

A coordinated sample design was developed for the 2014-2024 NSDUHs. It is state based, with an independent, multistage area probability sample within each state and the District of Columbia. In this approach, successively smaller geographic areas are selected within each state. In the smallest geographic areas, address lists are constructed and serve as the dwelling unit (DU) sampling frames, where DUs can be households or units (rooms, beds, or people) within an eligible group quarter. Then DUs and people within those DUs are randomly selected to participate. Each respondent having a known probability of selection allows NSDUH to make estimates for the population. The 2014-2024 NSDUH sample design provides a sufficient number of completed interviews to support both state and national estimates.

For the NSDUH coordinated sample design, states were the first level of stratification. Each state was further stratified into approximately equally populated state sampling regions (SSRs). As shown in Figure 2.1, there were five stages of selection within each SSR to create the multistage area probability sample. First, census tracts were selected within each SSR (Stage 1), then census block groups were selected within those census tracts (Stage 2), and then smaller area segments (i.e., a collection of census blocks) were selected within the census block groups (Stage 3). Next, DUs were selected within segments to receive a screener (Stage 4), and within each selected DU, up to two residents who were at least 12 years old were selected for the interview (Stage 5). However, for the 2024 NSDUH, the selection of smaller area segments within census block groups (Stage 3) was eliminated. Selecting the DU samples from larger geographic areas was expected to increase precision.

The coordinated sample design for 2014-2024 includes a 50 percent overlap in sampled areas within each successive 2-year period for 2014-2024 (e.g., 2014-2015, 2015-2016, 2016-2017). DUs that were not sampled the first year are eligible for selection the following year. There is no planned overlap of sampled residents and no longitudinal follow-up for individuals. However, people may be selected in consecutive years if they move and their new residence is selected.

Figure 2.1 2024 NSDUH Sample Selection with Hybrid ABS and FE Frame



ABS = address-based sampling; DU = dwelling unit; eListing = electronic listing; FE = field enumeration; SSR = state sampling region.
NOTE: Although the selection of census blocks (Stage 3) was not used in the 2024 NSDUH, the selection of DUs and people are still referred to as "Stage 4" and "Stage 5," respectively, for comparability with the 2014-2023 NSDUHs.

For the 2024 NSDUH, a hybrid field enumeration and address-based sampling (ABS) approach was used to construct DU frames within sampled areas. Census block groups, or "segments," were evaluated using a set of ABS coverage criteria. If the census block group met all coverage criteria,[3] the ABS frame was used. If the census block group failed one or more coverage criteria, field enumeration was used to construct the DU frame.[4] To improve efficiencies and data quality, an electronic listing (eListing) application has been used since 2023 to enumerate DUs in field enumeration segments and to locate sampled DUs during data collection.

[3] See the *2024 National Survey on Drug Use and Health (NSDUH) Methodological Resource Book, Section 2: Sample Design and Experience Report* (CBHSQ, forthcoming b) for specifics about the ABS coverage criteria.
[4] Some large census block groups required subsampling to make field enumeration feasible.

Table 2.1 provides the target and achieved sample allocations for the 2024 NSDUH. Adolescents aged 12 to 17 years and young adults aged 18 to 25 years were oversampled. See the *2024 National Survey on Drug Use and Health (NSDUH) Methodological Resource Book, Section 2: Sample Design and Experience Report* (Center for Behavioral Health Statistics and Quality [CBHSQ], forthcoming b) for more details on the sample design.

Table 2.1 2024 Target and Achieved Sample Allocation, by Age Group

Sample	12 to 17	18 to 25	26 or Older	26 to 34	35 to 49	50 or Older
Target	16,877 (25%)	16,877 (25%)	33,753 (50%)	10,126 (15%)	13,501 (20%)	10,126 (15%)
Achieved	13,985 (20%)	16,744 (24%)	39,512 (56%)	11,278 (16%)	15,379 (22%)	12,855 (18%)

NOTE: Percentages of the total sample are shown in parentheses.
NOTE: Achieved sample sizes are based on the reported age in the interview. These sample sizes differ from those in response rate tables in Chapter 3 because counts in the response rate tables are based on ages reported in the household screener.
Source: SAMHSA, Center for Behavioral Health Statistics and Quality, National Survey on Drug Use and Health, 2024.

2.1.2 Special Features of the 2024 Sample Design

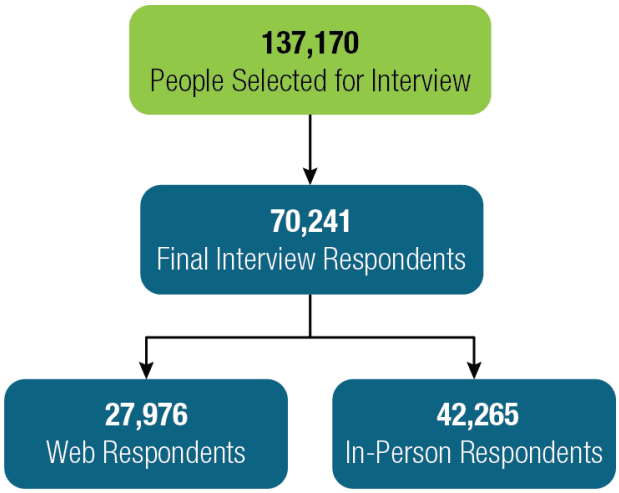
After completion of the NSDUH interview, a randomly selected subsample of adult respondents were invited to participate in an additional follow-up survey for the Mental Illness Calibration Study (MICS). MICS was conducted as part of the 2023 and 2024 NSDUHs. The goal of MICS was to fit a new prediction model for serious mental illness (SMI) among adults aged 18 or older. These new data are being used to create updated model-based estimates of SMI and other mental illness categories at the national and domain levels (e.g., by age group and race/ethnicity). Respondents randomly selected for MICS were asked to participate in a follow-up clinical interview conducted using diagnosis criteria in the *Diagnostic and Statistical Manual of Mental Disorders,* 5th edition (DSM-5; American Psychiatric Association, 2013). The 2023 and 2024 MICS samples were designed to yield 2,000 clinical interviews per year. For more information on the MICS sample design and results, see the 2024 Sample Design and Experience Report (CBHSQ, forthcoming b).

2.1.3 Sample Results for the 2024 NSDUH

The final sample selection and interview results for 2024 are shown in Figure 2.2. More information on the 2024 national screening and interview results can be found in Chapter 3.

In 2024, the actual sample sizes in the 12 largest states ranged from 1,441 to 4,517. In the remaining states, the actual sample sizes ranged from 815 to 1,459. For specific sample sizes by state, see the 2024 Sample Design and Experience Report (CBHSQ, forthcoming b).

Figure 2.2 2024 Sample Selection and Interview Results



2.2 Data Collection Methodology and Questionnaire Changes for 2024

2.2.1 Multimode Data Collection Procedures

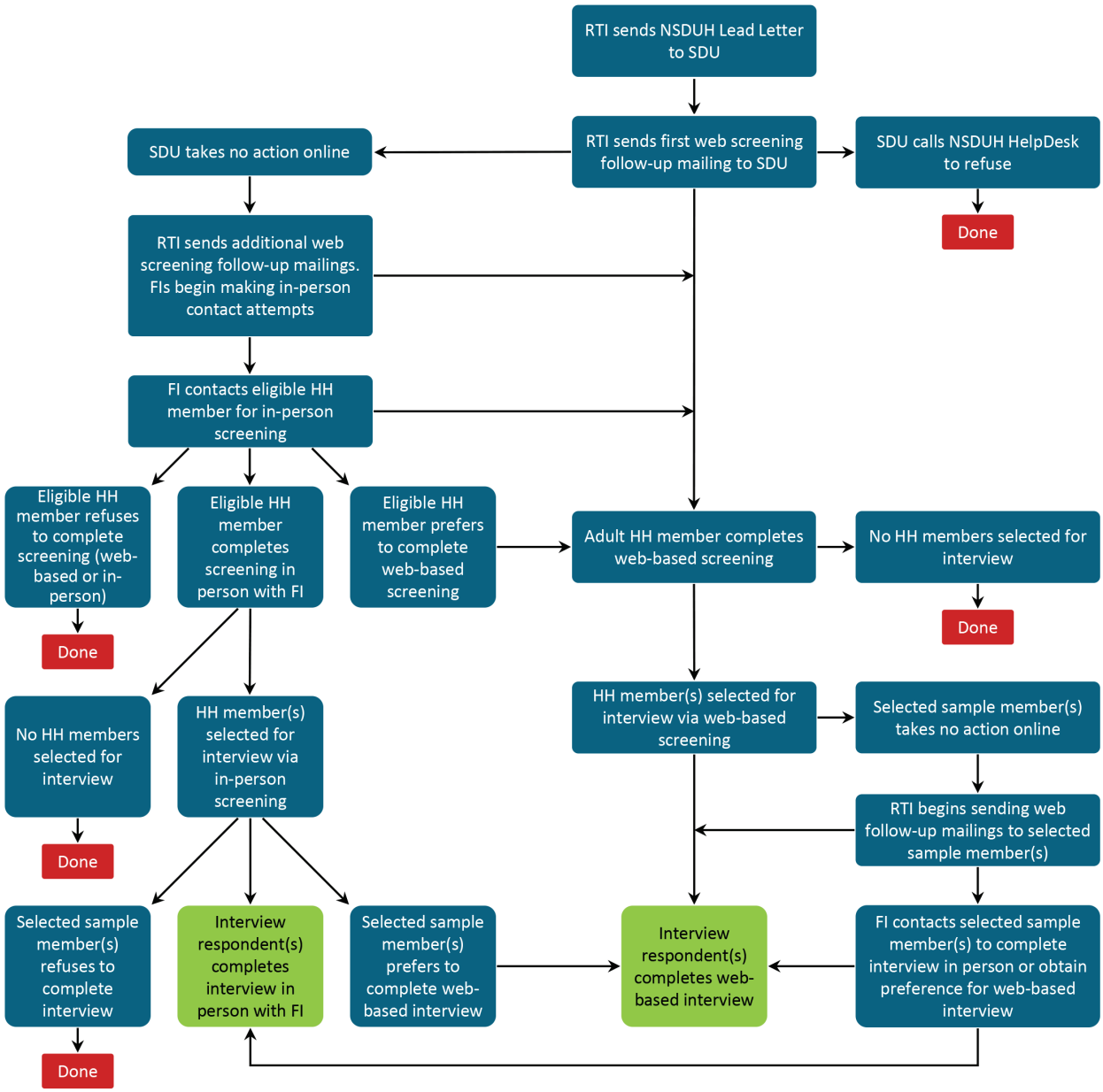
The 2024 NSDUH used multimode data collection throughout the year, in which respondents completed the survey in person or via the web. Figure 2.3 presents a flowchart of the multimode data collection procedures.

In-person data collection began after potential respondents were first given the opportunity to complete the survey via the web. As shown in Table 2.2, the start of in-person data collection in each quarter of 2024 followed the start of web-based data collection by slightly more than 1 week in Quarter 1 and by 3 to 5 days in the remaining quarters. Throughout 2024 data collection, sampled individuals could choose to participate via the web. Therefore, even after being contacted in person, respondents could complete screenings or interviews online instead of doing so in person with a field interviewer (FI). Where available, respondents also could switch data collection modes.

Regardless of the interview mode, data collection for completed responses in NSDUH had the following set of basic steps:

- making contact with the sample dwelling unit (SDU);
- screening of the SDU to identify residents aged 12 or older and determining whether zero, one, or two members would be selected to complete the interview;
- obtaining consent from SDU members aged 18 or older and/or permission from parents or adult guardians and respondent assent from adolescents aged 12 to 17 who were selected for an interview;
- administering the NSDUH questionnaire to consenting respondents; and
- verifying the validity of screening and interview data.

Figure 2.3 Multimode Data Collection Procedures



FI = field interviewer; HH = household; RTI = RTI International, the NSDUH contractor; SDU = sample dwelling unit.

Table 2.2 Schedule of 2024 Data Collection Dates, by Mode and Quarter

Quarter	Web-Based Data Collection (Start and Finish)	In-Person Data Collection (Start and Finish)
1	January 5-March 31, 2024	January 11-March 31, 2024
2	April 1-June 30, 2024	April 5-June 30, 2024
3	July 1-September 30, 2024	July 3-September 30, 2024
4	October 1-December 14, 2024	October 3-December 14, 2024

Source: SAMHSA, Center for Behavioral Health Statistics and Quality, National Survey on Drug Use and Health, 2024.

For both data collection procedures, communications with potential respondents stressed confidentiality. Consequently, respondents' names were not collected with the interview data. For web-based data collection, the website's https encryption provided sufficient security for information entered from compatible devices via any Internet connection (e.g., public Wi-Fi, cellular phone, at-home Wi-Fi).

After completion of the interview, a subset of adult respondents was selected for MICS based on the respondents' psychological distress scores in the main NSDUH interview (see Section 3.4.9). These respondents were asked to participate in a follow-up clinical interview using diagnostic criteria in the DSM-5. See Section 2.1.2 for additional information on MICS.

2.2.1.1 Initial Contact Procedures

Introductory letters were sent to all SDUs with mailable addresses.[5] The initial introductory letter was written in English with a Spanish version printed on the back. SDU members were given the option to complete the data collection procedures via the web, or adult SDU members could call the NSDUH contractor to request in-person interviewing for their SDU. The letters provided the website address to access the online screening and a unique participant code specific to each SDU. SDU members needed to use this code to participate via the web, which allowed tracking of which SDUs had responded and which had not.

2.2.1.2 Household Screening and Selection Procedures

Household screening could be completed in English or Spanish. For web-based data collection, these were the only languages available.[6] Because participation via the web required respondents to be able to read the questions online, SDU members who were blind or unable to read English or Spanish were not eligible to respond to the screener via the web. SDU members who did not have Internet access or access to an Internet-compatible device (e.g., smartphone, tablet, computer) also were ineligible for screening via the web.

For both modes of data collection, an adult resident aged 18 or older needed to consent to be the screening respondent and provide basic data on characteristics of all the household members aged 12 or older who lived at the address for most of the calendar quarter. Using the demographic data collected, a preprogrammed selection algorithm selected zero, one, or two individuals for the interview. The screening concluded if no members of the SDU were selected for an interview. If an adolescent aged 12 to 17 was selected for the main interview, permission from a parent or adult guardian and assent from the selected adolescent were needed for the adolescent to complete the interview.

[5] If a street address could not be determined, SDUs were not sent an introductory letter.
[6] See the *2023 National Survey on Drug Use and Health (NSDUH) Methodological Resource Book, Section 8: Data Collection Final Report* (CBHSQ, 2025b) for details on in-person screening for households in which members spoke only Spanish or when a language other than English or Spanish was encountered.

2.2.1.3 Interview Procedures

The main NSDUH interview could be completed in English or Spanish. If a person who was selected did not speak English or Spanish, the interview was not conducted. As with household screening, SDU members who were blind or unable to read English or Spanish were not eligible to respond to the interview via the web. SDU members who did not have Internet access or access to an Internet-compatible device also were ineligible for interviewing via the web.

If one or both of the SDU members selected for an interview were adolescents aged 12 to 17, verbal parental permission and adolescent assent were required before the adolescent could participate in the interview.

- Web Interviews
  - The parent or adult guardian and the adolescent were required to call a toll-free number together to speak with an RTI project representative before proceeding with the interview.
- In-Person Interviews
  - If an adolescent was selected and a parent or adult guardian was available, the FI attempted to obtain permission from the parent or adult guardian for the adolescent to complete the interview. If a parent or adult guardian and the selected adolescent were at home, the FI obtained assent from the selected adolescent after obtaining parental permission.
  - If the parent or adult guardian was at home and the adolescent was not, the FI returned at a later date to obtain adolescent assent after having obtained parental permission.

Web and in-person versions of the 2024 questionnaire had the same content. However, all web interview questions were self-administered regardless of the topic, and there was no option to listen to the question rather than read it. For in-person data collection, some questions about demographic characteristics and questions about household composition, health insurance coverage, and the respondent's personal and family income were interviewer administered using computer-assisted personal interviewing (CAPI). All other questions, including those about more sensitive topics, were self-administered using audio computer-assisted self-interviewing (ACASI) for in-person interviews. ACASI allowed in-person respondents to complete the interview if they had limited reading or visual ability.

Both versions of the interview began with demographic questions, followed by questions about sensitive topics, then by additional demographic questions such as immigration, current school enrollment, and employment and workplace issues. Questions about household composition, the respondent's health insurance coverage, and the respondent's personal and family income were also included at the end of the interview.

Interview respondents who completed the interview received a $30 incentive as a token of appreciation. Respondents who completed the interview in person received the money in cash.

Respondents who completed the web-based interview selected a preferred method for receiving the incentive: an electronic gift code sent to a designated email address or a physical gift card delivered to the SDU.

2.2.2 Notable Questionnaire Changes for 2024

In 2024, NSDUH was conducted using multimode data collection, and the two forms of the questionnaire for web or in-person administration were kept as identical as possible. Hence, all notable changes to the questionnaire described in this section were made to both forms. Differences between modes were primarily to accommodate self-administration via the web. For example, the web questionnaire did not include audio recordings of questions. Instead, pronunciations were spelled out visually on several screens, particularly for hallucinogens, inhalants, and prescription drug introduction screens, to help adolescents and respondents with a lower reading level understand the questions accurately.[7]

There were a large number of changes for the 2024 NSDUH, including dropping the coronavirus disease 2019 (COVID-19) section and all COVID-19-related questions. The following additional questions were removed from the questionnaire:

- Core Demographics
  - Removed questions about whether respondents were members of a reserve component currently serving full time in an active-duty status.
- Health
  - Removed questions about sexually transmitted diseases in the past 12 months.
- Youth Experiences[8]
  - Removed the questions about how adolescents' friends would feel about them using certain substances.
  - Removed the question about whether adolescents participated in a program or meeting in the past year for their own or a family member's substance use.
  - Removed the question about whether adolescents participated in pregnancy or sexually transmitted disease prevention programs in the past year.
  - Removed the question about whether adolescents had been exposed to any substance use prevention messages outside of school in the past year.
  - Removed the question about frequency of religious service attendance.

[7] For more details on adaptations for web interviews starting with the 2020 NSDUH, see Section 2.2.2.4 in the *2020 National Survey on Drug Use and Health (NSDUH): Methodological Summary and Definitions* (CBHSQ, 2021b).

[8] The question about whether adolescents spoke with at least one parent in the past year about the dangers of substance use was removed from the 2024 questionnaire. It has been added back to the 2025 questionnaire.

- Emerging Issues
  - Removed questions about marijuana vaping.
- Back-End Demographics
  - Removed the question about the state in which the respondent lived 1 year ago.
- Employment
  - Removed the question about workplace drug and alcohol use policies and access to workplace education and support.
  - Removed the questions about workplace drug and alcohol testing and respondent preferences for workplace testing.
- Household Roster
  - Removed the question about specific relationships of siblings in the household.

The following new questions were added to the 2024 questionnaire:

- Marijuana
  - Added a question about lifetime marijuana vaping to replace the question that was dropped from the emerging issues section.
- Hallucinogens
  - Added a question asking about the most recent use of psilocybin and subsequent consistency check questions.
- Pain Relievers Screener
  - Added a series of five "OTHER, Specify" questions for any use of other prescription pain relievers to capture details on whether respondents used opioid or nonopioid pain relievers.
- Alcohol and Drug Treatment
  - Added a follow-up question for respondents who reported inpatient or outpatient treatment but did not report any substances for which they received treatment.
  - Added a question about the use of overdose reversal medicine in the past year.
- Youth Experiences
  - Added questions about the General Anxiety Disorder Scale (GAD-7) for adolescents.
- Mental Health
  - Added questions about the General Anxiety Disorder Scale (GAD-7) for adults.

- Back-End Demographics
  - Added a question to capture information on the state or U.S. territory in which respondents were born.

There were other changes to the questionnaire as well. Some may have an effect on comparability with specific previous estimates, and these changes are discussed in Chapter 3.

The 2024 questionnaire included the following wording changes to existing items:

- Alcohol
  - Revised the introduction to update and simplify the definition of a drink.
  - Added help text to the introduction to show an updated and simplified list of example beverages.
- Cocaine
  - Revised the phrase "cocaine, in any form" to "any form of cocaine" to be more consistent with questions in other sections.
- Heroin
  - Revised the introduction to include example forms of heroin and ways of use.
- Inhalants
  - Revised the phrase "for kicks or to get high" to "for fun or to get high." Questions about the last use of inhalants in the prior substance use section later in the questionnaire also had this revision.
- Youth Experiences
  - Revised the question about parental limits on time watching television to include parental limits on time with tablets, smartphones, computers, or video games in addition to television.
  - Revised the question about how the respondent's parents would feel about the respondents smoking one or more packs of cigarettes a day to ask about smoking cigarettes every day.
  - Revised the question asking how adolescent respondents feel about their peers smoking one or more packs of cigarettes a day to ask about their peers smoking cigarettes every day. An introductory phrase was also added to this question.

- Alcohol and Drug Treatment[9]
  - Revised the introductory text about inpatient and outpatient alcohol and drug treatment to mention "any of the locations below" for response choices for specific locations.
  - Added text to response choices for each inpatient and outpatient alcohol and drug location to indicate that the treatment was "for alcohol or drug treatment."
  - Revised the heading in the grid format for respondents to report the locations where they received inpatient treatment from "Stayed overnight or longer for alcohol or drug treatment in…" to "LOCATION."
  - Revised the heading in the grid format for respondents to report the locations where they received outpatient treatment from "Received outpatient alcohol or drug treatment at…" to "LOCATION."
- Mental Health Services Utilization[9]
  - Revised the introductory text about inpatient and outpatient mental health treatment to mention "any of the locations below" for response choices for specific locations.
  - Added text to response choices for each inpatient and outpatient mental health treatment location to indicate that the treatment was "for mental health, emotions, or behavior."
  - Revised the heading in the grid format for respondents to report the locations where they received inpatient treatment from "Stayed overnight or longer for treatment in…" to "LOCATION."
  - Revised the heading in the grid format for respondents to report the locations where they received outpatient treatment from "Received outpatient treatment at…" to "LOCATION."

The 2024 questionnaire also included changes to existing items other than wording changes:

- Health
  - Updated the question about current pregnancy to be asked of female respondents aged 12 to 50. Before 2024, the question was asked of female respondents aged 12 to 44.
  - Updated the question about duration of the current pregnancy to ask about pregnancy length in weeks rather than months.
- Social Environment; Youth Experiences
  - For questions on attitudes about substance use in both the social environment section and the youth experiences section, updated the three-point scale ranging

[9] Changes to the alcohol and drug treatment and mental health service utilization sections were presented to respondents starting in June 2024.

from "Neither approve nor disapprove" to "Strongly disapprove" to a five-point scale ranging from "Strongly disapprove" to "Strongly approve."

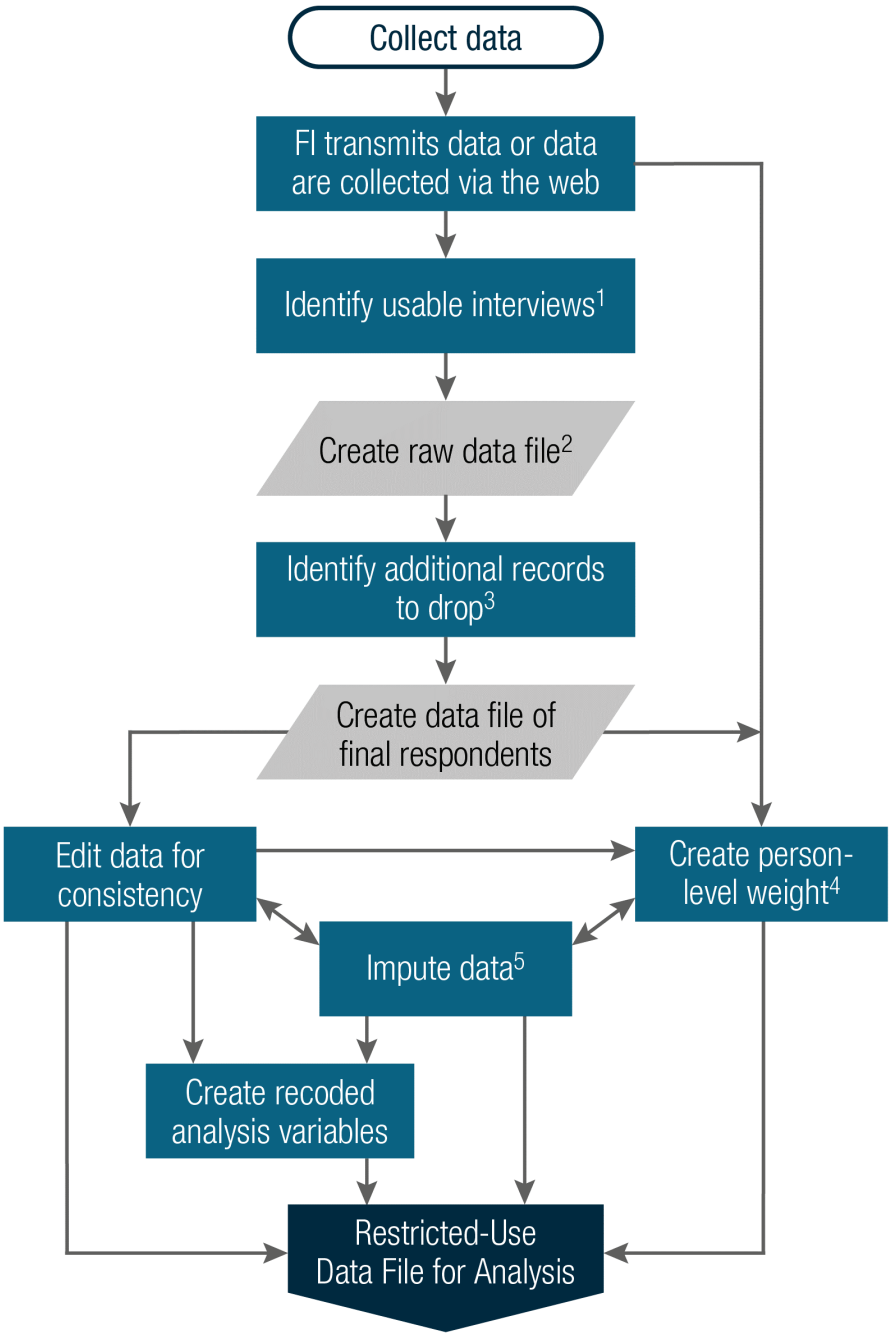
- Back-End Demographics
  - Updated residency questions to be asked of respondents who were born in a U.S. territory in addition to those who were born in the United States.

2.3 Data Processing

Survey data were either transmitted from FIs for in-person interviews or captured directly from web-based data collection. These data were initially processed to create a raw data file that consisted of one record for each usable interview (Section 2.3.1), with no logical editing or other corrections. Along with these steps, Figure 2.4 shows the processes applied to variables in the raw data file to create the final restricted-use file for analysis. These processes include the following:

- identification of additional records to treat as nonrespondents due to patterned responses in their substance use questions (Section 2.3.1);
- editing of data for internal consistency (Section 2.3.2);
- statistical imputation to replace missing, inexact, or nonspecific values in key variables after data had been edited (Section 2.3.3); and
- creation of analysis weights so that estimates would be representative of the target population (Section 2.3.4).

Figure 2.4 Main Data Processing Steps



[1] The usable interview rule removes interviews with excessive amounts of missing data in the initial set of substance use questions.
[2] The raw data file consists of one record for each interview with no logical editing or other corrections.
[3] Some interview records were not treated as final respondents because the patterns of answers in the substance use sections called into question the validity of the data. See Section 2.3.1 for details.
[4] Analysis weights are used in analyses to make the estimates representative of the target population.
[5] Statistical imputation replaces missing, inexact, or nonspecific values in key variables.

2.3.1 Criteria for Identifying Usable Interviews

A key step in the preliminary data processing procedures is to set the minimum item response requirements for interviews so that they can be used in weighting and further analysis (i.e., "usable" data). These procedures removed the data from interviews with too much missing data.

The following criteria were used to establish whether a 2024 interview could be included on the data files:

- The lifetime cigarette gate question CG01 must be answered as "yes" or "no."[10]
- In addition to the criterion for cigarettes, at least nine (9) of the following other substances must have "usable" data: (1) smokeless tobacco, (2) cigars, (3) alcohol, (4) marijuana, (5) cocaine (in any form), (6) heroin, (7) hallucinogens, (8) inhalants, (9) methamphetamine, (10) prescription pain relievers, (11) prescription tranquilizers, (12) prescription stimulants (i.e., independent of methamphetamine), and (13) prescription sedatives.

Further documentation on the specific usability criteria and reasons for certain substances not being included in the criteria (e.g., crack cocaine) can be found in the *2023 National Survey on Drug Use and Health (NSDUH) Methodological Resource Book, Section 10: Editing and Imputation Report* (CBHSQ, 2025c). Although the report is about 2023, the same criteria were used for the 2024 NSDUH.

In addition, if respondents were not paying attention to questions or were not taking the interview seriously for other reasons, they might use their computer keyboard or other input device (for web interviews) to enter nonsensical patterns of answers. For example, respondents could enter answers of "1" to all possible questions for several substances.[11] Thus, even among respondents who had sufficient data for their interviews to be considered usable, some interview records were not treated as final respondents because the patterns of answers in the substance use sections called into question the validity of the data. However, the number of records that are not treated as final respondents because of patterned responses is typically small (e.g., fewer than 10 otherwise usable interviews in 2024).

[10] A gate question is an initial question or set of questions that ask whether the behavior or characteristic of interest applies to the respondent. An affirmative response to a gate question leads to respondents being asked a series of other related questions. A response other than an affirmative one to all relevant gate questions results in respondents skipping additional questions on that topic and being routed to the next set of topics in the interview.

[11] For example, answers of "1" wherever possible in the sections for marijuana, cocaine, crack, heroin, and methamphetamine would mean that respondents used that substance in their lifetime, first used the substance at age 1, last used in the past 30 days, preferred to answer the 12-month frequency question as an average number of days per week in the past 12 months, used an average of 1 day per week in the past 12 months, and used on 1 day in the past 30 days.

2.3.2 Data Coding and Editing

Once a final respondent dataset is created, data coding and logical editing procedures were applied to all 2024 respondents, regardless of the following:

- mode of data collection (in person or via the web),
- language in which the interview was conducted (English or Spanish), or
- age of the respondent (12 to 17 or 18 or older).

Some questions were open-ended and asked respondents to type in a response. For instance, respondents might be asked to write in the names of other hallucinogens that they used in their lifetime. These typed responses were coded into numeric values for analyses. Data taken from these open-ended questions are referred to as "OTHER, Specify" data in this report and other NSDUH publications.

Logical editing involved using data from within a respondent's record to do the following:

- reduce the amount of item nonresponse (i.e., missing data) in interview records, including identification of items legitimately skipped;
- make related data elements consistent with each other; and
- identify inexact, nonspecific, or inconsistent responses needing to be resolved through statistical imputation procedures (see Section 2.3.3).

The computer-assisted interviewing (CAI) program included features to reduce the opportunity for respondents to enter answers that were inconsistent with their previous answers. These features included logic to skip respondents out of questions that did not apply to them and checks to alert respondents when they entered an answer that was inconsistent with a previous answer. For example, a consistency check was triggered if respondents reported that the last time they used Ecstasy was more recent than the last time they used any hallucinogen. In this way, the inconsistency could be resolved while the interview was in progress. However, inconsistencies could remain in the completed interviews. The CAI program did not include checks for every possible inconsistency in respondents' answers. Therefore, some logical editing of NSDUH data was required. See the 2023 Editing and Imputation Report (CBHSQ, 2025c) for the most recent documentation of editing procedures for NSDUH data.

2.3.2.1 General Principles of Editing NSDUH Data

Because the CAI logic controlled whether respondents were asked certain questions based on their answers to previous questions, an important aspect of editing the NSDUH data throughout the questionnaire involved identifying where questions had been legitimately skipped because they did not apply. The following are scenarios that might cause questions to be legitimately skipped:

- an event (e.g., use of a particular substance) occurring at least once in the respondent's lifetime, but the respondent previously reported the event never occurred;
- an event occurring in a particular time period (e.g., within the past 12 months), but the respondent previously reported the event occurred less recently; or

- respondents with a particular demographic characteristic (e.g., adults aged 18 or older), but the respondent was not part of that group.

These scenarios are represented by different codes in the edited variables.

Another important guideline in editing the data was that responses from one section (e.g., pain relievers) generally were not used to edit variables in another section (e.g., tranquilizers). For example, if a respondent specified the misuse of a tranquilizer as some other pain reliever the respondent misused in the past 12 months, then this "OTHER, Specify" response for pain relievers was not used to edit the data for tranquilizers. This principle of not using data in later sections to edit data in earlier sections has been important for maintaining comparable measures between years for outcomes of interest (e.g., substance use).

One exception to this principle of not editing across sections involved situations in which responses in one or more sections governed whether respondents were asked questions in a later section. For example, the alcohol and drug treatment section in 2024 was relevant only for respondents who reported the lifetime use of alcohol or other drugs, excluding tobacco products or nicotine vaping. Respondents who reported in the initial substance use sections that they had never used alcohol or drugs were not asked the questions in the alcohol and drug treatment section. In this situation, the responses from the earlier substance use sections were used to edit the substance use treatment variables to indicate that respondents were not asked the treatment questions because they reported they never used any of the relevant substances.

2.3.2.2 Editing of Prescription Drug Data

For alcohol, marijuana, cocaine, crack, heroin, hallucinogens, inhalants, and methamphetamine, there was only one section of questions for each drug. However, in the prescription drug questionnaire sections, respondents were first asked a series of screening questions about *any* use of specific prescription drugs in the past 12 months (i.e., use or misuse) or any lifetime use if they did not report past year use. Respondents were then asked about misuse in the past year of any of the specific prescription drugs they reported using in that period. The additional screener questions made the editing process more complex. See Chapter 4 for more details on the use and misuse of prescription drugs.

Consistent with the general editing principles, an important component of editing the prescription drug variables in 2024 involved assignment of codes to indicate when respondents were not asked inapplicable questions. For example, if respondents did not report any use of a particular prescription drug in the past 12 months, then the corresponding edited variables for misuse of that drug in the past 12 months[12] were assigned codes to indicate the questions did not apply.

[12] In the 2024 NSDUH, misuse was defined as use "in any way a doctor did not direct you to use [it or them]" and focused on behaviors that constitute misuse of prescription drugs. Examples of misuse were presented to respondents and included (1) use without a prescription of the respondent's own; (2) use in greater amounts, more often, or longer than told to take a drug; and (3) use in any other way a doctor did not direct the respondent to use a drug.

Because of the structure of the 2024 prescription drug questions, respondents were not asked about their most recent misuse of any prescription drug in that general category (e.g., most recent misuse of any pain reliever). Rather, variables for the most recent misuse of prescription pain relievers, tranquilizers, stimulants, and sedatives were created from respondents' answers to questions about the misuse of any prescription drug in the category in the past 30 days, misuse of specific prescription drugs in a given category in the past 12 months, and lifetime misuse of any prescription drug in the category. The following general principles were applied in creating the variables for the most recent misuse of any prescription drug in a given category in the 2024 data:

- Respondents who reported misuse of prescription drugs in the past 30 days were classified as having last misused prescription drugs in the past 30 days.
- Respondents who reported misuse of one or more specific prescription drugs in the past 12 months were classified as having last misused prescription drugs more than 30 days ago but within the past 12 months, provided they reported they did not misuse any drug in that category in the past 30 days.
- Respondents who reported lifetime (but not past year) misuse of prescription drugs were classified as having last misused prescription drugs more than 12 months ago, provided (1) they answered all applicable questions about misuse of specific prescription drugs in the past 12 months as "no"; or (2) they reported any use of prescription drugs in their lifetime and they explicitly reported they did not use any prescription drug in that category in the past 12 months.
- Respondents who reported they never used or never misused prescription drugs were classified as never having misused prescription drugs. (The coding of the variables for most recent use did not distinguish between respondents who never used prescription drugs and lifetime users who never misused prescription drugs.)

Some respondents provided indefinite information on when they last misused prescription drugs. For example, if respondents reported misuse of one or more specific prescription drugs in the past 12 months but they did not know or refused to report whether they misused any prescription drug in the past 30 days, it could be inferred these respondents misused prescription drugs in the past 12 months and potentially in the past 30 days. In these situations, a temporary "indefinite" value for the most recent period of misuse was assigned the variables created for the most recent misuse of pain relievers, tranquilizers, stimulants, and sedatives to indicate use at some point in the past 12 months. A final specific value was statistically imputed for most recent use in the past 30 days or more than 30 days ago but in the past 12 months.

In addition, respondents were instructed to report the use or misuse of prescription drugs and not to report the use or misuse of over-the-counter pain relievers, stimulants, and sedatives.[13] Therefore, if a respondent reported misuse in the past 12 months of only over-the-counter drugs or other substances that are not available by prescription, the respondent was logically

[13] This instruction was not included for prescription tranquilizers because no tranquilizers are available in the United States without a prescription.

inferred not to have misused any prescription drug in that category in the past 12 months. Similarly, beginning in 2024, if respondents reported in the pain reliever screener section that the only pain relievers they used in the past 12 months were over-the-counter or other nonprescription drugs, the respondents were logically inferred not to have used prescription pain relievers in the past 12 months.[14] Because these respondents answered "yes" to the questions about any use of pain relievers in the past 12 months or misuse in the past 12 months of any other prescription drug in a given category (e.g., tranquilizers), they were not asked about lifetime use or misuse of any prescription drug in that category; the CAI program handled these respondents as though they had used or misused prescription drugs in the past 12 months. Consequently, statistical imputation was used to assign a final value for whether these respondents misused prescription drugs more than 12 months ago or never in their lifetime.

2.3.3 Statistical Imputation

Imputation is defined as the use of statistical methods to replace missing values with nonmissing values. For many NSDUH variables, missing data are replaced with statistically imputed data after editing is complete. Missing or nonspecific values are imputed in NSDUH using two imputation methods: predictive mean neighborhood (PMN) and modified PMN (modPMN). Both methods replace the missing value somewhat randomly, informed by other information that is known about the respondent. See the 2023 Editing and Imputation Report (CBHSQ, 2025c) for the most recent documentation of imputation procedures for NSDUH data.

Examples of key NSDUH variables that underwent statistical imputation include the following:

- demographic characteristics,
- substance use,
- nicotine dependence,
- substance use disorder for drugs or alcohol,[15]
- psychological distress and impairment among adults,
- suicidal thoughts and behavior,
- major depressive episode,
- receipt of substance use treatment or other services for the use of alcohol or drugs, and
- receipt of mental health treatment or other services to help with mental health, emotions, or behavior.

In 2024, variables that were newly statistically imputed included those for lifetime marijuana vaping, lifetime use and most recent use of psilocybin, past year use and past year misuse of prescription opioids, receipt of overdose reversal medication, and symptoms of generalized anxiety disorder.

[14] "OTHER, Specify" questions were added to the pain reliever screener section of the questionnaire in 2024, allowing respondents to report which "other" pain relievers they had used in the past 12 months.
[15] In the 2024 NSDUH, drug use included use of marijuana, cocaine (including crack), heroin, hallucinogens, inhalants, or methamphetamine or any use of prescription stimulants, tranquilizers or sedatives (e.g., benzodiazepines), or pain relievers.

For the 2024 NSDUH, most variables that underwent statistical imputation (approximately 90 percent of imputed variables) required less than 5 percent of their records to be logically assigned or statistically imputed.

2.3.4 Development of Analysis Weights

This section discusses the general approach used to develop NSDUH person-level analysis weights for 2024.[16] The general weighting method used in 2024 is similar to what has been used in past NSDUHs and is described in Section 2.3.4.1. Starting with the 2021 NSDUH, additional procedures were applied that partially accounted for multimode data collection. These additional procedures are described in Section 2.3.4.2.[17]

2.3.4.1 General Weighting Approach

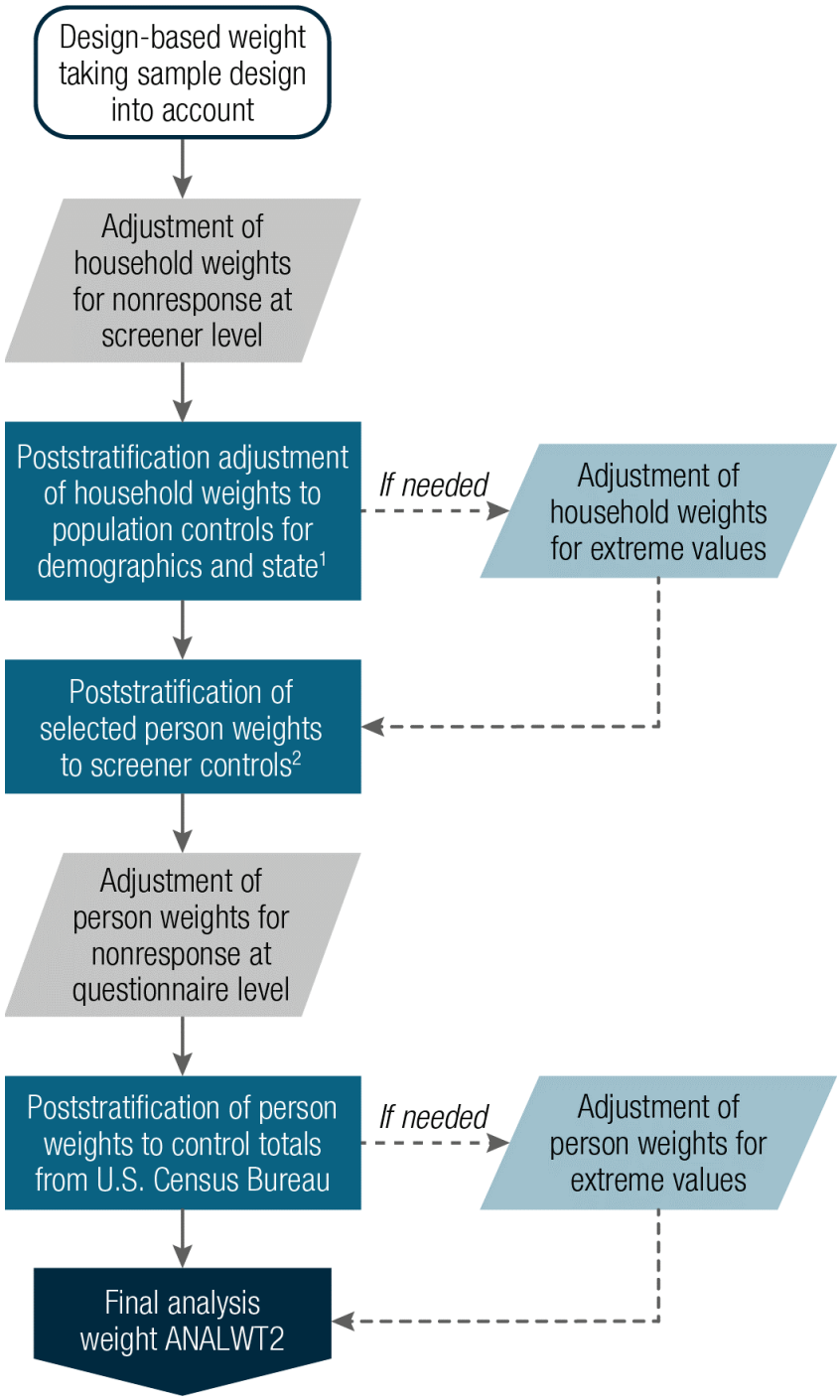
The general approach to developing NSDUH person-level analysis weights involves two types of components: design-based weights and weight adjustment factors. Design-based weights were created for the 2024 NSDUH to reflect probabilities of selection at the five sample stages described in Section 2.1.1 and Figure 2.1. The design-based weights subsequently underwent additional steps of adjustment to produce the final person-level analysis weights as shown in Figure 2.5.

More detailed information about the weighting procedures for 2024 will appear in the *2024 National Survey on Drug Use and Health Methodological Resource Book* (CBHSQ, forthcoming a). Until that volume becomes available, refer to the 2023 Methodological Resource Book (CBHSQ, 2025a) for general weighting information; however, specific information about the 2024 weighting procedures will not be available in the 2023 volume.

[16] Person-level analysis weights refer to the weights used to produce population estimates from final survey respondents' data. Other special weights are also produced for NSDUH (e.g., pair weights for analysis of data from pairs of responding household members), but this report does not discuss the creation of these special weights.

[17] Starting with the 2022 NSDUH, an adjustment was made to account for the proportions of interviews completed via the web or in person. These adjustments also were applied to the original 2021 weight to produce an updated version of that weight to facilitate comparisons between years. This updated 2021 analysis weight was used for all SAMHSA reports published in 2024 that include 2021 data.

Figure 2.5 Overview of Steps in the NSDUH Weighting Procedures



[1] Poststratification of household weights to meet population controls for various household-level demographics was done to obtain census-consistent estimates based on the household rosters from all screened households.

[2] This adjustment poststratified the weights of selected household members to conform to the adjusted roster estimates. This step took advantage of the separate screening and interviewing nature of the NSDUH design.

2.3.4.2 Additional Weighting Procedures for Multimode Data Collection

In addition to the general methodology and procedures described in Section 2.3.4.1, the 2024 weighting procedures accounted for multimode data collection in the following ways:

- inclusion of an implicit unknown eligibility assignment for web-based data collection,[18]
- inclusion of a screening mode indicator and its interactions in the DU-level nonresponse adjustment,
- inclusion of an interview mode indicator and its interactions in the person-level nonresponse adjustment,
- inclusion of educational attainment and interview mode in poststratification adjustment models,
- inclusion of two-way interactions of quarter with other demographic characteristics in the adjustment models as a means of partially accounting for quarterly variations in interview mode proportions (i.e., proportions of interviews in each quarter that were completed in person or via the web), and
- development of special break-off analysis weights for final respondents who progressed far enough but did not complete the full interview.

Response rates for the web and in-person modes differed at both the screening stage and interview stage. To account for the different response rates for the two modes, the screening mode indicator and its two-way interactions by state and by quarter were included in the DU nonresponse adjustment models. The interview mode indicator and its two-way interactions by state, quarter, age group, sex, and race/ethnicity were included in the person-level nonresponse adjustment models.

Educational attainment was included in poststratification adjustment models because a greater proportion of adults who were college graduates responded to multimode data collection compared with the surveys prior to 2020 that included only in-person data collection. This difference was driven by a higher percentage of college graduates among adult web respondents. The control totals for educational attainment were obtained by multiplying the American Community Survey (ACS) educational attainment proportions by the year-specific civilian, noninstitutionalized population estimates received from the U.S. Census Bureau. For the 2024 NSDUH, 2023 ACS data were used to calculate educational attainment proportions.

The 2024 weighting procedures also included two-way interactions of quarter by demographic variables in the person-level weight adjustment models. Including these interactions helped to account for quarterly variations in the proportions of interviews that were completed via the web or in person. However, further adjustments were needed to take into account overall annual variations in the proportions of interviews completed in each mode.

[18] Web-based data collection yielded more sample dwelling units (SDUs) with unknown eligibility because of nonresponse from SDUs, and in-person contact attempts were not successful for determining eligibility.

Specifically, differences in key substance use and mental health estimates between the web and in-person data collection modes (also known as "mode effects") could distort differences in estimates from 2021 to 2024, unless the analysis weights are adjusted to take into account these different proportions.[19] The unweighted proportion of NSDUH interviews completed in person increased from 45.4 percent in 2021 to 60.2 percent in 2024.

Therefore, for the NSDUH weights since 2022, mode has been included as a main effect in the person-level poststratification adjustment, with a 30 percent target for the web mode and a 70 percent target for the in-person mode to standardize the weighted proportions for each mode. This mode adjustment also was applied to the weights for 2021 to produce revised weights that allow estimates for 2021 to be compared with those in later survey years without differences in estimates being confounded by changes in proportions of interviews in each mode. The mode-adjusted weight for 2021 should be used for any comparison of estimates for 2021 with those for 2022-2024. See the 2022 Methodological Summary and Definitions report (CBHSQ, 2023) for more information.

The last weight adjustment accounted for the number of adult web respondents who provided usable information on their substance use (see Section 2.3.1) but did not complete the full interview (i.e., "break-offs"). Specifically, additional break-off analysis weights were created to analyze the unimputed outcomes for the mental health section and subsequent sections of the questionnaire. Table 2.3 shows a list of questionnaire sections that require the use of the break-off weight starting with the mental health section and whether any variables in that section were imputed. The main analysis weight rather than the break-off analysis weight was used for imputed variables from these sections (e.g., lifetime MDE). Nevertheless, the break-off analysis weights still were needed for variables that were not imputed (e.g., college enrollment).

Table 2.3 Questionnaire Sections That Require Use of the Break-Off Analysis Weight and Imputed Variables That Use the Main Analysis Weight

Questionnaire Section	Variables That Require the Break-Off Analysis Weight[1]	Imputed Variables That Require the Main Analysis Weight[2]
Mental Health	Some	Kessler-6 variables on psychological distress;[3] WHODAS variables on impairment due to psychological distress;[4] serious thoughts of suicide, suicide plans, and suicide attempts in the past year; receipt of medical attention or a hospital stay because of a suicide attempt in the past year; and symptoms of generalized anxiety disorder[5]
Adult Depression	Some	Lifetime and past year MDE and past year MDE with severe impairment
Consumption of Alcohol	All	N/A

[19] Analyses conducted for the 2021 NSDUH indicated that key substance use and mental health estimates were subject to mode effects. See Chapter 6 in the 2021 Methodological Summary and Definitions report (CBHSQ, 2022) for more information on these analyses.

Table 2.3 Questionnaire Sections That Require Use of the Break-Off Analysis Weight and Imputed Variables That Use the Main Analysis Weight (continued)

Questionnaire Section	Variables That Require the Break-Off Analysis Weight[1]	Imputed Variables That Require the Main Analysis Weight[2]
Emerging Issues	Some	Lifetime use and recency of use for kratom, synthetic marijuana, synthetic stimulants, vaping of flavoring, and illegally made fentanyl
Market Information for Marijuana	All	N/A
Back-End Demographics	Some	Immigrant status and immigrant age at entry to the United States
Education	All	N/A
Employment	Some	Employment status
Household Composition (Roster)	Some	Household size, number of people aged younger than 18, number of people aged 65 or older, other family in household, number of family members in household, and number of family members in household aged younger than 18
Proxy Information	All	N/A
Health Insurance	Some	Type of coverage (Medicare, Medicaid/CHIP, CHAMPUS, Private, Other)
Income	Some	Source of income (Social Security, Supplemental Security Income, food stamps, public assistance, welfare), months on welfare, personal income, and family income

MDE = major depressive episode; N/A = not applicable; WHODAS = World Health Organization Disability Assessment Schedule.

NOTE: The mental health and subsequent sections listed in this table are affected by interview break-offs. The break-off analysis weight should be used for analyses involving adult respondents and analyses involving the population aged 12 or older.

[1] A response of "All" indicates that the break-off analysis weight should be used for all variables from this section. A response of "Some" indicates that the break-off analysis weight is needed for use with variables in that section that were not imputed. Imputed variables in that section used the main analysis weight.

[2] Listed are specific variables or measures that were imputed. An overall composite measure may be imputed (e.g., past year MDE), but the individual source variables used to make a composite measure may not be imputed. Similarly, an overall composite measure itself may not be imputed (e.g., any mental illness, serious mental illness, poverty status), but all source variables were imputed. If no variables in a section were imputed, then the section is marked "N/A."

[3] See Section 3.4.9 for a list of the Kessler-6 items.

[4] See Appendix A in the *Results from the 2024 National Survey on Drug Use and Health: Detailed Tables* (Center for Behavioral Health Statistics and Quality, 2025h) for a list of the WHODAS items included in the NSDUH questionnaire.

[5] See Section 3.4.11 for a list of the symptoms of generalized anxiety disorder.

Source: SAMHSA, Center for Behavioral Health Statistics and Quality, National Survey on Drug Use and Health, 2024.

To create the break-off analysis weights, respondents were classified into two groups: (1) break-off respondents and (2) non-break-off respondents. Break-off respondents did not complete the adult depression section of the interview (for adults aged 18 or older) or the youth experiences section of the interview (for adolescents aged 12 to 17). Non-break-off respondents completed the interview or broke off the interview after the adult depression section (for adults) or the youth experiences section (for adolescents).

Relatively few adolescents aged 12 to 17 broke off the interview after the youth experiences section. Therefore, a break-off adjustment was not performed for interview data from adolescents aged 12 to 17. Nevertheless, a break-off weight was created for adolescents for use in analyses for people aged 12 or older that required the break-off analysis weight. For these analyses, the break-off weight that was created for adolescents was the same as the main weight.

3. Statistical Methods and Measurement

3.1 Target Population

Prevalence estimates of substance use and mental health conditions from the National Survey on Drug Use and Health (NSDUH) are designed to describe the target population of the survey—the civilian, noninstitutionalized population aged 12 years or older living in the United States. This population covers residents of households (people living in houses or townhouses, apartments, or condominiums; civilians living in housing on military bases; etc.) and people in noninstitutional group quarters (shelters, rooming or boarding houses, college dormitories, migratory workers' camps, halfway houses, etc.). The 2023 American Community Survey (ACS) 5-year estimates reported 332.4 million people of all ages living in the United States, of whom 324.3 million were living in households (U.S. Census Bureau, n.d.), or about 97.6 percent of the total population of the United States.[20] Further, 2023 ACS 5-year estimates indicated that 2.8 million people (or approximately 0.8 percent of the total U.S. population) were residents of college dormitories.[21] Thus, the civilian, noninstitutionalized population aged 12 years or older, those who could possibly be contacted for the survey, would be expected to include *more than* 98.4 percent of the total U.S. population aged 12 years or older.

People excluded from the target population for the survey include active-duty military personnel, people experiencing homelessness and not in shelters, and residents of institutional group quarters. People also are excluded during data collection when they cannot complete the survey in either English or Spanish or when they are not physically or mentally capable of completing the interview.

Some of these subpopulations may have very different estimates of mental disorders and substance use and therefore may have specific issues or needs. For example, active military personnel may be exposed to combat situations or stressors associated with extended overseas deployment. In addition, military personnel have been shown to have significantly lower rates of illicit drug use but higher rates of heavy alcohol use compared with their counterparts in the civilian population (Bray et al., 2009). Some people living in institutional group quarters, such as jails or prisons, residential substance use treatment or mental health facilities, nursing homes, and long-term hospitals, may have higher rates of mental disorders or substance use disorders (SUDs) compared with the general population. People with no fixed address (e.g., people experiencing homelessness and not in shelters) are another population shown to have higher than average rates of mental disorders and substance use problems (Bassuk et al., 2015; Solari et al., 2014).

[20] See the estimated total U.S. population and estimated number of people in households averaged over 5 years in Table B09019.
[21] See the estimated group quarters population by type averaged over 5 years in Table B26203.

3.2 Estimation and Statistical Significance

The sampling error of an estimate is the error caused by the selection of a sample instead of interviewing every person in the population. The sampling error may be reduced by selecting a large sample and/or by using efficient sample design and estimation strategies (such as stratification, optimal allocation, and ratio estimation). The use of probability sampling methods in NSDUH allows estimation of sampling error from the survey data.

Estimates based on NSDUH data are presented in national reports and tables on the NSDUH Data Collection web page. The national estimates, along with the associated standard errors (SEs, which are the square roots of the variances), were computed for all tables using SUDAAN®, a multiprocedure software package. This software uses a Taylor series linearization approach to account for the effects of NSDUH's complex sample design features in estimating the SEs (RTI International, 2020). The SEs can be used to identify unreliable estimates and also to test for the statistical significance of differences between estimates. The final, nonresponse-adjusted, and poststratified analysis weights were used in SUDAAN to compute unbiased, design-based estimates. See Section 2.3.4 for more information on the analysis weights.

3.2.1 Variance Estimation for Estimated Numbers of People

The variances and SEs of estimates of both means and proportions can be calculated reasonably well in SUDAAN using a Taylor series linearization approach. Estimates of proportions (e.g., the percentage of people who used drugs) for a domain can be thought of as a ratio. Domains are the subgroups for which the estimates were calculated. The numerator of the ratio is the estimated number of people in that domain who had the characteristic of interest (e.g., substance users), and the denominator of the ratio is the total number of people in the domain (including people with or without the characteristic of interest, such as substance users and nonusers). For most population domains, SUDAAN is used to calculate direct estimates of the number of people, the domain size, and the ratio. SUDAAN also is used to estimate the respective SEs.

For certain populations, the domain size is fixed during the weighting process (Table 3.1), meaning that the domain size has been forced to match the respective U.S. Census Bureau or ACS population estimates. For these fixed domains, the domain size is assumed to be free of sampling error induced by the NSDUH design. However, there still would be sampling error associated with the estimated number of people having the characteristic of interest (e.g., substance use). In these situations, an alternative SE estimation method is used for the number of people with the characteristic of interest (i.e., the numerator of the proportion). Using this alternative method, the SE is calculated by multiplying the domain size by the SE of the mean or proportion. See the *2023 National Survey on Drug Use and Health (NSDUH) Methodological Resource Book, Section 13: Statistical Inference Report* (Center for Behavioral Health Statistics and Quality [CBHSQ], 2025d) for more details on variance estimation in NSDUH.

Table 3.1 Demographic and Geographic Domains Shown in the NSDUH National Reports and Detailed Tables Using the Alternative Standard Error Estimation Method for Calculating Standard Errors of the Estimated Number of People (Totals), 2024

Main Effects	Two-Way Interactions[1]
Age Group	**Age Group × Sex**
12-17	(e.g., males aged 12 to 17)
18-25	
26-34	**Hispanic Origin × Age Group**
35-49	(e.g., Hispanic or Latino people aged 18 to 25)
50-64	
65 or Older	**Age Group × Geographic Region**
Collapsed Age Group Categories from Above[2]	(e.g., people aged 12 to 25 in the Northeast)
Sex	**Sex × Hispanic Origin**
Male	(e.g., not Hispanic or Latino males)
Female	
Hispanic Origin	
Hispanic or Latino	
Not Hispanic or Latino	
Race[3]	**White, not Hispanic or Latino**
Geographic Region	
Northeast	
Midwest	
South	
West	
Education (Aged 18 or Older)	
Less Than High School	
High School Graduate	
Some College/Associate's Degree	
College Graduate	

NOTE: The alternative standard error (SE) estimation method for the estimated number of people (totals), $\mathrm{SE}\left(\hat{Y}_d\right) = \hat{N}_d \mathrm{SE}\left(\hat{p}_d\right)$, is applied when the domain size estimates, $\hat{N}_d$, are among those forced to match their respective U.S. Census Bureau or American Community Survey (ACS) population estimates through the weight calibration process.

NOTE: The alternative SE estimation method does not affect the SEs for the corresponding means and proportions. These latter SEs are calculated directly in SUDAAN® (RTI International, 2020), whereas the alternative SE estimation method is computed outside of SUDAAN using the formula provided in the first note.

Table 3.1 Demographic and Geographic Domains Shown in the NSDUH National Reports and Detailed Tables Using the Alternative Standard Error Estimation Method for Calculating Standard Errors of the Estimated Number of People (Totals), 2024 (continued)

NOTE: This table shows only the domains and domain combinations used in the *Results from the 2024 National Survey on Drug Use and Health: Detailed Tables* (Center for Behavioral Health Statistics and Quality [CBHSQ], 2025h) and *Key Substance Use and Mental Health Indicators in the United States: Results from the 2024 National Survey on Drug Use and Health* report (CBHSQ, 2025g). Other domains and domain combinations (omitted here) also use this alternative SE estimation method, but they are not included in these specific reports or tables. For example, methodological studies or special requests often include a wider variety of domains and survey years. This variation requires the SE method to be assessed for each individual analysis. For a detailed list of domains for NSDUH forced to match their respective U.S. Census Bureau or ACS population estimates through the weight calibration process, see the *2024 National Survey on Drug Use and Health (NSDUH) Methodological Resource Book, Section 11: Person-Level Sampling Weight Calibration* report (CBHSQ, forthcoming d).

NOTE: The domains using the alternative SE estimation method for calculating the SE of the estimated number of people (total) are the same for both the main analysis weight and the break-off analysis weight (see Section 2.3.4 of this report for more details about these two weights).

[1] Unless otherwise noted, the domains for the two-way interactions are the same as the main-effect domains (including the collapsed age categories). Two-way interactions involving age group include the main-effect and collapsed age group categories. If age groups are listed in the two-way interaction columns, then only those age groups can be collapsed to form broader age categories.

[2] Main-effect age group categories shown in the table can be collapsed to form broader age group categories (e.g., 12 or older, 50 or older, 18 to 49, 26 to 49). Collapsed main-effect age group categories and two-way interactions with other main-effect demographic or geographic domains shown (e.g., males aged 50 or older) also use the alternative SE estimation method because the collapsed main effects will sum to the census totals for the category being defined. However, broader age groups that include only a subset of the main-effect age groups (e.g., 12 to 20, 21 or older, 15 to 44), age groups finer than the main-effect age groups (e.g., 12 to 13, 18 to 20), or two-way interactions of these types of collapsed age categories with other main-effect domains (e.g., females aged 15 to 44) should not use the alternative SE estimation method.

[3] Race is included as a main effect in this table for completeness; however, racial groups include all people within a given racial category, regardless of whether they are Hispanic or not Hispanic. In contrast, groups other than Hispanic in the national report tables and detailed tables are indented under the "Non-Hispanic" ethnicity row heading. For example, the domain for White people in national reports and tables is actually non-Hispanic White people and is therefore a two-way interaction. Thus, any additional domains crossed with non-Hispanic White people (e.g., White people aged 18 to 25) represent three-way interactions not using the alternative SE estimation method.

Source: SAMHSA, Center for Behavioral Health Statistics and Quality, National Survey on Drug Use and Health, 2024.

For the 2024 NSDUH, a "mixed-method" approach was used for estimated numbers of people (i.e., totals) in the national reports and tables to improve the accuracy of the associated SEs and to better reflect that some domain sizes were forced to match the U.S. Census Bureau or ACS population estimates.[22] The mixed-method approach assigns the method of SE calculation based on whether the domain size was fixed or not, as described above. The set of domains with a fixed domain size was restricted to main effects and two-way interactions to maintain continuity between years.[23] Table 3.1 includes the NSDUH national report and table domains that employed the alternative SE estimation method, including the main domains and the two-way interactions. However, Table 3.1 does not include an exhaustive list of domains and

[22] This "mixed-method" approach applies only to SEs for estimated numbers of people. All SEs for proportions and means are calculated directly in SUDAAN.

[23] In some years, not all of the race domains in Table 3.1 are forced to fully match the U.S. Census Bureau population estimates due to some models not converging. Even when race domains do not fully match the U.S. Census Bureau population estimates, the sampling variation for these domains is considered negligible. Therefore, the same race domains are considered fixed for every year.

interactions that are included in NSDUH national reports and detailed tables. For domains not included in Table 3.1, SEs for the estimates of totals are calculated directly in SUDAAN.

Tables 6.2 and 6.5 in the *Results from the 2024 National Survey on Drug Use and Health: Detailed Tables* (CBHSQ, 2025h) are examples where the SEs were calculated using the mixed method. These tables present estimates of any mental illness (AMI) and serious mental illness (SMI), respectively, among adults aged 18 or older within the domains of sex, Hispanic origin and race, and current employment. The estimated numbers of adults with AMI or SMI among the total population and age group (age group is the main effect), males and females (age group by sex interaction), and people who were Hispanic or not Hispanic (age group by Hispanic origin interaction) used the alternative SE estimation method to calculate the SEs for the estimated totals. The SEs for all other estimated numbers of people, including current employment, were calculated directly in SUDAAN. Racial or ethnic groups presented in the 2024 Detailed Tables are among people who were not Hispanic, unless noted otherwise. Because the domain for White people is actually for White people who were not Hispanic, it is a two-way interaction. Therefore, age group for White people is considered a three-way interaction, and the SEs by age group for White people were calculated directly in SUDAAN. Current employment is also not a fixed domain, and the SE of the estimated number of people would be calculated directly in SUDAAN.

3.2.2 Suppression Criteria for Unreliable Estimates and Rules for Presentation of Estimates

Direct estimates from NSDUH that are designated as unreliable are not shown in reports or tables and are replaced by asterisks (*). This section first describes the criteria used to define estimates as being unreliable, followed by guidelines for the presentation of estimates or related information in NSDUH national reports and tables.

Beginning with SAMHSA publications that include 2024 NSDUH data, direct estimates from NSDUH are defined as unreliable based on the following (Table 3.2):

- prevalence (for proportion estimates),
- relative standard error (RSE, defined as the SE divided by the estimate), and
- sample size.

The suppression criteria for the prevalence rates, shown at the top of Table 3.2, were changed for data products using 2024 NSDUH data, primarily for greater simplicity. The new suppression criteria for 2024 were also applied to estimates from 2021 to 2023 that were included in national reports and tables for the 2024 NSDUH. Consequently, some estimates from 2021 to 2023 that were suppressed in prior years may be published in reports and tables for the 2024 NSDUH. For documentation of the suppression rules used in earlier reports and tables, see Chapter 3 in the *2023 National Survey on Drug Use and Health (NSDUH): Methodological Summary and Definitions* (CBHSQ, 2024b).

Table 3.2 Summary of 2024 NSDUH Suppression Rules

Estimate	Suppress if any of the following conditions are met:
Estimated prevalence rate, $\hat{p}$, with unweighted sample size of denominator, *n*	(1) The estimated prevalence rate, $\hat{p}$, is 0 or 1, or (2) $\frac{SE(\hat{p})/\hat{p}}{-\ln(\hat{p})} > .175$ when $\hat{p} \le .5$, or $\frac{SE(\hat{p})/(1-\hat{p})}{-\ln(1-\hat{p})} > .175$ when $\hat{p} > .5$, or (3) *n* < 50. Note: Rule (1) applies in the rare situations where estimated percentages are calculated as 100.0 percent because the numbers of respondents in the numerator and denominator are identical by chance. It does not apply in situations where reported percentages are exactly 100.0 percent because the numbers of respondents in the numerator and denominator have been forced to be identical.
Estimated number (numerator of $\hat{p}$)	The associated estimated prevalence rate, $\hat{p}$, is suppressed.
Means other than percentages (e.g., mean age at first use), $\bar{x}$, with sample size, *n*	(1) $RSE(\bar{x}) > .5$, or (2) *n* < 10.

RSE = relative standard error; SE = standard error.
Source: SAMHSA, Center for Behavioral Health Statistics and Quality, National Survey on Drug Use and Health, 2024.

3.2.2.1 Suppression Rules for Proportions

Under the current rules, direct estimates are suppressed under any of the following specific situations:

1. Prevalence estimates are suppressed if they are exactly 0 or 1 (or expressed in percentages as 0 or 100 percent), not counting the exception noted in Table 3.2.
2. Proportion estimates $(\hat{p})$, or rates, are suppressed if[24]

$$\frac{SE(\hat{p})/\hat{p}}{-\ln(\hat{p})} > .175 \text{ when } \hat{p} \le .5$$

or

[24] The computational formula in Rule 2 was derived from $RSE[-\ln(\hat{p})]$: $RSE[-\ln(\hat{p})] \equiv SE[-\ln(\hat{p})]/[-\ln(\hat{p})]$ for $\hat{p} \le 0.5$. The Taylor series linearization of the numerator $SE[-\ln(\hat{p})]$ is $SE[-\ln(\hat{p})] = \sqrt{var[-\ln(\hat{p})]}$, which approximately equals $\sqrt{(-1/\hat{p})^2 \, var(\hat{p})}$ by Taylor series linearization, which in turn equals $SE(\hat{p})/\hat{p}$. The same principles apply for the computational formula when $\hat{p} > 0.5$, except that $\hat{p}$ is replaced with $1-\hat{p}$.

$$\frac{\mathrm{SE}(\hat{p})/(1-\hat{p})}{-\ln(1-\hat{p})} > .175 \text{ when } \hat{p} > .5$$

3. Proportion estimates are also suppressed if the unweighted sample size of the denominator, *n*, is less than 50.

Under Rule 1, prevalence estimates of exactly 0 or 1 are suppressed because they will disclose information about all respondents in the domain.

Rule 2 suppresses prevalence estimates with large estimated SEs that indicate low precision. Rather than a simple RSE, the suppression rule for proportions based on $\mathrm{RSE}[-\ln(\hat{p})]$ is used because it ensures that the strictness of the rule is similar for both large and small values of $\hat{p}$. Additionally, the separate formulas for $\hat{p} \le .5$ and $\hat{p} > .5$ produce a symmetric suppression rule; that is, if $\hat{p}$ is suppressed, $1-\hat{p}$ also will be suppressed. In contrast, a commonly used rule that suppresses estimates when $\mathrm{RSE}(\hat{p}) > .5$ is stringent when $\hat{p}$ is small (e.g., 0.01) but much less so when $\hat{p}$ is large (e.g., 0.49). In addition, a rule based only on $\mathrm{RSE}(\hat{p}) > .5$ is asymmetric in the sense that suppression occurs only in terms of $\hat{p}$; that is, there is no complementary rule for $(1-\hat{p})$. This would mean that changing the way that the question is formulated would result in different suppression decisions, even when the information is the same.

Under Rule 3, estimates are also suppressed if the sample size is less than 50 in order to protect against the instability of the estimated SE for small sample sizes. Unstable estimated SEs cause Rule 2 to be unreliable.

3.2.2.2 Suppression Rules for Estimated Totals and Means

Estimates of totals (i.e., estimated numbers of people) are suppressed if the corresponding prevalence rates are suppressed. Because of this rule, data users may encounter some unexpected results after applying the suppression rules. For instance, estimated percentages, $\hat{p}$, with the same numerator (and therefore the same corresponding total estimate) may be suppressed differently based on the size of the denominator.

Consider a situation where estimates of the misuse of prescription drugs in the past year are presented both among the population aged 12 or older and among people who used prescription drugs for any reason in the past year. Because the associated percentages have different denominators, $\hat{p}$ may not be suppressed for the population aged 12 or older but could be suppressed for the percentage among past year prescription drug users. In this situation, the estimated total would be displayed for the population aged 12 or older, but the same estimated total that is associated with the suppressed percentage among past year users would be suppressed.

Estimates of means other than percentages (e.g., mean of age at first use, mean number of days of use in the past 30 days or past 12 months) are suppressed if the RSEs of the estimates

were larger than .5 or if the sample size was smaller than 10 respondents. This rule is based on an empirical examination of the estimates of mean age at first use and their SEs for various sample sizes. Although arbitrary, a sample size of 10 appears to provide sufficient precision and still allow reporting by age at first use for many substances.

3.2.2.3 Rounding Rules and Format for Presentation of Certain Unsuppressed Estimates

In addition to the suppression rules, the NSDUH national tables and reports present rounded estimates and apply specific formats to estimates in order to provide additional confidentiality protection. Table 3.3 describes these rounding and presentation rules.

Table 3.3 Rounding Rules and Format for 2024 NSDUH Tables and Reports

Estimate Type	Rounding and Presentation Format
Estimated Totals and Standard Errors of Totals	▪ Total estimates are rounded to the nearest thousand people. ▪ Estimated numbers less than 500 people are shown as "<1" in tables to indicate that the number rounds to less than 1,000 people. ▪ Estimated numbers greater than or equal to 500 but less than 1,000 people are shown as "1" in tables because they round to 1,000 people.
Estimated Percentages	▪ Estimated percentages are rounded to the nearest tenth of a percent. ▪ Estimated percentages less than 0.05 are shown as "<0.1" in tables. ▪ Estimated percentages greater than or equal to 0.05 but less than 0.1 are shown as "0.1" in tables because they round to 0.1 percent.
Standard Errors of Percentages	▪ Estimated standard errors of percentages are rounded to the nearest hundredth of a percent. ▪ Estimated standard errors of percentages less than 0.005 are shown as "<0.01" in tables. ▪ Estimated standard errors of percentages greater than or equal to 0.005 but less than 0.01 are shown as "0.01" in tables because they round to 0.01.
Unweighted Sample Size	▪ Unweighted sample sizes are rounded to the nearest 10 people. ▪ Unweighted sample sizes less than 100 people are shown as "<100" in tables.
P Values	▪ *P* values for the test of differences between estimates are rounded to the nearest ten thousandth.

3.2.3 Statistical Significance of Differences

This section describes the methods used to compare prevalence estimates. Customarily, one way that observed differences between estimates are assessed is in terms of their statistical significance. Statistical significance is based on the probability that a difference as large as that observed would occur due to random variability in the estimates if there were no differences in the prevalence estimates being compared; this probability is known as the *p* value. The significance of observed differences is generally reported at the 0.05 and 0.01 levels when the *p* value is defined as less than or equal to the designated significance level. Although significance tests are often used to distinguish whether a difference is "real" or simply occurring

due to sampling, it is important to note that (1) a "real" difference does not necessarily mean a policy-relevant difference, and (2) tests are based on probability and may give a false impression of certainty.

Section 3.2.3.1 presents the general procedures for testing for statistically significant differences between pairs of estimates. These procedures were applied to all pairwise significance tests between estimates for 2 years of data[25] (e.g., between 2023 and 2024) and between subgroups within a single year (e.g., between adolescents aged 12 to 17 and young adults aged 18 to 25 in 2024). These procedures also applied to pairwise tests of annual averages between subgroups using pooled data (e.g., between adolescents aged 12 to 17 and young adults aged 18 to 25 using pooled data from 2023 and 2024). Section 3.2.3.2 presents information on testing between years, including linear tests of trends that are conducted for outcomes with 4 or more consecutive years of comparable data. Sections 3.2.3.3 through 3.2.3.5 present information about significance testing among categorical subgroups.

3.2.3.1 General Pairwise Significance Testing Procedures

For comparison of a pair of estimated percentages, the null hypothesis (no difference between percentages) was tested against the alternative hypothesis (there is a difference between percentages) using the standard *t* test[26] (with the appropriate degrees of freedom) for the difference in proportions test. Under the null hypothesis, the test statistic *t* is a random variable approximately following a *t* distribution for moderate to large sample sizes. Therefore, calculated values of *t*, along with the appropriate degrees of freedom, can be used to determine the corresponding probability level (i.e., *p* value). Whether testing for differences in estimates between years, from different populations within the same year, or from different populations in pooled data, the two percentages, in general, will not be independent. SUDAAN was used to compute estimates of *t* along with the associated *p* values using the analysis weights and accounting for the sample design, as described in Chapter 2.

A similar procedure was used for estimated numbers of people with a characteristic of interest, except when the alternative SE calculation was used (see Section 3.2.1). Whenever the SE for an estimated number of people was calculated outside of SUDAAN using the alternative SE estimation method, the corresponding test statistics also were computed outside of SUDAAN.

Under the null hypothesis, a test statistic with known variances asymptotically follows a standard normal *Z* distribution. However, because the variances of the test statistic are estimated, its distribution is more accurately described by the *t* distribution for finite sample sizes. As the degrees of freedom approach infinity, the *t* distribution approaches the *Z* distribution. Because most tests performed for the 2024 NSDUH had 750 degrees of

[25] Pairwise significance testing between 2 years of data is not the same as linear tests of trends that involve testing for changes across 4 or more consecutive years of comparable data, including the baseline year.

[26] The test statistic *t* is a ratio with the numerator defined as the difference between two prevalence estimates. The denominator is the standard deviation of this difference due to sampling variability. The denominator also takes into account when estimates are not independent.

freedom,[27] the *t* tests performed produce approximately the same numerical results as if a *Z* test had been performed.

If data users perform independent *t* tests for the difference of proportions using published estimates and SEs, the test results will typically be similar to tests performed in SUDAAN. However, testing results may differ for two reasons: (1) the correlation between estimates is included in SUDAAN tests, whereas it is not included in independent *t* tests; and (2) the reduced number of significant digits shown in the published estimates may cause rounding errors in the independent *t* tests.

The 2023 Statistical Inference Report (CBHSQ, 2025d) includes more details on these procedures, including detailed formulas. The Statistical Inference Report and the *2023 National Survey on Drug Use and Health: Public Use File Data Users' Guide* (CBHSQ, 2025e) provide examples of code for use in several software applications to calculate tests of differences.

3.2.3.2 Significance Testing of Estimates in Different Years

For the 2024 NSDUH national tables and some reports, statistical testing was conducted between estimates from different years (e.g., past month alcohol use in 2024 vs. the estimate in 2023). Where testing for the 2024 Detailed Tables (CBHSQ, 2025h) involved 3 years or more of comparable data for 2021-2024, pairwise testing was conducted between 2024 estimates and estimates in each of the previous years (i.e., 2021 vs. 2024, 2022 vs. 2024, and 2023 vs. 2024).

These statistical tests between estimates from different years for the 2024 Detailed Tables indicated whether an estimate in the first year being compared was lower than, greater than, or similar to the corresponding estimate in the second year being compared. Estimates from 2021 through 2024 are not comparable with estimates from previous years due to methodological changes to the survey, as described in Section 6 of the *2021 National Survey on Drug Use and Health (NSDUH): Methodological Summary and Definitions* (CBHSQ, 2022).

NSDUH estimates for each year, along with estimates of differences between years, are computed using SUDAAN to take the NSDUH sample design into account, including the overlapping area segments between successive survey years. When estimates are compared between adjacent years (e.g., 2023 and 2024), there is a small positive correlation because of the overlap in sampled areas (see Section 2.1.1). Ignoring the overlap may result in a slightly more conservative test outcome, with a greater likelihood of assuming no difference between the two estimates. When there is no overlap in the sample across years (e.g., between 2022 and 2024), the estimates are independent.

[27] The degrees of freedom for most statistical tests are typically calculated as the number of primary sampling units (variance replicates) minus the number of strata. Because there are two replicates per stratum, 750 degrees of freedom equal the number of strata in the national sample for 2024. However, the degrees of freedom are smaller for some statistical comparisons; specifically, the degrees of freedom are reduced for estimates on the average number of days people used substances. Details can be found in the *2023 National Survey on Drug Use and Health (NSDUH) Methodological Resource Book, Section 13: Statistical Inference Report* (CBHSQ, 2025d).

Small differences in estimates between years can be statistically significant because of NSDUH's large sample sizes. These large sample sizes in each year reduce the size of the variances and increase the likelihood that the *t* test will yield a statistically significant difference. As stated previously, however, small differences between estimates, even when not explained by sampling variability, are not necessarily relevant from a policy perspective.

Caution is needed when interpreting changes across years in the estimated numbers of people with a characteristic of interest. Respondents with large analysis weights can greatly influence the estimated number in a given year when the number of people in the population with that characteristic is relatively small (e.g., past month heroin users). Large analysis weights for some respondents in a single year can result in the estimated numbers of people with a given characteristic showing an increase between Year 1 and Year 2 (i.e., the year that had the respondents with large analysis weights). The potential for these kinds of year-to-year variations in estimated numbers of people also underscores the importance of reviewing changes across a larger range of years when possible, especially for outcome measures corresponding to a relatively small proportion of the total population.

Caution also is needed when interpreting changes in estimated numbers of people. A change in the estimated number of people with a characteristic of interest could reflect a change in the size of the overall population. Therefore, changes in estimated numbers of people should be considered in conjunction with the corresponding estimated percentages because percentages will control for changes in both the number of people with the characteristic of interest and the total number of people in the population.

Linear trend testing for outcomes of interest with four comparable NSDUH data points also was resumed for the 2024 NSDUH. Linear trend tests were conducted using SUDAAN (e.g., from 2021 to 2024 for cigarette use in the past month). Linear trend testing indicates whether estimates have decreased, increased, or showed no change across the overall period of interest. See the 2019 Statistical Inference Report (CBHSQ, 2021a) for additional details on how to compute linear trend tests.

3.2.3.3 Significance Testing among Categorical Subgroups in a Single Year

In addition to statistical tests of estimates across years, statistical tests were conducted among population subgroups within a single year. For more details on statistical testing among population subgroups, see the 2023 Statistical Inference Report (CBHSQ, 2025d).

For subpopulations defined by three or more levels of a categorical variable (e.g., age group, race/ethnicity), starting with a test of whether there is any distinction at all between levels is recommended to first control the error level for multiple comparisons. For this purpose, log-linear chi-square tests of independence of the subgroups and the prevalence variables were conducted using SUDAAN.

If Shah's Wald *F* test (transformed from the standard Wald chi-square) indicated overall significant differences, then the significance of each particular pairwise comparison of interest was tested using SUDAAN analytic procedures to properly account for the sample design, as

described in Section 3.2.3.1 (RTI International, 2020). This two-step procedure protected against inappropriate inferences being drawn due to the number of pairwise differences tested.[28]

For example, if the overall log-linear chi-square test of independence among standard age groups is statistically significant, then the following pairwise tests would be conducted between age group estimates: (1) 12 to 17 vs. 18 to 25, (2) 12 to 17 vs. 26 or older, and (3) 18 to 25 vs. 26 or older. When there is no overlap across groups, as is the situation for age groups, the estimates are independent.

The pairwise test results are generally not published in the national reports and tables, but they are used in NSDUH reports to determine whether to highlight a particular difference across subgroups (e.g., by age group). The differences in age groups (and differences among most population subgroups) were deemed to be statistically significant if the *p* value was less than 0.05.

3.2.3.4 Testing among Levels of Categorical Subgroups: Age Adjustment

Age adjustment is a method used to standardize age distributions across subgroups. The adjustment is used to estimate what the prevalence would be if the age distribution of each subgroup category was the same. These age-adjusted estimates are relative measures and not the actual prevalence estimates. NSDUH reports such as *Behavioral Health by Race and Ethnicity: Results from the 2021-2023 National Surveys on Drug Use and Health* (CBHSQ, 2024d) show age-adjusted estimates for subgroups. These estimates were created by adjusting the weights (Klein & Schoenborn, 2001) so that the age distribution matched that of the 2000 U.S. standard population. This adjustment was in accordance with federal agency best practices (Surveillance, Epidemiology, and End Results Program, n.d.).

The testing procedures discussed in Section 3.2.3.3 also apply to the testing of age-adjusted estimates. Specifically, for subpopulations defined by three or more levels of a categorical variable (e.g., race/ethnicity), overall log-linear chi-square tests of independence of the subgroups are first performed for the age-adjusted estimates. If the test result indicates statistical significance, then the significance of each particular pairwise comparison of interest is tested.

3.2.3.5 Significance Testing among Categorical Subgroups in Pooled Data from Multiple Years

Some NSDUH reports present estimates using pooled data from 2 or more survey years to improve the precision of estimates. These estimates represent annual averages across the

[28] Other statistical methods have been used for comparisons of pairwise differences across three or more levels of a categorical variable once an overall test (such as Shah's *F*) suggests there are differences. Although a Bonferroni adjustment can be applied to every pairwise difference (i.e., and not just to the pairwise difference with the lowest *p* value, which is sometimes recommended instead of Shah's *F* as an alternative overall test), this is an overly conservative procedure. For example, if a *p* value of .05 is set as the criterion for statistical significance and there are three pairwise comparisons, then the Bonferroni-adjusted *p* value for statistical significance becomes .017 (i.e., .05 divided by 3 equals .017).

number of years of data being pooled. For example, estimates in the 2021-2023 Behavioral Health by Race and Ethnicity report (CBHSQ, 2024d) were based on pooled 2021-2023 NSDUH data and reflected annual averages across those 3 years.

The testing procedures discussed for single-year data in Section 3.2.3.3 also apply to testing of estimates among subgroups using pooled data. Specifically, for subgroups defined by three or more levels of a categorical variable (e.g., age group, race/ethnicity), overall log-linear chi-square tests of independence of the subgroups are first performed. If the test result indicates statistical significance, then the significance of each particular pairwise comparison of interest is tested.

3.3 Other Information on Data Accuracy

The accuracy of survey estimates can be affected by nonresponse, coding errors, computer processing errors, errors in the sampling frame, reporting errors, and other errors not due to sampling. They are sometimes referred to as "nonsampling errors." These types of errors and their impact can be reduced through data editing, statistical adjustments for nonresponse, close monitoring and periodic retraining of interviewers, and improvement in various quality control procedures.

Although these types of errors often can be much larger than sampling errors, measurement of most of these errors is difficult. However, the effects of some types of these errors can be examined through proxy measures, such as response rates, and from other research studies.

3.3.1 Screening and Interview Response Rate Patterns

All dwelling units (DUs[29]) in the sample are screened to confirm eligibility and to select zero, one, or two members to participate in the survey. The weighted screening response rate (SRR[30]) is defined as the weighted number of successfully screened DUs[31] divided by the weighted number of eligible DUs (as defined in Table 3.4). Of the 927,672 eligible sampled DUs in 2024, 203,743 were screened successfully, for a weighted SRR of 21.91 percent (Table 3.4).

For the 2024 NSDUH, DU eligibility was imputed for sampled DUs whose members did not initiate the web screening interview and that were not visited by a field interviewer (i.e., DUs with unknown eligibility). Thus, the weighted SRR reflects an American Association for Public Opinion Research (AAPOR) standard definition (AAPOR, 2016), which assumes that some sampled DUs with unknown eligibility are actually eligible.

[29] A dwelling unit (DU) in NSDUH refers to either a housing unit or a group quarters listing unit, such as a dormitory room or a shelter bed.
[30] The DU weight is the number of DUs in the population represented by each sampled DU. Thus, the weighted screening response rate (SRR) is the estimated population-level SRR.
[31] A successfully screened DU is one in which all screening questionnaire items were answered by an adult resident of the DU and either zero, one, or two DU members were selected for the NSDUH interview.

Table 3.4 Weighted Percentages and Sample Sizes for the 2023 and 2024 NSDUHs; by Screening Results

Screening Result	Sample Size 2023	Sample Size 2024	Weighted Percentage 2023[1]	Weighted Percentage 2024[1]
Total Sample	874,764	1,009,122	100.00	100.00
Ineligible Cases	69,793	81,450	7.76	7.50
Eligible Cases	804,971	927,672	92.24	92.50
Ineligibles	69,793	81,450	7.76	7.50
Vacant	10,571	8,075	16.61	10.96
Not a Primary Residence	1,665	905	3.74	1.28
Not a Dwelling Unit	1,500	1,669	2.32	1.97
All Military Personnel	429	466	0.57	0.59
Other, Ineligible[2]	55,628	70,335	76.76	85.20
Eligible Cases	804,971	927,672	92.24	92.50
Screening Complete[3]	198,246	203,743	24.36	21.91
No One Selected	102,295	106,594	12.47	11.25
One Selected	55,689	56,745	6.91	6.18
Two Selected	40,262	40,393	4.98	4.48
Screening Not Complete	606,725	723,929	75.64	78.09
No One Home/No Contact Made	201,466	213,494	24.97	22.38
Respondent Unavailable/Web Nonrespondent	273,430	372,293	34.20	40.11
Physically/Mentally Incapable	1,219	1,302	0.15	0.15
Language Barrier – Hispanic	3,504	3,671	0.47	0.45
Language Barrier – Other	2,611	2,331	0.36	0.29
Refusal	76,001	69,872	9.48	7.69
Other, Access Denied[4]	47,764	60,241	5.93	6.93
Other, Eligible	354	435	0.03	0.04
Segment Not Accessible	0	0	0.00	0.00
Screener Not Returned	237	235	0.03	0.02
Fraudulent Case	139	55	0.02	0.01
Electronic Screening Problem	0	0	0.00	0.00

[1] Weighted percentages are computed using design-based weights and do not reflect adjustments to standardize the weighted proportion of interviews that were completed in person or via the web.

[2] Examples of "Other, Ineligible" cases are those in which all residents lived in the dwelling unit (DU) for less than half of the calendar quarter and dwelling units listed in error.

[3] The screening response rate (SRR) is calculated as the weighted number of successfully screened DUs divided by the weighted number of eligible DUs, $SRR = \frac{\sum w_{DU} complete_{DU}}{\sum w_{DU} eligible_{DU}}$, where w_{DU} is the inverse of the unconditional probability of selection for the DU and excludes all adjustments for nonresponse and poststratification defined in Section 2.3.4 of this report.

[4] "Other, Access Denied" includes all dwelling units to which the field interviewer was denied access, including locked or guarded buildings, gated communities, and other controlled access situations.

Source: SAMHSA, Center for Behavioral Health Statistics and Quality, National Survey on Drug Use and Health, 2023 and 2024.

In successfully screened DUs, eligible DU members who were selected were then asked to complete the interview. The number of respondents who completed the interview and the number of people who were selected were used to calculate the unweighted interview response rate (IRR) for NSDUH. The weighted IRR is defined as the number of people represented by the respondents divided by the number of people represented by the individuals who were selected (Table 3.5). In the screened DUs, a total of 137,170 people were selected, and completed interviews were obtained from 70,241 of these sampled people, for a weighted IRR of 51.54 percent. In an effort to maximize the IRR, all respondents were offered a $30 incentive to encourage them to complete the 2024 NSDUH interview. To be considered a completed interview, a respondent for the 2024 NSDUH needed to provide enough data to pass the usability criteria described in Section 2.3.1.

Among people who were selected for the interview but did not complete it, the most common reasons for not responding were (1) refusal to participate by the respondent or by the parent or adult guardian of the adolescent respondent (23.6 percent) and (2) did not participate because the residents were not available, never at home, or did not respond to the web survey (20.3 percent) (Table 3.5). Among demographic subgroups, the weighted IRR was higher among people aged 26 or older (52.8 percent), females (54.0 percent), Black people (54.2 percent), people in the South (52.8 percent), and residents of nonmetropolitan areas (53.2 percent) than among their corresponding counterparts (Table 3.6).

The overall weighted response rate (ORR) of 11.3 percent is defined as the product of the weighted SRR and weighted IRR, or ORR = SRR × IRR.

Table 3.5 Final Interview Code: Weighted Percentages and Sample Sizes, 2023 and 2024

Final Interview Code	12 or Older Sample Size 2023	12 or Older Sample Size 2024	12 or Older Weighted Percentage 2023[1]	12 or Older Weighted Percentage 2024[1]	12 to 17 Sample Size 2023	12 to 17 Sample Size 2024	12 to 17 Weighted Percentage 2023[1]	12 to 17 Weighted Percentage 2024[1]	18 or Older Sample Size 2023	18 or Older Sample Size 2024	18 or Older Weighted Percentage 2023[1]	18 or Older Weighted Percentage 2024[1]
Total	135,737	137,170	100.00	100.00	31,358	31,211	100.00	100.00	104,379	105,959	100.00	100.00
70 - Interview Complete	67,679	70,241	50.45	51.54	14,305	14,013	46.09	45.34	53,374	56,228	50.89	52.15
71 - No One at Dwelling Unit/Web Nonrespondent	18,259	21,149	11.88	13.39	5,389	6,599	16.58	20.27	12,870	14,550	11.41	12.71
72 - Respondent Unavailable	10,193	9,733	7.04	6.87	2,437	2,288	8.28	7.86	7,756	7,445	6.92	6.77
73 - Break-Off	621	602	0.56	0.49	14	13	0.04	0.03	607	589	0.61	0.53
74 - Physically/Mentally Incapable	1,437	1,368	1.45	1.41	342	308	1.20	1.00	1,095	1,060	1.47	1.45
75 - Language Barrier – Hispanic	758	790	0.69	0.76	141	158	0.51	0.53	617	632	0.71	0.78
76 - Language Barrier – Other	553	451	0.80	0.73	77	64	0.30	0.31	476	387	0.85	0.77
77 - Refusal	28,974	25,915	24.52	22.21	2,962	2,556	9.14	8.19	26,012	23,359	26.09	23.60
78 - Parental Refusal	5,235	4,694	1.54	1.36	5,235	4,694	16.62	15.08	-	-	0.00	0.00
91 - Fraudulent Case	24	32	0.01	0.02	10	15	0.03	0.06	14	17	0.01	0.02
Other[2]	2,004	2,195	1.06	1.22	446	503	1.22	1.33	1,558	1,692	1.05	1.20

NOTE: Some eligible and selected people at the dwelling unit screening stage were later determined to be ineligible based on information obtained at the interviewing stage. These ineligible people are not included in the table.

[1] Weighted response rates are computed using design-based weights and do not reflect adjustments to standardize the weighted proportion of interviews that were completed in person or via the web.

[2] "Other" includes eligible person moved, data not received from field, too dangerous to interview, access to building denied, computer problem, and interviewed wrong dwelling unit member.

Source: SAMHSA, Center for Behavioral Health Statistics and Quality, National Survey on Drug Use and Health, 2023 and 2024.

Table 3.6 Interview Response Rates and Sample Sizes for the 2023 and 2024 NSDUHs; by Demographic Characteristics

Demographic Characteristic	Selected People 2023	Selected People 2024	Completed Interviews 2023	Completed Interviews 2024	Weighted Response Rate 2023	Weighted Response Rate 2024
Total	135,737	137,170	67,679	70,241	50.45%	51.54%
Age						
12-17	31,358	31,211	14,305	14,013	46.09%	45.34%
18-25	34,315	34,276	16,285	16,837	46.69%	47.87%
26+	70,064	71,683	37,089	39,391	51.54%	52.82%
Sex						
Male	66,404	67,219	31,324	32,601	47.70%	48.96%
Female	69,333	69,951	36,355	37,640	53.05%	54.02%
Race/Ethnicity						
Hispanic	27,938	27,725	13,567	13,721	47.81%	48.40%
Non-Hispanic, White	78,498	79,480	39,130	40,781	51.32%	52.46%
Non-Hispanic, Black	15,034	15,925	7,985	8,649	53.44%	54.17%
Non-Hispanic, All Other Races	14,267	14,040	6,997	7,090	46.18%	48.74%
Region						
Northeast	26,396	24,842	13,000	12,331	52.45%	51.97%
Midwest	31,678	32,588	15,615	16,659	51.14%	51.66%
South	43,718	44,895	22,234	23,672	50.63%	52.85%
West	33,945	34,845	16,830	17,579	48.09%	48.94%
County Type						
Large Metropolitan	62,802	63,692	30,822	32,159	49.47%	50.87%
Small Metropolitan	51,748	52,397	25,925	27,152	51.21%	52.04%
Nonmetropolitan	21,187	21,081	10,932	10,930	52.79%	53.25%

NOTE: Estimates are based on demographic information obtained from screener data and are not consistent with estimates on demographic characteristics presented in the *Results from the 2024 National Survey on Drug Use and Health: Detailed Tables* (Center for Behavioral Health Statistics and Quality [CBHSQ], 2025h).

NOTE: The weighted interview response rate (IRR) is calculated as the weighted number of respondents divided by the weighted number of selected people, defined as $IRR = \frac{\sum w_i complete_i}{\sum w_i selected_i}$, where w_i is the design-based weight or the inverse of the probability of selection for the person and includes dwelling until-level nonresponse and poststratification adjustments (see Section 2.3.4.1). However, w_i does not include person-level weight adjustments (see Section 2.3.4.1) and does not reflect adjustments to standardize the weighted proportion of interviews that were completed in person or via the web (see Section 2.3.4.2).

Source: SAMHSA, Center for Behavioral Health Statistics and Quality, National Survey on Drug Use and Health, 2023 and 2024.

3.3.1.1 Bias Due to Unit Nonresponse

Maximizing NSDUH response rates is intended to minimize bias in estimates due to different characteristics of respondents and nonrespondents. Surveys on substance use and mental health may be particularly vulnerable to nonresponse bias if people are less likely to participate in the survey who recently used drugs, used drugs frequently, or had mental disorders. For example, the NSDUH response rate in 2023 among young adults aged 18 to 25 was lower than that among adults aged 26 or older (Table 3.6); young adults in 2023 were more likely than adults aged 26 or older to use many of the substances discussed in the *Key Substance Use and Mental Health Indicators in the United States: Results from the 2023 National Survey on Drug Use and Health* report (CBHSQ, 2024e) and to have AMI in the past year.

However, potential sources of nonresponse bias in one direction (e.g., bias that would decrease estimates) could be offset by corresponding sources of bias in the opposite direction (e.g., bias that would increase estimates), such that overall effects on prevalence because of nonresponse bias could be minimal. For example, the NSDUH response rate in 2023 among females was higher than that among males; NSDUH findings have shown adult females to be more likely than adult males to have a major depressive episode (MDE), AMI, or SMI in the past year. See the *Differences in Past Year Mental Health among Young Adults Aged 18 to 25* (CBHSQ, 2024/2025) for more information. Further research on this important topic with recent NSDUH data would be useful.

3.3.2 Item Nonresponse and Inconsistent Responses

3.3.2.1 Item Nonresponse

Among survey participants, item response rates were generally very high for most mental health and drug use items. With the use of multimode data collection in 2024, however, item nonresponse rates were higher than what they were in survey years prior to 2020 when data were collected through in-person interviews only. The increase is due to web respondents discontinuing the survey prior to completion (i.e., breaking off). See Section 2.3.4 for details.

Prior to the introduction of the web mode of data collection in NSDUH, item nonresponse was predominantly caused by responses of "don't know" or "refused" either in the specific question or in earlier questions that governed skip logic.[32] In 2024, item nonresponse for usable interviews among adults was driven by survey break-offs. Item nonresponse tended to be lower

[32] As an example of skip logic, respondents were asked questions regarding alcohol or drug use treatment only if they previously reported lifetime use of any of these substances. If respondents did not report lifetime use, then questions about locations where substance use treatment was received were skipped. If respondents did not report lifetime use for any substances but did not know or refused to report lifetime use for some substances, then variables corresponding to these skipped questions had missing data.

for drug use items because of the usability criteria described in Section 2.3.1 and higher for mental health items.[33]

Occurrences of missing data in mental health items (e.g., mental health treatment, MDE) are important to examine because these questions occurred later in the NSDUH interview when respondents could break off but their interviews could remain usable. Furthermore, not all mental health variables were statistically imputed for 2024 (see Section 3.4.10).

As an example of how break-offs can introduce bias, rates of missing data for the receipt of mental health treatment in the past 12 months and the types of missing data (e.g., responses of "don't know" or "refused," no response because respondents broke off the interview) varied by age group and mode of interview in 2024. Among adult respondents aged 18 or older, item nonresponse was generally lower for in-person respondents for the receipt of mental health treatment at specific locations in the past 12 months or the use of prescription medication in that period to help with their mental health.[34] Among adult web respondents, the majority of the missing data was from interview break-offs. Break-offs contributed less to item nonresponse for adolescents aged 12 to 17 than for adults. Unlike adults, item nonresponse was generally lower for adolescents who completed the survey via the web than it was for those who were interviewed in person.

The questions about the receipt of substance use treatment also occurred later in the NSDUH interview than the questions about substance use but earlier in the interview than the mental health items; respondents could break off before these questions and their interviews could remain usable.[35] In addition, respondents could have been skipped out of the substance use treatment questions because information was unknown for whether they used alcohol or drugs in their lifetime. Because not all of the substance use treatment variables were statistically imputed for 2024 (see Section 3.4.8), potential bias due to missing data is relevant to NSDUH estimates for substances for which treatment was received.

For questions that ask for an opinion or a perception, such as the perceived risk of harm from the use of different substances or the perceived availability of substances, responses of "don't know" may be a valid indication of respondents not having an opinion or not having enough knowledge about these topics. Some questions in the NSDUH interview also gave respondents the option of choosing responses such as "I'm not sure" and "I don't want to answer" instead of using function keys (as is the practice elsewhere in the interview) for answers of "don't know" or "refused," respectively. See Section 3.4.13 for more information on these response choices for suicidal thoughts and behavior among adolescents and the reporting of associated estimates.

[33] People who started the survey but broke off the interview before completing a minimum number of drug use questions were not kept as final respondents because of the usability criteria described in Section 2.3.1.
[34] See Section 3.4.10.1 for the definition of mental health treatment.
[35] See Section 3.4.8.1 for the definition of substance use treatment.

3.3.2.2 Effects of Missing Data on Estimates

When statistical imputation was not used to replace missing values with nonmissing values (see Section 2.3.3), NSDUH estimates were based on variables that have some missing data. Generally, observations with missing values are excluded from standard NSDUH analyses, including a portion (but not all) of the analyses used to create the annual detailed tables.

Since 2021, additional variables have been statistically imputed to mitigate the effect of increased rates of item nonresponse due to break-offs. See Table 2.3 for a list of measures that were statistically imputed in the mental health and later sections of the interview for the 2024 NSDUH. However, some mental health or other variables in later sections of the 2024 interview were not statistically imputed. There may be bias in estimates from these variables due to the presence of missing data. Specifically, there may be bias when respondents with missing data are excluded from the analysis.

For estimated numbers of people with a given characteristic, there will always be a negative bias if there are missing values in the domain variable(s), the outcome variable, or both. For example, consider an analysis that estimates the percentage and number of adolescents between the ages of 12 and 17 who were exposed to school-based substance use prevention messages. This analysis includes a domain variable for adolescents who attended school in the past 12 months (including those who were homeschooled) and multiple outcome variables, which consist of whether adolescents received substance use prevention messages in various school settings. Respondents may have missing data for either the domain of school attendance or the outcomes of exposure to school-based prevention activities. Respondents with missing data who were exposed to these messages would not be included in the estimated number of exposed adolescents. Assuming that some of these adolescents with missing data were exposed to prevention messages, there would be negative bias in the population estimates.

When a population mean or proportion is estimated, there may or may not be bias, and the bias can be either negative or positive. The direction and magnitude of the bias for means and proportions depend on how different the item respondents are from the item nonrespondents with respect to the outcome of interest. For example, if the "true" sources where respondents with missing data obtained the last prescription pain relievers they misused matched the distribution among respondents who did not have missing data, then excluding missing data (and decreasing the number of respondents in the denominator) would not be expected to affect the estimated percentages of people in the population who obtained their prescription pain relievers from specific sources. However, if the source of prescription pain relievers among respondents with missing data was skewed in favor of a particular source (e.g., through prescriptions from more than one doctor), then excluding these respondents with missing data might introduce bias in published estimates.

These missing data effects are discussed in more detail in the 2023 Statistical Inference Report (CBHSQ, 2025d).

3.3.2.3 Inconsistent Responses

In order to minimize respondent confusion, inconsistent responses, and item nonresponse, the NSDUH instrument for both in-person and web-based interviews was programmed to skip respondents out of inapplicable questions based on their previous answers. This skip logic reduced the potential for inconsistent data by limiting respondents' opportunity to provide answers that were inconsistent with previous answers. For example, if respondents did not report that they last used alcohol in the past 30 days, they were not asked questions about their use of alcohol in the past 30 days. In some situations, respondents were asked to resolve an inconsistency between two answers. However, programming within the in-person and web instruments could not eliminate all occurrences of missing or inconsistent data. For example, respondents could report they last used any form of cocaine more than 12 months ago but they last used crack cocaine at some point within the past 12 months; both answers logically cannot be true.

These missing or inconsistent responses in the substance use data are first resolved (where possible) through a logical editing process (e.g., logically inferring more recent reported use of crack cocaine applies to any cocaine). When the missing or inconsistent responses are not resolved through logical editing, they may be resolved through imputation using statistical methodology. Editing and imputation of missing responses are potential sources of measurement error.

Not all inconsistencies in the data are resolved through editing or imputation. Inconsistencies could remain when they arise between questions from different questionnaire sections. Respondents could indicate in the cocaine section that they last used cocaine more than 12 months ago and also indicate in the special drugs section that they last used a needle to inject cocaine at some point in the past 12 months. According to the general principle of not editing across sections of the interview (see Section 2.3.2), this information from the later special drugs section was not used to edit the variable for the most recent use of cocaine. In situations such as these, data users will need to decide how to handle these inconsistencies in their analyses.

For more information on editing and statistical imputation, see Sections 2.3.2 and 2.3.3.

3.3.3 Validity of Self-Reported Substance Use

Most estimates of substance use, including those produced for NSDUH, are based on self-reports of use. This section focuses on the validity of NSDUH respondents' self-reports of substance use and is not intended to provide a comprehensive discussion of issues associated with the validity of any self-report in NSDUH. Factors such as the length of time between an event and the interview date or respondents' interpretation of a question also can affect respondents' recall or reporting, independent of the potential sensitivity of the topic covered by a question. An additional issue discussed in this section is the reporting of sensitive topics in web surveys.

Survey questions about topics such as substance use are considered to be sensitive because respondents may think the questions are intrusive ("none of your business"), pose risks for

negative social or legal consequences if their answers were to become known, or require them to provide socially undesirable answers (Tourangeau & Yan, 2007). Although studies generally have supported the validity of self-report data for sensitive topics, the potential for these data to be biased (underreported or overreported) is well documented. The bias varies by several factors, including the mode of administration, the setting, perceptions of privacy, the population under investigation, and for substance use, the type of drug (Aquilino, 1994; Brener et al., 2006; CBHSQ, 2012; Harrison & Hughes, 1997; Lindberg & Scott, 2018; Tourangeau & Smith, 1996; Tourangeau & Yan, 2007; Turner et al., 1992). NSDUH utilizes widely accepted methodological practices for increasing the accuracy of self-reports, such as encouraging privacy through self-administration of questions about sensitive topics—including audio computer-assisted self-interviewing (ACASI) for in-person data collection—and providing assurances that individual responses will remain confidential. Comparisons using these methods within NSDUH data (collected in person) have shown they reduce reporting bias (Gfroerer et al., 2002).

A special study cosponsored by the Substance Abuse and Mental Health Services Administration (SAMHSA) and the National Institute on Drug Abuse (NIDA) examined the validity of NSDUH self-report data on drug use among people aged 12 to 25. The study found urine and hair specimens can be collected with a relatively high response rate in a general population survey, and most adolescents and young adults reported their recent drug use accurately in self-reports (Harrison et al., 2007).[36]

3.3.3.1 Reporting of Sensitive Behaviors in Web Surveys

Prior studies spanning multiple decades (e.g., Aquilino, 1994; Lindberg & Scott, 2018; Tourangeau & Smith, 1996) have established that respondents are more likely to report sensitive behaviors when questions are self-administered than when they need to report their answers to interviewers. With the use of multimode data collection since the 2020 NSDUH, including some interviews being completed via the web, it is important to examine respondent willingness to report sensitive behaviors via the web. Given the use of the web for interactions that involve the sharing of sensitive data (e.g., Social Security numbers, credit card numbers), people may be expected to accurately report information about their substance use in a web-based survey if they feel assured that the privacy of their answers will be protected. However, the reliability of self-reports in surveys administered via the web remains an open and evolving question.

Kreuter and colleagues (2008) assessed social desirability bias in the reporting of potentially sensitive academic information using interviewer administration via computer-assisted telephone interviewing (CATI) and self-administration using interactive voice recognition (IVR) and web administration. Under a randomized experimental design, web administration increased the reporting of socially undesirable academic information such as a cumulative grade

[36] Results of this study showed an 84.6 percent agreement between self-reported tobacco use in the past 30 days and urine drug test results for tobacco. For marijuana, there was 89.8 percent agreement between self-reported use in the past 30 days and urine drug test results, although this agreement was dominated by people who reported no use and tested negative (82.9 percent).

point average below 2.5 compared with CATI, with IVR yielding results between those of web administration and CATI. Web respondents also were less likely than CATI respondents to falsely deny socially undesirable outcomes relative to information from external data sources. Although this study focused on academic outcomes rather than substance use, it is consistent with other literature showing that self-administered data collection modes—including web-based data collection—yield increased reports of sensitive behaviors compared with interviewer-administered modes.

Cernat and colleagues (2016) used data from the 2010 to 2012 waves of the Health and Retirement Study to compare estimates of depression symptoms from the Center for Epidemiologic Studies Depression Scale (CES-D) in interviewer-administered and self-administered web data collection modes. Depression estimates based on the CES-D were lower in interviewer-administered modes than in the web mode. However, the study did not compare the web data collection mode with other self-administration modes.

Some research suggests that web surveys can provide respondents with a greater sense of anonymity and privacy for reporting on sensitive topics such as substance use (Lygidakis et al., 2010; Miller & Sønderlund, 2010). Beckalova and Drapalova (2022) noted the following limitations of web surveys for collecting data on substance use:

- To reach populations of substance users who may not be commonly found in general population surveys, web surveys are often based on nonprobability samples, such that inferences cannot be made about substance use in the broader population.
- Web surveys may have higher rates of item nonresponse and break-offs because an interviewer is not present to answer respondents' questions or keep respondents motivated to complete the interview.
  - NSDUH requires respondents to have answered a minimum set of substance use questions in order for their interviews to be considered usable (see Section 2.3.1).
  - Nevertheless, a higher rate of break-offs has been noted among adult NSDUH respondents who completed their interviews via the web (see Section 2.3.4). A special break-off analysis weight has been used to create estimates for specific measures that were not statistically imputed.
- Web survey respondents could be prone to "satisficing," in which they answer questions in a manner to reduce the amount of time to complete the survey. However, a study in four northern European countries with high Internet coverage that compared survey administration via the web or a paper questionnaire did not support the assumption that web survey respondents are more prone to satisficing behavior (Clement et al., 2023).

The Monitoring the Future (MTF) study for 2023 uses a probability sample and since 2021 has used a web-based questionnaire to collect data from 8th, 10th, and 12th graders. Students since 2021 have used their own electronic devices (rather than tablets provided by the study) to connect to the web and complete the survey during class time. Analyses found that substance use trends were replicated when the sample for 2021 was restricted to students who reported that all their classes were in school (46 percent of the sample). Investigators concluded that

web administration of the survey at home produced results similar to those for students who completed the web survey in school (Johnston et al., 2022).

One issue for web-based data collection in NSDUH concerns privacy in settings such as within households. As noted previously, respondents' perceptions of privacy can influence the reporting of sensitive behaviors, as described by Brener and colleagues (2006) and in a report comparing adolescent substance use estimates across national surveys (CBHSQ, 2012).

For the 2024 NSDUH, field interviewers asked in-person respondents to find a private location to complete the survey. Web respondents were asked to be in a private location within the home and to affirm before starting an interview that they were in a private location. Unlike in-person interviews, however, no privacy ratings were available for web-based interviews to indicate whether interviews remained private throughout the entire interview, or if not, the extent of time for which the interview was less than private and who else might have been present. It also is not known whether NSDUH web respondents perceived the web mode to be a more private method for answering sensitive questions compared with in-person data collection in a private setting using ACASI, but with a field interviewer present. As web-based interviewing is increasingly used for collecting survey data—including data on sensitive topics such as substance use—recent methodological research comparing data collection via the web with other modes that collect self-administered data would be useful for establishing the factors that encourage or discourage the reporting of sensitive behaviors via the web.

3.3.3.2 Issues for Self-Reporting of Prescription Drug Misuse

The emphasis on past year rather than lifetime misuse of specific prescription drugs in the NSDUH questionnaire appears to affect the validity of estimates for lifetime misuse of prescription psychotherapeutic drugs (see Section C in the 2015 Methodological Summary and Definitions report; CBHSQ, 2016). Respondents in 2024 who did not misuse prescription psychotherapeutic drugs in the past 12 months were asked only about lifetime misuse of general categories of prescription psychotherapeutic drugs (e.g., prescription pain relievers). Respondents also did not have cues for recalling misuse more than 12 months ago of drugs no longer available by prescription in the United States (e.g., sedatives containing methaqualone, such as those with the brand names Quaalude® or Sopor®). Field testing of these prescription drug questions suggested that the emphasis on the past year misuse of specific prescription drugs can result in underreporting of lifetime misuse of prescription psychotherapeutic drugs. For more information, see the *National Survey on Drug Use and Health: 2012 Questionnaire Field Test Final Report* and the *National Survey on Drug Use and Health: 2013 Dress Rehearsal Final Report* (CBHSQ, 2014b, 2014c). For this reason, estimates of lifetime misuse of prescription psychotherapeutic drugs are not included in the 2024 national reports and tables.

The prescription drug questions in 2024 allowed respondents to report any use or misuse in the past 12 months for specific medications within a given psychotherapeutic category (e.g., the benzodiazepine tranquilizers Xanax®, Xanax® XR, generic alprazolam, and generic extended-release alprazolam). These details were presented to respondents to aid them with recall and recognition. However, respondents could have difficulty knowing or remembering whether they took a generic or brand name drug or what type of formulation they took (i.e., immediate

release or extended release). Therefore, recall of the use or misuse of prescription drugs containing a given active ingredient has been assumed to be more accurate than recall of the exact drugs respondents took. For example, respondents who took the generic benzodiazepine alprazolam (brand name drug Xanax®) could recognize the drug by its brand name and report use or misuse of "Xanax." This issue may be especially relevant for respondents who misused prescription drugs by taking them without a prescription of their own. However, self-reports of the use or misuse of Xanax® or alprazolam are equivalent analytically for estimating the use or misuse of tranquilizers containing alprazolam, even if respondents may have misreported the exact drug they used or misused in the past year. Therefore, 2024 NSDUH estimates for the use or misuse of prescription psychotherapeutic drugs in the past year are reported for overall psychotherapeutic drug categories (e.g., tranquilizers) or for subtypes of related drugs (e.g., benzodiazepine tranquilizers, tranquilizers containing alprazolam), but they are generally not reported for specific individual prescription drugs from the NSDUH questionnaire.[37]

3.3.4 Changes to Questions about Race

Starting with the 2022 NSDUH, revisions were made to question QD05, which asked respondents to report the race category that best described them. For the 2022 NSDUH, the response options for the race question were rearranged to be in alphabetical order, as per Office of Management and Budget (OMB) guidelines. The response option for American Indian or Alaska Native was moved to the top of the list; in prior years, White had been the first response option. Also, respondents were no longer offered a response option for "other" race.[38] See Section 2.2.2 in the 2022 Methodological Summary and Definitions report (CBHSQ, 2023).

Starting with the 2023 NSDUH, the race and ethnicity questions in the in-person questionnaire became self-administered instead of being administered by an interviewer. The race questions were placed in a self-administered demographic section because some questions that were added for 2023 were considered sensitive and better suited for self-administration. The 2022-2024 NSDUH data were evaluated to look for patterns for both item nonresponse and sample sizes due to these changes.

3.3.4.1 Item Nonresponse

The change from interviewer administration to self-administration for the race question in the 2023 in-person questionnaire increased the overall unweighted number and percentage of respondents who did not know or refused to answer the race question from about 800 respondents in 2022 (1.1 percent) to about 1,000 in 2023 (1.5 percent). This change was driven by the increase in missing data among respondents who completed the interview in person; the missing data percentages were consistent between years for those who completed the survey via the web because these questions were always self-administered for web respondents. In 2024, the percentage of respondents who did not know or refused to answer

[37] The exception was for the pain reliever OxyContin® (an extended-release formulation of oxycodone).
[38] In the revised version of the race question, Native Hawaiian, Samoan, and Other Pacific Islander were part of a single category rather than three categories as in years prior to 2022. However, more detailed information was collected in a new follow-up question and corresponding "OTHER, Specify" question.

the race question continued to be higher than the percentage from 2022. Specifically, the numbers and percentages of respondents with missing data for race in 2024 were about 1,250 (1.8 percent) regardless of mode; about 250 (0.9 percent) for web; and about 950 (2.3 percent) for in person. More information on item nonresponse can be found in Section 3.3.2.1. For additional information on changes to the race question for the 2022 NSDUH questionnaire that also increased the percentage of respondents who did not know or refused to answer the race question, see the 2022 Methodological Summary and Definitions report (CBHSQ, 2023).

3.3.4.2 Changes to the Numbers of Non-Hispanic American Indian or Alaska Native Respondents

With the reordering of the response options in the 2022 NSDUH questionnaire and change of the in-person questions to being self-administered in 2023, the unweighted number of in-person non-Hispanic American Indian or Alaska Native respondents more than doubled from 2021 to 2023 (around 460 to 1,130), and the corresponding unweighted percentage increased over this period (from 1.5 to 2.6 percent of in-person respondents). In 2024, the number and percentage of in-person respondents who reported that they were American Indian or Alaska Native were similar to those in 2023 (1,000 people and 2.4 percent). As for item nonresponse, there was little change for web respondents between 2022 and 2024.

NSDUH data that are published in the 2024 national reports and tables are weighted to reflect U.S. Census Bureau or ACS population estimates for specific racial or ethnic groups. As noted in Section 2.3.4, the weighting process for 2022-2024 did not change compared with the procedures for 2021. The NSDUH weighted population counts were consistent with those from prior years. Despite the weighted population size for non-Hispanic American Indian or Alaska Native people being consistent between years, caution is nevertheless advised in comparing outcomes for this group from 2021 with outcomes from 2022 and later years due to the change since 2022 in the unweighted number of respondents for this group.

3.4 Measurement Issues

Several measurement issues that are presented in the 2024 NSDUH data products are discussed in this section. Issues addressed include the methods for measuring the use of inhalants, central nervous system (CNS) stimulant use, use and misuse of fentanyl (including illegally made fentanyl [IMF]), nicotine vaping, experiences among adolescents aged 12 to 17, the initiation of substance use or misuse of prescription drugs, SUDs, and substance use treatment. Additionally, this section discusses the mental health measures for AMI, SMI, receipt of mental health treatment, symptoms of generalized anxiety disorder (GAD), MDE, and suicidality. Starting with Section 3.4.14, topics are discussed for modes of marijuana use, recovery, and the definition of county type.

This section also discusses how missing data were handled analytically to produce the estimates found in the national reports and tables for the 2024 NSDUH. Refer to Section 3.3.2 for a discussion of potential bias in estimates because of missing data.

Except where noted, estimates for measures described in this section used the main analysis weight described in Section 2.3.4. The section specifically mentions when estimates used the break-off analysis weight because the measures were not statistically imputed.

3.4.1 Inhalant Use

Questions for inhalant use were revised for the 2024 NSDUH. The introductory language for the inhalants section was revised for brevity, and the individual inhalant question wordings were revised from asking about use of inhalants "for kicks or to get high" to "for fun or to get high." The wording update was intended to make the questions more relatable and understandable, especially for younger respondents. Review of the final data from all 12 months of 2024 indicated that this change in questionnaire wording affected the reporting of the use of inhalants, especially among adolescent respondents aged 12 to 17.

Consequently, estimates in 2024 data products based on questions from the inhalants section of the interview were considered not to be comparable with those in data products from prior years for the following:

- lifetime, past year, and past month use of any inhalant;
- lifetime use of all specific inhalants (e.g., amyl nitrite, poppers, locker room odorizers, or Rush; correction fluid, degreaser, or cleaning fluid; gasoline or lighter fluid);
- past year initiation of inhalant use, including risk for initiation, age at first use, and the date of first use (see Section 3.4.6); and
- past year and past month frequency of use (i.e., number of days used).

Estimates in 2024 data products also were considered not to be comparable with those in data products from prior years for the use of illicit drugs other than marijuana in the lifetime, past year, and past month because a relatively large proportion of people who used illicit drugs other than marijuana used inhalants.

The change in question wording for inhalants did not appear to affect the comparability of estimates between 2024 and prior years for the overall use of any illicit drug in the lifetime, past year, and past month and for inhalant use disorder in the past year. However, investigations into comparability were limited and may not cover other comparisons that data users want to make.

The variables for the measure of inhalant use, initiation of inhalant use, and frequency of use were statistically imputed and have no missing data. Variables for specific lifetime inhalants (e.g., amyl nitrite, poppers, locker room odorizers) were not imputed and contain missing data.

3.4.2 CNS Stimulant Misuse

CNS stimulants are a group of drugs that include cocaine, methamphetamine, and prescription stimulants. These drugs act in similar ways to stimulate the brain. They produce stimulant effects, such as increased alertness, wakefulness, or energy. They can also produce physical side effects of rapid or irregular heartbeat or increased blood pressure and body temperature (NIDA, 2023, 2024a, 2024c).

An aggregate measure for CNS stimulant misuse was created for inclusion of estimates in national reports and tables for the 2024 NSDUH. Because this aggregate measure includes the misuse of prescription stimulants in addition to the use of cocaine or methamphetamine, it was defined as CNS stimulant misuse.

CNS stimulant misuse data for 2024 were available for the past year and past month, and data for any use of CNS stimulants were available for the past year. Because of potential measurement issues for the lifetime misuse of prescription drugs (see Section 3.3.3.2), lifetime estimates for CNS stimulant misuse and any use were not presented in national reports and tables for the 2024 NSDUH. Measures for CNS stimulant misuse in the past year or past month periods and any use of CNS stimulants in the past year were created according to the most recent time when respondents used or misused these substances. Because the measures were statistically imputed for cocaine use, methamphetamine use, and prescription stimulant misuse for the past year and past month, the aggregate measures for CNS stimulant misuse in those periods had no missing data for 2024. Similarly, the aggregate measure for any CNS stimulant use in the past year had no missing data because the measure for any past year use of prescription stimulants was statistically imputed. Section 3.4.7.2 also describes the creation of measures for CNS stimulant use disorder.

3.4.3 Use and Misuse of Fentanyl, Including Illegally Made Fentanyl

Data from the Centers for Disease Control and Prevention's (CDC's) National Vital Statistics System indicate that overdose deaths involving synthetic opioids other than methadone have been the leading drugs for opioid-involved overdose deaths since the end of 2016 (Ahmad et al., 2025). This involvement of synthetic opioids in fatal overdoses has been attributed to fentanyl and fentanyl analogues that are made illegally in clandestine laboratories rather than by the pharmaceutical industry. These forms of IMF and analogues have been mixed with heroin as an adulterant, substituted for heroin entirely, or sold as counterfeit prescription drugs (Ciccarone, 2019). Although the age-adjusted death rates for drug overdose deaths involving any opioid or synthetic opioids other than methadone decreased between 2022 and 2023 (Garnett & Miniño, 2025), age-adjusted rates of drug overdose deaths from 2016 to 2021 more than tripled for deaths involving fentanyl (Spencer et al., 2023).

From 2015 to 2021, NSDUH asked about the use and misuse of prescription forms of fentanyl but did not ask specifically about the use of IMF. However, respondents could specify in the special drugs section of the NSDUH interview that fentanyl was another drug that they injected with a needle. Because many prescription fentanyl products that are available by prescription in the United States (e.g., Duragesic®, Fentora®, generic fentanyl) are in forms such as skin patches or "lollipops" that would require extraction of the fentanyl before it can be injected, most reports in NSDUH of the injection of fentanyl are assumed to involve the injection of IMF. However, reports of fentanyl injection that require respondents to specify it as some other drug that they injected are likely to yield underestimates of IMF use (Kroutil et al., 2010). Moreover,

heroin that contains IMF as an adulterant could be used in ways other than injection, such as sniffing (Mars et al., 2018).[39]

Therefore, beginning with the 2022 NSDUH, questions were included in the emerging issues section that specifically asked about the use of IMF. Respondents aged 12 or older were asked whether they ever used IMF and, if so, how long it had been since they last used it. The questionnaire explained that IMF is fentanyl that people cannot get from a doctor or pharmacy and that IMF can come in forms such as powder, pills, blotter paper, or mixed with heroin or other drugs. If respondents reported that they had ever used IMF, they also were asked whether they ever used a needle to inject it. If they reported ever having used IMF with a needle, and if they had previously indicated that they used IMF in any way within the past 12 months, they were asked how long it had been since they last injected it. Respondents who reported using IMF with a needle in their lifetime and reported last using it in any way more than 12 months ago were not asked when they last used a needle to inject IMF; logically, these respondents last injected IMF more than 12 months ago.

Estimates for any use of IMF in the lifetime, past year, and past month used the main analysis weight because the variables for the lifetime and most recent use of any IMF were statistically imputed. However, estimates in the 2024 Detailed Tables (CBHSQ, 2025h) for the use of IMF with a needle used the break-off analysis weight because variables for the use of IMF with a needle were not statistically imputed.

Beginning in 2022, national reports and tables included estimates for aggregate measures that included past year use of prescription fentanyl products or IMF. The measure for *any use* of fentanyl in the past year included reports of any use of prescription fentanyl or use of IMF in the past year. The measure for fentanyl *misuse* included misuse of prescription fentanyl or use of IMF in the past year.

3.4.4 Nicotine Vaping

Questions on vaping of nicotine were included in the nicotine section of the 2024 questionnaire. Respondents aged 12 or older were asked whether they ever vaped nicotine with an e-cigarette or other vaping device, including devices that may also be called vapes, vape pens, or mods. Respondents were asked to consider any device that heats a liquid containing nicotine into a vapor.

Follow-up questions if respondents reported nicotine vaping in their lifetime were patterned after similar questions for tobacco products and included questions for the first time respondents vaped nicotine (age at first use, and if applicable, the year and month of first use), when respondents last vaped nicotine, and the number of days that respondents vaped nicotine in the past 30 days, if they last vaped nicotine in that period. Associated measures for initiation of nicotine vaping, most recent nicotine vaping, and the frequency of nicotine vaping in the past

[39] See the fentanyl drug fact sheet.

30 days were all statistically imputed and do not contain missing data. Nicotine vaping measures are comparable for 2022-2024.[40]

3.4.5 Youth Experiences

This section focuses on changes to existing questions in the youth experiences section for the 2024 NSDUH (see Section 2.2.2) and implications for analyses. Section 2.2.2 also lists questions from this section that were dropped from the 2024 questionnaire. Analyses were conducted using data from the first 6 months of 2024 to verify that deletion of these questions did not affect how respondents answered subsequent questions.

Starting with the 2024 NSDUH, response options were revised for questions corresponding to how respondents aged 12 to 17 think their parents would feel about them using substances. Response options also were revised for questions corresponding to how respondents would feel about their peers using substances. Two new response options were added for "strongly approve" and "somewhat approve." The response option order was also reversed to begin with "strongly disapprove" as a first option and "strongly approve" as the last option. Prior to 2024, response options for these questions were limited to "neither approve nor disapprove," "somewhat disapprove," and "strongly disapprove." The following behaviors were asked about in these questions:

- smoking cigarettes every day,
- trying marijuana or any cannabis product,
- using marijuana or any cannabis product once a month or more, and
- having one or two alcoholic beverages nearly every day.

Also beginning in 2024, the question measuring screen time was revised to ask respondents aged 12 to 17 how often their parents limited the amount of time spent with television, tablets, smartphones, computer, or video games. Prior to 2024, respondents were asked specifically how often their parents limited the amount of time they watched television in the past 12 months.

Because of these changes for 2024, these measures are not considered comparable with those from prior years.

Respondents with missing data for these measures were excluded from the analyses to produce published estimates for the 2024 NSDUH. For a discussion on how procedures for handling missing data may bias estimates, see Section 3.3.2.

In addition, some questions in the youth experiences section of the survey ask for an opinion or a perception. As mentioned previously, for example, respondents aged 12 to 17 were asked what they thought their parents would think about them using or trying cigarettes, marijuana,

[40] Nicotine vaping measures in 2022 through 2024 are not comparable with corresponding measures from 2021 because the nicotine vaping questions in 2021 were asked in the later emerging issues section. For additional details, see Chapter 3 in the 2022 Methodological Summary and Definitions report (CBHSQ, 2023).

or alcohol. If respondents do not know what their parents would think, a response of "don't know" can be an informative and valid response.[41] Also, analyses of data from these types of questions that exclude responses of "don't know" could include bias. See Section 3.3.2 for more information on bias.

3.4.6 Initiation of Substance Use or Misuse

In NSDUH, initiation refers to the first use or misuse of a particular substance. For prescription psychotherapeutic drugs (pain relievers, tranquilizers, stimulants, and sedatives), initiation refers to the first time misuse ever occurred.[42] All of the initiation variables used to create published estimates for the 2024 NSDUH underwent statistical imputation to remove missing values (see Section 2.3.3). Therefore, these variables were not subject to the kinds of potential bias because of missing data described in Section 3.3.2.

The NSDUH questionnaire collected year and month of first use of a substance for recent initiates—that is, people who used a particular substance for the first time at their current age or the year before their current age. Month, day, and year of birth also were obtained directly or were statistically imputed for item nonrespondents as part of the data postprocessing. Additionally, the date of the interview was recorded for both in-person and web respondents.

Past year initiation referred to respondents whose first use of a substance (or misuse for psychotherapeutic drugs) was within the 12 months prior to their interview date.[43] Past year initiation was determined by self-reported past year use, the age at first use, the year and month of recent new use, and the interview date.

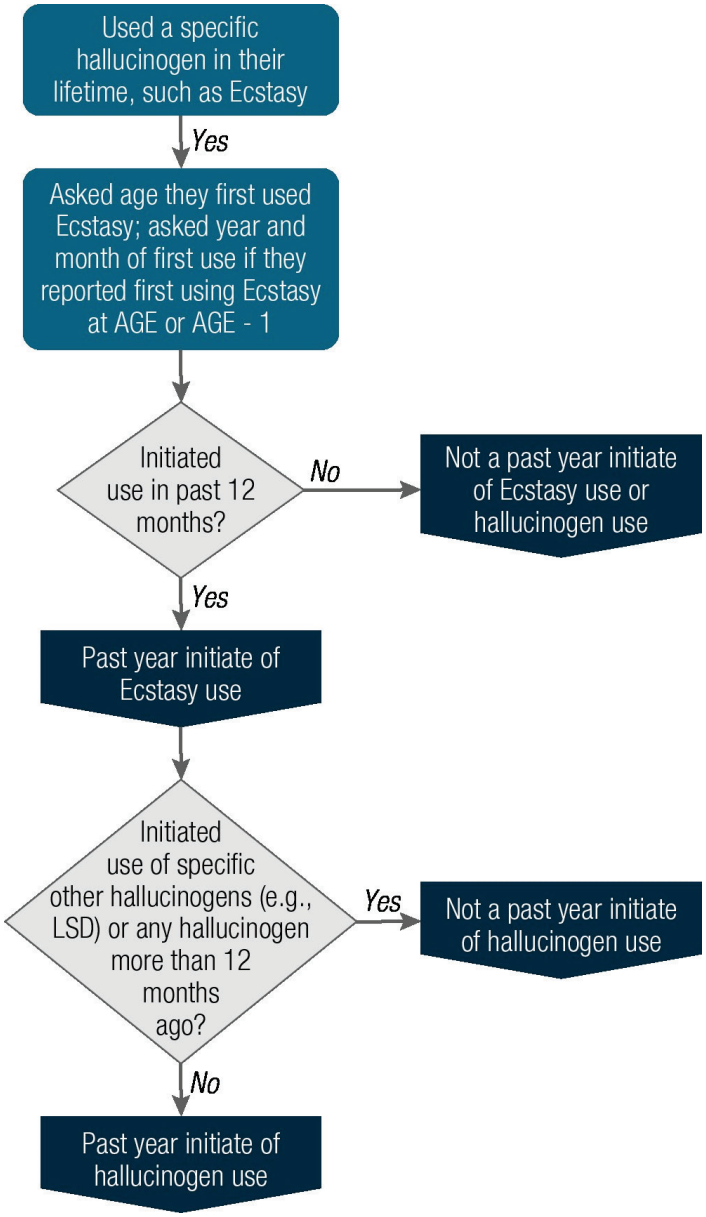
Calculations of estimates of past year initiation did not take into account whether respondents initiated substance use while a resident of the United States. This method of calculation allowed for direct comparability with other standard measures of substance use because the populations of interest for the measures will be the same (i.e., both measures examined all possible respondents and were not restricted to those initiating substance use only in the United States).

One important note for initiation estimates is the relationship between the main categories and subcategories of substances (e.g., hallucinogens would be a main category, and lysergic acid diethylamide [LSD], phencyclidine [PCP], and Ecstasy would be subcategories in relation to hallucinogens). As shown in Figure 3.1, an individual can initiate use of any hallucinogen, LSD, PCP, or Ecstasy only once. Respondents who initiated use of any hallucinogen more than

[41] Section 3.4.13 discusses reporting of estimates for adolescent suicidality when adolescents did not know how to answer the questions.
[42] Respondents beginning in 2022 were asked about *any* use of prescription psychotherapeutic drugs. Any use includes use of medication as directed with a prescription of the individual's own or misuse of prescription psychotherapeutics. However, respondents were not asked when they first used psychotherapeutics for any reason. Therefore, initiation for psychotherapeutics in NSDUH refers to the first time people misused these medications rather than the first time they used these medications for any reason.
[43] For brevity, "misuse" is not repeated whenever the text refers to first use. Terms such as "past year use" and "first use" used in the remainder of this chapter for substance use in general refer to *misuse* for prescription psychotherapeutic drugs.

12 months ago by definition were not past year initiates of hallucinogen use, even if they initiated use of LSD, PCP, or Ecstasy in the past year. Similar principles applied to other main categories and subcategories, such as cocaine and crack, a main category and a subcategory, respectively.

Figure 3.1 Relationship between Past Year Initiation of Use of Subcategories of Hallucinogens and Past Year Initiation of Any Hallucinogen Use: Example for Ecstasy



A related issue applied to initiation estimates for the aggregate substance use categories that include prescription drug measures, such as past year initiation of the use of any illicit drug or

the misuse of prescription psychotherapeutic drugs.[44] For example, people who first misused prescription stimulants in the past 12 months but who first misused prescription pain relievers more than 12 months prior to the interview date would be past year initiates for the misuse of stimulants. These people would not be past year initiates for the misuse of prescription psychotherapeutic drugs or illicit drugs because they had already misused pain relievers more than 12 months ago. Because of the potential for respondents to underreport lifetime (but not past year) misuse of prescription psychotherapeutic drugs (see Section 3.3.3.2), lifetime (but not past year) misusers of prescription drugs could be misclassified as past year initiates for illicit drugs if they reported past year initiation of another illicit drug (e.g., heroin) but failed to report their lifetime misuse of a prescription psychotherapeutic drug (e.g., pain relievers).[45] Section 4.6.3 discusses additional issues for the measurement of initiation of benzodiazepine misuse. For these reasons, the 2024 national reports and tables do not show past year initiation estimates for the aggregate categories of illicit drugs,[46] prescription psychotherapeutic drugs, opioids, benzodiazepines, the aggregate category for tranquilizers or sedatives, or CNS stimulants.

NSDUH does not collect initiation data for pipe tobacco. In addition, NSDUH does not collect initiation data on marijuana vaping or vaping of flavoring. Thus, the 2024 national reports and tables do not show initiation estimates for the aggregate category of any tobacco product use, any tobacco product use or nicotine vaping, marijuana vaping, or vaping of flavoring.

In addition to estimates of the percentage and number of people initiating use of a substance in the past year, 2024 NSDUH estimates were created for the following:

- risk for initiation, and
- mean age at first use or misuse among people who initiated use or misuse of a substance in the past year.

People were classified as being at risk for initiation in the past 12 months if they did not use a given substance in their lifetime or if they used it for the first time in the past year. People who first used the substance more than 12 months ago were no longer considered to be at risk for initiation. NSDUH can identify people at risk for initiation of use of marijuana, cocaine, crack, heroin, hallucinogens, LSD, PCP, Ecstasy, inhalants, methamphetamine, cigarettes, nicotine vaping, smokeless tobacco, cigars, and alcohol and those at risk for initiation of daily cigarette use based on responses from the respective substance use questionnaire sections.

NSDUH cannot identify people at risk for initiation of use for the aggregate categories of either tobacco products or tobacco products or nicotine vaping because respondents were not asked

[44] Other potential aggregate categories for initiation that included prescription drugs were tranquilizers or sedatives as a combined category, benzodiazepines, opioids (i.e., heroin or prescription pain relievers), and CNS stimulants (i.e., cocaine, methamphetamine, or prescription stimulants).
[45] This example where respondents reported past year initiation of heroin use but failed to report lifetime misuse of prescription pain relievers also would apply to the measure for past year initiation of opioid misuse.
[46] The *Results from the 2024 National Survey on Drug Use and Health: Detailed Tables* (CBHSQ, 2025h) also do not show initiation estimates for the aggregate category of illicit drugs other than marijuana.

when they first used pipe tobacco. NSDUH also cannot identify people at risk for initiation of illicit drug use, use of illicit drugs other than marijuana, misuse of prescription psychotherapeutic drugs (i.e., pain relievers, tranquilizers, stimulants, or sedatives), benzodiazepines, misuse of opioids, misuse for the aggregate category for tranquilizers or sedatives, and misuse of CNS stimulants.

Estimates were computed for the mean age at first use or misuse among people who initiated use or misuse of a given substance in the past year. Unless specified otherwise, estimates of the mean age at initiation in the past 12 months were restricted to people aged 12 to 49 so that these mean age estimates were not influenced by those few respondents who were past year initiates and were aged 50 or older. As a measure of central tendency, means are influenced by the presence of extreme values in the data. Therefore, constraining the mean age estimates to past year initiates aged 12 to 49 was expected to increase the utility of these results to health researchers and analysts by providing a less biased picture of the substance use initiation behaviors among the civilian, noninstitutionalized population in the United States. This constraint was applied only to estimates of mean ages at first use and did not affect estimates for the 2024 NSDUH of the numbers of new users or associated percentages (e.g., the percentage of past year users who initiated use in the past year).

For the 2024 NSDUH, questions for inhalant use were revised from asking about use of inhalants "for kicks or to get high" to "for fun or to get high" (see Section 2.2.2). This change in the questionnaire wording appeared to have affected how respondents answered questions about the use of inhalants, especially among adolescent respondents aged 12 to 17. Consequently, estimates in 2024 data products for past year initiation of inhalant use, including risk for initiation and the mean age at first use among past year initiates, were considered not to be comparable with those in data products from prior years. See Section 3.4.1 for more details on this change to the inhalant use questions.

3.4.6.1 Initiation of Use of Substances from the Nicotine Section through the Methamphetamine Section

For cigarettes, nicotine vaping, smokeless tobacco, cigars, alcohol, marijuana, cocaine, crack cocaine, heroin, hallucinogens, inhalants, and methamphetamine, respondents were classified as past year initiates if there were fewer than 365 days between the interview date in 2024 and the statistically imputed month, day, and year of first use of the relevant substance. (If respondents reported first use in calendar year 2023, their first use could be within 365 days of the interview date.) The total number of past year initiates can be used in the estimation of different percentages. For these substances, denominators for the percentages vary according to whether rates are being estimated for (1) all people in the population (or all people in a subgroup of the population, such as people in a given age group), (2) people who are at risk for initiation because they have not used the substance of interest prior to the past 12 months, or (3) past year users of the substance. The 2024 Detailed Tables (CBHSQ, 2025h) show all three of these percentages.

3.4.6.2 Initiation of Misuse of Prescription Psychotherapeutic Drugs

Respondents were asked about the initiation of misuse of prescription psychotherapeutic drugs only for the individual prescription drugs they misused in the past 12 months. Asking respondents to recall their first misuse of any prescription drug in an overall category (e.g., pain relievers) would require them to think about all prescription drugs that could have been available to them when they initiated misuse. However, some of these drugs may no longer have been available when respondents were interviewed.

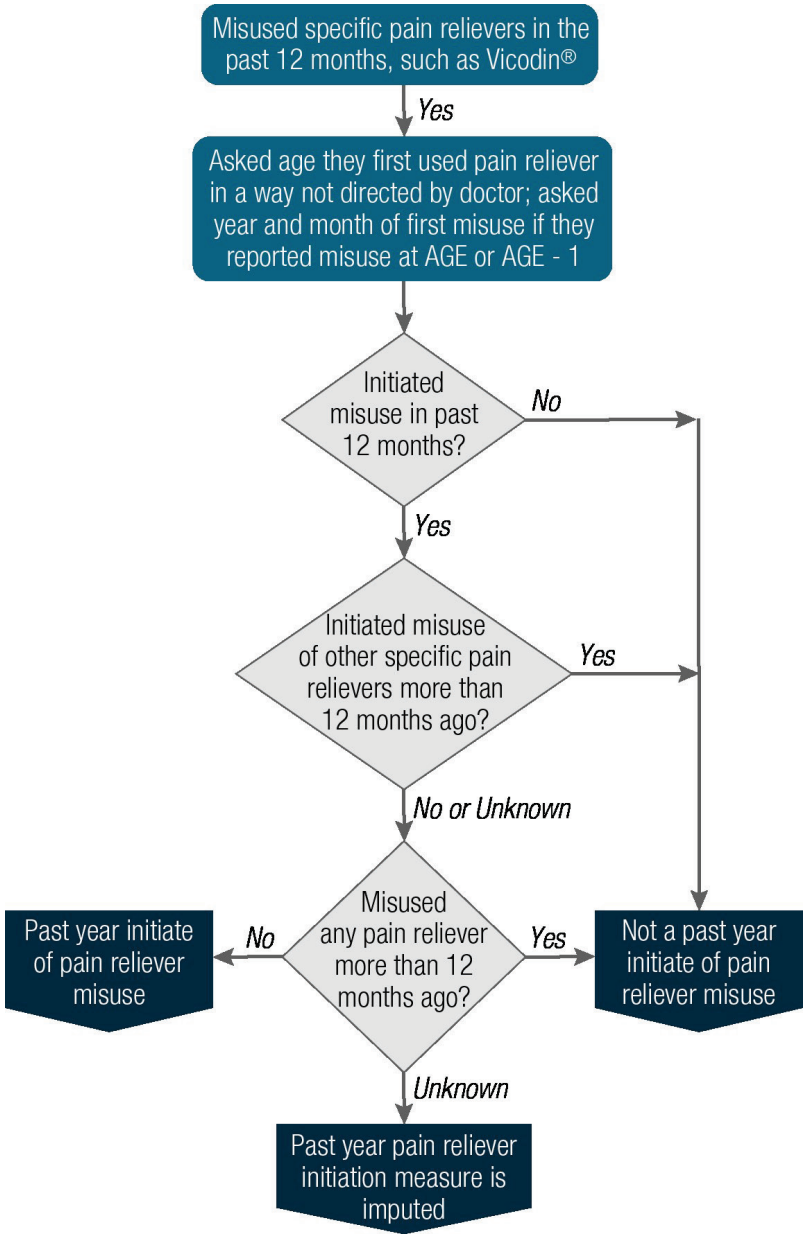
Figure 3.2 uses prescription pain relievers as an example to illustrate the steps for identifying past year initiates of the misuse of prescription drugs. These steps are the same for the other prescription drug categories. If respondents reported they first misused one or more prescription pain relievers at an age or in a year and month *more than 12 months* prior to the interview date, they logically did not initiate the misuse of pain relievers in the past year. Respondents who reported that they first misused all specific pain relievers in the past 12 months were asked a general question for whether they misused any pain reliever more than 12 months ago; "any pain reliever" in this follow-up question was not necessarily limited to the specific pain relievers that respondents were asked about in the questionnaire.[47] The answers to these follow-up questions determined whether these respondents who reported initiation of misuse for all specific pain relievers they misused in the past 12 months were past year initiates for the misuse of any pain reliever or if initiation of misuse in the past year needed to be resolved through imputation (see Section 2.3.3). See the *2023 National Survey on Drug Use and Health (NSDUH) Methodological Resource Book, Section 10: Editing and Imputation Report* (CBHSQ, 2025c) for more information.

Field testing in 2013 of redesigned questions for the initiation of misuse of prescription drugs indicated that estimated numbers and percentages of people who initiated the misuse of prescription drugs in the past year were similar based on the redesigned questions and questions from the main survey that measured past year initiation. For more information, see the 2013 Dress Rehearsal Final Report (CBHSQ, 2014c). Therefore, the 2024 national reports and tables present estimates for people who initiated the misuse of prescription pain relievers, tranquilizers, stimulants, or sedatives in the past year.

In addition, the total number of past year initiates of misuse of any psychotherapeutic drug in a category can be used in the estimation of percentages among (1) all people in the population (or all people in a subgroup of the population, such as those in a given age group) and (2) people who were past year users of the substance. The 2024 Detailed Tables (CBHSQ, 2025h) show estimates for these two percentages.

[47] Respondents also were asked the follow-up question if the sum of the reports of past year initiation plus missing data for initiation equaled the number of specific drugs they misused in the past year (and there were no reports of initiation of misuse more than 12 months prior to the interview date).

Figure 3.2 Illustration of the Steps for Identifying Past Year Initiates of the Misuse of Prescription Pain Relievers



As noted in Section 3.3.3, however, respondents who last misused prescription psychotherapeutic drugs in a category more than 12 months ago may underreport misuse, especially if they were not presented with examples of drugs formerly available by prescription in the United States but no longer available at the time respondents were interviewed. These respondents who did not report misuse occurring more than 12 months ago would be misclassified as still being "at risk" for initiation of misuse of prescription drugs in that psychotherapeutic category.[48] For this reason, percentages for initiation of misuse of psychotherapeutic drugs among people who were at risk for initiation were not created for the 2024 Detailed Tables (CBHSQ, 2025h).

3.4.7 Substance Use Disorders

The NSDUH questionnaire included questions designed to measure dependence on nicotine in the form of cigarettes. The questionnaire also included questions to measure SUDs for alcohol and drugs. Estimates for nicotine dependence are included in the 2024 Detailed Tables. Estimates for SUD and SUD severity for alcohol and drugs are included in national reports and tables for the 2024 NSDUH.

3.4.7.1 Nicotine Dependence

For nicotine, questions pertaining to dependence were based on the Nicotine Dependence Syndrome Scale (NDSS; Shiffman et al., 1995, 2004) and the Fagerström Test of Nicotine Dependence (FTND; Fagerström, 1978; Heatherton et al., 1991). Nicotine dependence was not measured for respondents who did not smoke cigarettes in the past month but used other products containing nicotine.

To identify patterns of nicotine dependence within the 2024 NSDUH data, questions measured dependence on nicotine through the use of cigarettes. Respondents were classified as being dependent if they met either the NDSS or the FTND classifications for dependence. The 2024 NSDUH contained 19 NDSS questions addressing five aspects of dependence: (1) smoking drive (compulsion to smoke driven by nicotine craving and withdrawal), (2) nicotine tolerance, (3) continuous smoking, (4) behavioral priority (i.e., preferring smoking over other reinforcing activities), and (5) stereotypy (i.e., fixed patterns of smoking). The 2024 NSDUH contained two mutually exclusive questions (DRCGE19a and DRCGE19b) addressing the FTND measure of dependence. These questions ask respondents who reported smoking cigarettes in the past month if the first cigarette they smoked was within 30 minutes of waking up on the days they smoked.

Missing data for nicotine dependence were statistically imputed for past month cigarette smokers. See Section 2.3.3 for more information on imputation.

[48] People who never misused prescription drugs remain "at risk" of initiation. Therefore, respondents who underreported the lifetime (but not past year) misuse of prescription drugs could be misclassified as still being at risk of initiation for the misuse of prescription drugs.

3.4.7.2 Substance Use Disorder for Drugs and Alcohol

SUD estimates for drugs and alcohol in the 2024 NSDUH were based on the criteria in the *Diagnostic and Statistical Manual of Mental Disorders*, 5th edition (DSM-5; American Psychiatric Association, 2013). Respondents were asked SUD questions separately for any drugs or alcohol they used in the 12 months prior to the survey.[49] SUD questions for drugs applied to marijuana, cocaine (including crack), heroin, hallucinogens, inhalants, methamphetamine, and *any* use of psychotherapeutic drugs, including prescription pain relievers, tranquilizers, stimulants, or sedatives. Beginning in 2021, NSDUH respondents who reported *any* use of prescription psychotherapeutic drugs in the past year (i.e., not just misuse of prescription drugs) were asked the respective SUD questions for that category of prescription drugs. More information on how the current SUD questions were developed based on DSM-5 criteria can be found in the 2020 Methodological Summary and Definitions report (CBHSQ, 2021b).

Users of NSDUH reports and tables also should be aware that questions about the use of IMF appear after SUD questions (see Section 3.4.3 for additional information about the IMF questions). For this reason, overall SUD, drug use disorder, and opioid use disorder measures do not capture disorders arising solely from the use of IMF.

DSM-5 includes the following SUD criteria (as measured in the 2024 NSDUH):

1. The substance is often taken in larger amounts or over a longer period than intended.
2. There is a persistent desire or unsuccessful efforts to cut down or control substance use.
3. A great deal of time is spent in activities necessary to obtain the substance, use the substance, or recover from its effects.
4. There is craving, or a strong desire or urge, to use the substance.
5. There is recurrent substance use that results in a failure to fulfill major role obligations at work, school, or home.
6. There is continued substance use despite having persistent or recurrent social or interpersonal problems caused by or exacerbated by the effects of the substance.
7. Important social, occupational, or recreational activities are given up or reduced because of substance use.
8. There is recurrent substance use in situations in which it is physically hazardous.
9. Substance use is continued despite knowledge of having a persistent or recurrent physical or psychological problem that is likely to have been caused or exacerbated by the substance.

[49] NSDUH respondents in 2024 were asked the respective questions for alcohol use disorder or marijuana use disorder if they reported use of these substances on 6 or more days in the past year. Respondents were asked SUD questions for other substances if they reported any use in the past year.

10. There is a need for markedly increased amounts of the substance to achieve intoxication or the desired effect, or there is a markedly diminished effect with continued use of the same amount of the substance (i.e., tolerance).
11. There are two components of withdrawal symptoms, either of which meet the overall criterion for withdrawal symptoms:

    a. There is a required number of withdrawal symptoms that occur when substance use is cut back or stopped following a period of prolonged use.[50]
    b. The substance or a related substance is used to get over or avoid withdrawal symptoms.[51]

Table 3.7 shows how these 11 DSM-5 SUD criteria apply to substances in NSDUH. For prescription opioids, tranquilizers, stimulants, and sedatives, Table 3.7 also shows how these criteria apply if respondents misused these prescription drugs, or if they simply used but did not misuse them in the past year. For consistency with the DSM-5 criteria, NSDUH respondents were classified as having an SUD if they met two or more of the applicable criteria in the 12-month period before the interview.

For alcohol, marijuana, cocaine, heroin, and methamphetamine, all 11 criteria applied (see Table 3.7). For hallucinogens and inhalants, only the first 10 criteria applied; the withdrawal criterion is not applicable to hallucinogens and inhalants.

For the prescription drugs shown in Table 3.7, the applicable DSM-5 criteria for classifying respondents as having an SUD depends on whether respondents misused prescription drugs or used but did not misuse prescription drugs in the past year. If respondents misused prescription drugs in the past year, all 11 criteria shown in Table 3.7 applied. However, if respondents used but did not misuse prescription drugs in the past year, only the first 9 criteria shown in Table 3.7 applied. Criteria 10 (tolerance) and 11 (withdrawal) do not apply to respondents who used but did not misuse these prescription drugs in the past year; tolerance and withdrawal can occur as normal physiological adaptations when people use these prescription drugs appropriately under medical supervision (Hasin et al., 2013).

[50] For alcohol, for example, withdrawal symptoms include (but are not limited to) trouble sleeping, hands trembling, hallucinations (seeing, feeling, or hearing things that are not really there), or feeling anxious.
[51] For alcohol use disorder, for example, this criterion involves the use of alcohol, sedatives, or tranquilizers to get over or avoid alcohol withdrawal symptoms.

Table 3.7 DSM-5 Substance Use Disorder Criteria for Substances and Types of Use in the 2024 NSDUH

Criterion[1]	Alcohol	Marijuana	Cocaine[2]	Heroin[3]	Hallucinogens	Inhalants	Methamphetamine[2]	Rx Opioids, Use but Not Misuse[3]	Rx Opioids, Misuse[3]	Rx Tranquilizers, Use but Not Misuse[4]	Rx Tranquilizers, Misuse[4]	Rx Stimulants, Use but Not Misuse[2]	Rx Stimulants, Misuse[2]	Rx Sedatives, Use but Not Misuse[4]	Rx Sedatives, Misuse[4]
1: Substance is often taken in larger amounts, longer than intended	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•
2: Unsuccessful efforts to cut down/control use	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•
3: A great deal of time is spent obtaining, using, recovering	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•
4: Craving/strong urge to use	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•
5: Recurrent use resulting in failure to fulfill major role obligations at work/school/ home	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•
6: Continued use despite social problems	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•
7: Important social/ occupational/ recreational activities given up or reduced because of use	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•
8: Recurrent use in physically hazardous situations	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•
9: Continued use despite physical, psychological problems	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•

Table 3.7 DSM-5 Substance Use Disorder Criteria for Substances and Types of Use in the 2024 NSDUH (continued)

Criterion[1]	Alcohol	Marijuana	Cocaine[2]	Heroin[3]	Hallucinogens	Inhalants	Methamphetamine[2]	Rx Opioids, Use but Not Misuse[3]	Rx Opioids, Misuse[3]	Rx Tranquilizers, Use but Not Misuse[4]	Rx Tranquilizers, Misuse[4]	Rx Stimulants, Use but Not Misuse[2]	Rx Stimulants, Misuse[2]	Rx Sedatives, Use but Not Misuse[4]	Rx Sedatives, Misuse[4]
10: Increased amount of substance is needed to achieve same effect	•	•	•	•	•	•	•	–	•	–	•	–	•	–	•
11a: Withdrawal symptoms[5]	•	•	•	•	–	–	•	–	•	–	•	–	•	–	•
11b: The same or related substance is taken to avoid withdrawal symptoms	•	•	•	•	–	–	•	–	•	–	•	–	•	–	•

• = criterion applies; – = criterion does not apply.
DSM-5 = *Diagnostic and Statistical Manual of Mental Disorders*, 5th edition; Rx = prescription.
[1] The criterion wording is based on NSDUH questions since 2021.
[2] These substances contribute to central nervous system stimulant use disorder (see Table 3.8).
[3] These substances contribute to opioid use disorder (see Table 3.8).
[4] These substances contribute to prescription tranquilizer or sedative use disorder (see Table 3.8).
[5] Withdrawal symptoms and requirements differ by substance.
Source: SAMHSA, Center for Behavioral Health Statistics and Quality, National Survey on Drug Use and Health, 2024.

Table 3.8 lists the substances that were included in aggregate SUD measures in the 2024 NSDUH. Questions are asked about SUD symptoms that respondents specifically attributed to their use in the past year of cocaine, heroin, methamphetamine, prescription pain relievers, prescription tranquilizers, prescription stimulants, and prescription sedatives. However, DSM-5 groups these substances into a smaller set of categories for classifying people as having SUDs.

- Cocaine, methamphetamine, and prescription stimulants are included in an overall category for stimulants (referred to in NSDUH as CNS stimulants to differentiate between stimulants that are available by prescription and those that are not).
- Heroin and prescription pain relievers that are opioids are included in an overall category for opioids.
- Prescription tranquilizers and sedatives are included in an overall category for sedatives, hypnotics, and anxiolytics (referred to in NSDUH as prescription tranquilizer or sedative use disorder).

Table 3.8 Substances Included in Aggregate Substance Use Disorder Measures in the 2024 NSDUH

Substance	Substance Use Disorder	Drug Use Disorder	Prescription Drug Use Disorder	Tranquilizer or Sedative Use Disorder	Opioid Use Disorder[1]	CNS Stimulant Use Disorder
Alcohol[2]	•	–	–	–	–	–
Marijuana (Cannabis)[2]	•	•	–	–	–	–
Cocaine[3]	•	•	–	–	–	•
Heroin[3]	•	•	–	–	•	–
Hallucinogens[3]	•	•	–	–	–	–
Inhalants[3]	•	•	–	–	–	–
Methamphetamine[3]	•	•	–	–	–	•
Prescription Pain Relievers[3,4]	•	•	•	–	•	–
Prescription Opioids[5]	–	–	–	–	•	–
Prescription Tranquilizers[3]	•	•	•	•	–	–
Prescription Stimulants[3]	•	•	•	–	–	•
Prescription Sedatives[3]	•	•	•	•	–	–

• = included; – = not included; CNS = central nervous system.

NOTE: For NSDUH, respondents were only considered to have the aggregate disorder noted in the columns if they were defined as having a disorder for at least one of the substances marked in that column.

[1] The opioid use disorder measure also does not capture symptoms that might have arisen solely from the use of illegally made fentanyl. Beginning with the 2024 NSDUH, opioid use disorder refers to the use of prescription opioids or heroin rather than the use of prescription pain relievers or heroin.

[2] NSDUH respondents were asked the respective questions for alcohol use disorder or marijuana (cannabis) use disorder if they reported using these substances on 6 or more days in the past year.

[3] Respondents were asked the respective substance use disorder questions for these substances if they reported any use of these substances in the past year.

[4] For consistency with 2021-2023, creation of the aggregate measures in 2024 for prescription drug use disorder, drug use disorder, and substance use disorder included data from respondents who were classified as having a prescription pain reliever use disorder.

[5] The prescription opioid category is a subset of the prescription pain reliever category. However, columns for prescription drug use disorder, drug use disorder, and substance use disorder were not marked for prescription opioids as per footnote 4.

Source: SAMHSA, Center for Behavioral Health Statistics and Quality, National Survey on Drug Use and Health, 2024.

The 2024 Key Substance Use and Mental Health Indicators report (CBHSQ, 2025g) focused on aggregate SUD measures for consistency with the substance categories in the DSM-5 criteria. The 2024 Detailed Tables (CBHSQ, 2025h) and appendix tables in the 2024 Key Substance Use and Mental Health Indicators report continue to show estimates for SUDs that people attributed to the use of these specific substances.

3.4.7.2.1 Prescription Pain Reliever and Opioid Use Disorder Changes

As discussed in Section 2.2.2, new "OTHER, Specify" questions for any use of other prescription pain relievers were added to the 2024 NSDUH questionnaire to capture details on whether respondents used opioid or nonopioid pain relievers. Section 4.4 contains more information on how these new questions were used to create measures for the use and misuse of prescription opioids for 2024. Section 4.4 also discusses how measures for the use and misuse of prescription opioids were created for 2021-2023 without the availability of these "OTHER, Specify" questions for any use of other prescription pain relievers.

Using this information for the use and misuse of prescription opioids, a new measure for prescription opioid use disorder was created and reported in 2024 data products. Beginning with the 2024 NSDUH, respondents were not counted as having an opioid use disorder in the past year if they used prescription pain relievers but not prescription opioids in the past year and they did not use heroin in the past year.[52] Respondents also were not counted as having an opioid use disorder if they did not meet the full SUD criteria individually for heroin or prescription opioids. To allow trends in opioid use disorder to be presented for 2021-2024 in NSDUH data products, similar measures of opioid use disorder were created for 2021-2023. Because "OTHER, Specify" data for any use of other pain relievers were not available for these prior years, respondents were assumed not to have used prescription opioids in the past year if they reported the use of only "other" pain relievers (see Section 4.4.5 and Figure 4.7). A measure of prescription pain reliever use disorder in the past year continued to be created for 2024 for use in creating measures for any SUD, drug use disorder, and prescription drug use disorder, but this measure was not used to estimate opioid use disorder (Table 3.8).

3.4.7.2.2 Substance Use Disorder Severity

The severity of an SUD for a particular substance (or category of substances) according to DSM-5 criteria was determined by the number of individual criteria that a respondent met for a particular substance (or category of substances). The number of criteria required for each severity level was the same for every substance, regardless of how many criteria were applicable for the substance.

- People were classified in the mild SUD category if they met two or three applicable SUD criteria for that substance.
- People were classified in the moderate SUD category if they met four or five applicable SUD criteria for that substance.
- People were classified in the severe SUD category if they met six or more applicable SUD criteria for that substance.

In addition to the three levels of SUD severity for individual substances, measures of severity were created for aggregate SUD categories listed in Table 3.8. SUD severity measures for these aggregate SUD categories were defined from the maximum severity level (i.e., mild, moderate,

[52] In NSDUH data products before 2024, respondents who had a heroin use disorder or prescription pain reliever use disorder were classified as having an opioid use disorder.

or severe) across the multiple SUDs that were included in the category. For example, if people had a moderate alcohol use disorder and a mild marijuana (cannabis) use disorder as their only SUDs in the past year, then they were classified as having moderate SUD. SUD severity measures for prescription pain reliever use disorder and prescription opioid use disorder were also updated based on the changes described in Section 3.4.7.2.1 to be consistent with their respective SUD variables. SUD severity measures for CNS stimulant use disorder, opioid use disorder, and tranquilizer or sedative use disorder in the 2024 Key Substance Use and Mental Health Indicators report (CBHSQ, 2025g) focused on whether people had an SUD that they attributed to illicit drug use within that category (e.g., heroin use or the misuse of prescription opioids), or they had an SUD that they attributed only to the use but not misuse of prescription drugs (e.g., use but not misuse of prescription opioids).

3.4.7.2.3 Imputation Issues for Substance Use Disorder Measures

Missing values in the DSM-5 SUD data for drugs and alcohol were replaced using statistical imputation (see Section 2.3.3). Consequently, the variables for specific SUDs (e.g., marijuana [cannabis] use disorder; Table 3.7) as well as the composite SUD variables (e.g., drug use disorder; Table 3.8) were not subject to the kinds of potential biases due to missing data described in Section 3.3.2.

If measures for SUD and SUD severity needed to be imputed for a given substance, only the final SUD and severity outcomes were imputed; individual SUD criterion variables for a given substance were not. Further documentation of imputation procedures for SUD measures other than the SUD measures with changes for 2024 can be found in the 2023 Editing and Imputation Report (CBHSQ, 2025c). Information on changes to imputation procedures for SUD measures for 2024 is expected to appear in the 2024 Editing and Imputation Report (CBHSQ, forthcoming c).

3.4.7.2.4 Miscellaneous Data Issues for Substance Use Disorder Measures

For alcohol and marijuana, respondents were asked the SUD questions if they reported substance use on 6 or more days in the past year or if they reported any substance use in the past year but had missing data for the frequency of use in the past year. Therefore, inconsistencies could occur where respondents were classified as having an alcohol or marijuana use disorder but their statistically imputed frequency of use was fewer than 6 days in the past year. However, this situation was uncommon. For example, about 41,000 respondents (unweighted) reported past year alcohol use in 2024.[53] Of these, fewer than 50 respondents had missing frequency data and were asked the alcohol use disorder questions, but their final statistically imputed frequency of use indicated they used alcohol on 5 or fewer days in the past year.

For methamphetamine, cocaine, and heroin, respondents were asked the respective SUD questions if they reported past year use in the corresponding substance use sections *or* if they reported use in the past year in the special drugs section (i.e., use of methamphetamine,

[53] This number does not include respondents whose status as past year alcohol users was unknown based on their questionnaire responses but who were statistically imputed to be past year alcohol users.

cocaine, or heroin with a needle in the past year or smoking or sniffing of heroin in the past year). Thus, the questionnaire logic allowed some respondents to be asked the SUD questions for these drugs even if they did not report past year use when they were asked previously about their most recent use of methamphetamine, cocaine, crack cocaine, or heroin in the respective main drug sections. For example, about 600 respondents in 2024 were asked the questions about methamphetamine use disorder because they reported past year use when asked directly about their most recent use of methamphetamine. Fewer than 15 additional respondents were asked these questions because they reported past year use of methamphetamine with a needle in the special drugs section despite not having previously reported past year use of methamphetamine.

Missing or incomplete responses were not statistically imputed in the special drugs section for the use of cocaine, heroin, or methamphetamine with a needle or for smoking or sniffing heroin. Therefore, if respondents were statistically imputed to have last used cocaine, heroin, or methamphetamine more than 12 months ago and there was nothing from the special drugs section to indicate use of these substances in the past year, then the final statistically imputed recency was used to infer that SUD questions did not apply.

3.4.8 Substance Use Treatment

The substance use treatment questions underwent considerable revisions for the 2022 NSDUH. Revisions for 2022 were intended to reflect contemporary changes in the delivery of substance use treatment services. The following is a summary of key changes to these questions:

- Respondents who reported any lifetime use of prescription drugs (pain relievers, tranquilizers, stimulants, or sedatives) were eligible to be asked questions about the receipt of substance use treatment. Before 2022, respondents who reported lifetime use but not misuse of prescription drugs were not asked about the receipt of substance use treatment unless they reported lifetime use of alcohol or other drugs (marijuana, cocaine, heroin, hallucinogens, inhalants, or methamphetamine).
- All lifetime users of alcohol or drugs were asked whether they received specific types of treatment services in the past 12 months. Before 2022, respondents who reported the use of alcohol or illicit drugs were not asked about the specific types of treatment they received in the past 12 months unless they first reported that they received any substance use treatment in their lifetime and, if so, whether they received substance use treatment in the past 12 months. These changes meant that more respondents beginning in 2022 were eligible to be asked the questions about the types of treatment they received in the past 12 months.
- Additional inpatient and outpatient locations were provided for respondents to report where they received treatment in the past 12 months.
- The question about the receipt of treatment over the phone or through video (i.e., telehealth treatment) was revised to emphasize that the treatment was from a therapist or other healthcare professional.
- Questions about medications for alcohol use disorder (MAUD) or medications for opioid use disorder (MOUD) were added to the section.

New follow-up questions were added to the 2024 NSDUH for respondents who reported that they received treatment in an inpatient or outpatient location but they did not report any substances for which they received treatment (see Section 2.2.2). These respondents were given a second opportunity to specify the substances for which they received inpatient or outpatient treatment or to enter "None" if they did not receive treatment.

Starting in June 2024, additional changes were included in the questionnaire to improve respondent understanding of whether they received treatment for their use of alcohol or drugs at specific inpatient or outpatient locations (see Section 2.2.2). The following changes were made:

- Introductory text for questions about inpatient and outpatient treatment was revised to mention "any of the locations below" for response choices for specific locations.
- For each inpatient and outpatient location that was presented to respondents, text was added to indicate that the treatment was "for alcohol or drug treatment." For example, the response option for a mental health treatment center as an outpatient was revised to mention a mental health center as an outpatient "for alcohol or drug treatment."
- In the grid format for respondents to report the locations where they received inpatient treatment, the heading "Stayed overnight or longer for alcohol or drug treatment in…" was replaced with "LOCATION."
- In the grid format for respondents to report the locations where they received outpatient treatment, the heading "Received outpatient alcohol or drug treatment at…" was replaced with "LOCATION."

Because of the changes in 2022, the definition for the receipt of substance use treatment changed. Therefore, estimates from 2022 based on these outcomes should not be compared with estimates from 2021.

Analysis of the data from June to December 2024 indicated that the changes implemented in June 2024 not only reduced the number of respondents who did not report the specific substances for which they received inpatient or outpatient treatment (see Section 3.4.8.3) but also affected reporting of the receipt of treatment in inpatient and outpatient locations. As a consequence of these changes, substance use treatment estimates overall and for inpatient or outpatient locations from 2024 are not comparable with those from 2022 and 2023. The substance use treatment estimates from 2024 for the receipt of MAUD or MOUD, treatment via telehealth, and treatment in a prison, jail, or juvenile detention center (see Section 3.4.8.1) are still comparable with estimates from 2022 and 2023. For measures of the receipt of specific other services described in Section 3.4.8.2 that were available for 2021-2024, estimates also remain comparable with estimates from 2022 and 2023.[54] Work will be performed for the 2025 NSDUH to allow trends to be presented for the receipt of substance use treatment from 2022 to 2025.

[54] The question about receipt of overdose reversal medicine such as Narcan® or naloxone was added for 2024. Estimates based on this question are not available for prior years.

Questions for whether respondents perceived an unmet need for substance use treatment and the reasons for respondents not receiving treatment were also revised for the 2022 NSDUH. Due to the additional changes for 2024 that were described previously, estimates from 2024 for the perceived unmet need for substance use treatment and reasons for not receiving treatment also are not considered comparable with corresponding estimates from prior years. Specific changes are discussed in Section 3.4.8.5 in the context of the measurement of these outcomes.

3.4.8.1 Locations and Types of Services for Substance Use Treatment

Receipt of substance use treatment includes the receipt of treatment in the past year for the use of alcohol or drugs in an inpatient location, in an outpatient location, via telehealth, or in a prison, jail, or juvenile detention center. The definition also includes the receipt of MAUD or MOUD.

Inpatient treatment was defined as treatment in locations where people stayed overnight or longer and included the following locations:

- a hospital,
- a residential drug or alcohol rehab or treatment center,
- a residential mental health treatment center, or
- some other place the respondent stayed overnight or longer.

Outpatient treatment was defined as treatment people received in locations that did not require them to stay overnight or longer. These measures included the following locations:

- a drug or alcohol rehab or treatment center as an outpatient;
- a mental health treatment center as an outpatient;
- the office of a therapist, psychologist, psychiatrist, or other substance use treatment professional;
- a general medical clinic or doctor's office;
- a hospital as an outpatient;
- a school health or counseling center; or
- some other place as an outpatient.

MAUD and MOUD questions were revised and asked in the alcohol or drug treatment section for 2022 in the context of questions about other substance use treatment in the past 12 months. Therefore, estimates for MAUD and MOUD since 2022 are not comparable with MAUD and MOUD estimates from the 2021 NSDUH that were based on questions in the emerging issues section of the questionnaire. However, the 2024 MAUD and MOUD estimates are comparable with the 2022 and 2023 estimates.

As mentioned previously, however, estimates from 2024 based on variables for any receipt of substance use treatment in the past year and receipt of treatment in inpatient and outpatient locations are not comparable with estimates from 2022 and 2023. Estimates from 2024 for the receipt of treatment via telehealth and treatment in a prison, jail, or juvenile detention center remain comparable with estimates from 2022 and 2023.

Variables for the receipt of substance use treatment in the past year in inpatient locations, in outpatient locations, via telehealth, or in a prison, jail, or juvenile detention center were statistically imputed. Variables also were statistically imputed for the receipt of MAUD and MOUD in the past year. Consequently, variables since 2022 for the receipt of substance use treatment in the past year do not have missing data.

3.4.8.2 Receipt of Other Services for Substance Use

The 2024 NSDUH also collected information on the receipt of other services in the past year for the use of alcohol or drugs, including the following:

- services from a support group,
- services from a peer support specialist or recovery coach,
- services in an emergency room or emergency department,
- withdrawal management services, or
- receipt of overdose reversal medicine such as Narcan® or naloxone.

NSDUH did not classify these other services as "substance use treatment." For example, the subject matter experts who worked with SAMHSA to develop the revised substance use treatment questions for 2022 thought that services in an emergency room or emergency department and withdrawal management services were primarily intended to treat acute medical conditions, such as an overdose or withdrawal symptoms, rather than to engage patients in a longer term process of change to promote recovery from SUDs. Experts also did not consider services from a support group or recovery coach to be substance use treatment because of issues related to licensure or reimbursement for services.

However, these other services were included in a separate aggregate measure presented in the 2024 Detailed Tables (CBHSQ, 2025h) that was created to cover the receipt of substance use treatment or other services. A question about the receipt of overdose reversal medicine was added to the questionnaire, and the associated measure for the receipt of overdose reversal medicine was included in this aggregate measure beginning in 2024.

Because of the changes to the inpatient and outpatient questions in June 2024 and the addition of this new question on receipt of overdose reversal medicine in 2024, the estimates from 2024 for the aggregate measure of the receipt of substance use treatment or other services are not comparable with those from prior years. However, estimates from 2024 for specific services from a support group through withdrawal management services that were mentioned previously remained comparable with estimates from 2022 and 2023. Variables for the receipt of other services also were statistically imputed, so these variables do not have missing data.

3.4.8.3 Substances for Which People Received Treatment or Other Services

If respondents in the 2024 NSDUH reported the use of alcohol or drugs in their lifetime and reported that they received substance use treatment in the past year, then they were asked to report the substances for which they received treatment. Questions varied depending on the locations in which people received treatment or whether people received treatment via telehealth. Except for the receipt of overdose reversal medicine, respondents who received

other services in the past year, as described in Section 3.4.8.2, also were asked to report whether they received these services for alcohol use only, drug use only, or both alcohol and drug use.

For MAUD and MOUD, the questions applied specifically to the use of alcohol or opioids. Respondents who reported lifetime use of alcohol were asked whether they received medication in the past 12 months that was prescribed to them to help them cut back on or stop their use of alcohol. Examples of medications shown to respondents that are prescribed as a part of MAUD included the following:

- acamprosate (also known as Campral®),
- disulfiram (also known as Antabuse®),
- naltrexone pills (also known as ReVia® or Trexan®), and
- injectable naltrexone (also known as Vivitrol®).

Respondents who reported the lifetime use of heroin or prescription pain relievers were asked whether they received medication in the past 12 months to help them cut back on or stop their "drug use," but this question was asked only if respondents reported lifetime use of the drugs mentioned previously. Examples of medications shown to respondents that are prescribed as a part of MOUD included the following:

- methadone,
- buprenorphine or buprenorphine-naloxone pills or film taken by mouth (also known as Suboxone®, Zubsolv®, Bunavail®, or Subutex®),
- injectable buprenorphine (also known as Sublocade®),
- buprenorphine implants (also known as Probuphine®),
- naltrexone pills (also known as ReVia® or Trexan®), and
- injectable naltrexone (also known as Vivitrol®).

Because of this list of drugs in the questionnaire, respondents who used prescription pain relievers but not heroin in their lifetime and who reported receiving prescription medication in the past year to help reduce or stop their use of drugs were assumed to have received MOUD.

3.4.8.3.1 Substances for Which People Received Treatment as Inpatients or Outpatients

Respondents who reported receiving treatment in inpatient or outpatient locations in the past year were asked to report whether they received treatment for the specific substances that they used in their lifetime; respondents were asked about treatment only for the specific substances they reported using in their lifetime. Respondents could be asked about the receipt of inpatient or outpatient treatment for the following substances:

- alcohol,
- marijuana,
- cocaine,
- heroin,
- hallucinogens,
- inhalants,
- methamphetamine,
- prescription pain relievers,
- prescription tranquilizers,
- prescription stimulants, and
- prescription sedatives.

For example, if respondents reported the use of alcohol, marijuana, cocaine, heroin, and prescription pain relievers in their lifetime, and they reported the receipt of treatment in one or more outpatient locations in the past year, then they were asked whether they received treatment as an outpatient for their use of these five specific substances. These respondents were not asked whether they received treatment as an outpatient for their use of hallucinogens, inhalants, methamphetamine, prescription tranquilizers, prescription stimulants, or prescription sedatives. The same logic applied to questions for the receipt of treatment as an inpatient in the past year for the use of specific substances respondents used in their lifetime.

In addition to these substances, respondents who reported substance use treatment as an inpatient or an outpatient were asked whether they received treatment in the respective setting for their use of "some other drug." Respondents were asked to specify the other drug or drugs for which they received treatment. However, respondents might not report that they received treatment as an inpatient or an outpatient for their use of a particular substance (e.g., heroin), but they could specify it as "some other drug" for which they received treatment. In this situation, respondents were logically inferred to have received treatment in the respective setting for their use of that substance.

In 2024, relatively large proportions of people who reported that they received inpatient or outpatient treatment in the past 12 months did not indicate the specific substance(s) for which they received treatment in these locations, including treatment for the use of some other drug. Stated another way, these reports of inpatient or outpatient treatment were not substantiated by reports of treatment for the use of specific substances. Specifically, about one quarter of respondents who reported inpatient treatment in the past year did not report the specific substances for which they received treatment as inpatients. Among respondents who reported outpatient treatment in the past year, about one third did not report the specific substances for which they received treatment as outpatients. A "substance unspecified" category was created for these respondents. If respondents in this "substance unspecified" group did not actually receive substance use treatment, then estimates for any substance use treatment and for inpatient or outpatient substance use treatment could be overestimates.

As mentioned previously, new follow-up questions were added to the 2024 NSDUH for respondents who reported that they received treatment in an inpatient or outpatient location but they did not report any substances for which they received treatment. These respondents were given a second opportunity to specify the substances for which they received inpatient or outpatient treatment or to enter "None" if they did not receive treatment. For example, respondents could enter "None" if they made a mistake in their previous reports of inpatient or outpatient treatment. If respondents entered "None" on follow-up, it logically would follow that

they did not receive treatment. Consequently, data from these respondents could overestimate the percentages of people who received any substance use treatment, inpatient treatment, or outpatient treatment if these respondents had not received treatment for their use of alcohol or drugs. For the 2024 NSDUH reports and tables, substance use treatment estimates from 2024 do not incorporate information from these new follow-up questions.

As was mentioned in Section 3.4.8, additional questionnaire changes were included in June 2024 to improve respondent understanding of whether they received treatment for their use of alcohol or drugs at specific inpatient or outpatient locations. Analysis of the data from June to December 2024 indicated that these changes reduced the number of respondents in the "substance unspecified" categories. As a consequence of these changes, however, estimates from 2024 for any treatment for the use of specific substances and estimates for inpatient and outpatient treatment for the use of specific substances are not comparable with those from prior years. Work will be performed for the 2025 NSDUH to allow trends to be presented for the receipt of substance use treatment from 2022 to 2025.

3.4.8.3.2 Substances for Which People Received Treatment via Telehealth; in a Prison, Jail, or Juvenile Detention Center; or through Other Services

Respondents who reported (1) substance use treatment via telehealth; treatment in a prison, jail, or juvenile detention center; or the receipt of other substance use services (see Section 3.4.8.2), except for the receipt of overdose reversal medicine; and (2) lifetime use of alcohol and drugs were asked for these types of treatment whether they received treatment for their use of alcohol only, drugs only, or both alcohol and drugs. Respondents were not asked these follow-up questions if they reported lifetime use of only alcohol or only drugs. Whether respondents received other services for the use of alcohol only, drugs only, or both alcohol and drugs did not apply to the question about overdose reversal medicine because this question was specific to drug use.

If respondents reported the lifetime use of alcohol, and they answered "no" to all questions about their use of drugs in their lifetime, then they were logically inferred to have received these types of treatment for their use of alcohol only. Similarly, if respondents reported the lifetime use of one or more drugs, and they answered "no" for their lifetime use of alcohol, then they were logically inferred to have received these types of treatment for their use of drugs only.

Edited variables for the receipt of these types of treatment for the use of alcohol only, drugs only, or both alcohol and drugs retained missing values if respondents reported the lifetime use of only alcohol or only drugs but they had missing data for the lifetime use of some substances. For example, if respondents reported the lifetime use of only alcohol, but they did not know or refused to report whether they had ever used some drugs, then it could not be determined unambiguously that these respondents were lifetime users of only alcohol.

Variables for the substances for which respondents received treatment in these locations were not statistically imputed. However, the procedures for creating recoded variables for the receipt of telehealth treatment for the use of alcohol or the use of drugs that were reported in 2024

NSDUH data products reduced the amount of missing data by incorporating data from statistically imputed variables for the lifetime use of alcohol and drugs. Any remaining respondents with missing data were excluded from the analyses to produce published estimates for the 2024 NSDUH. See Section 3.3.2 for a discussion of the potential bias in estimates because of missing data.

Estimates for the substances for which people received treatment via telehealth or in a prison, jail, or juvenile detention center remain comparable with estimates from 2022 and 2023. Estimates for the substances for which people received other services also remain comparable with estimates from 2022 and 2023.

3.4.8.4 Need for Substance Use Treatment

Before 2022, NSDUH data products included substance use treatment at a "specialty facility" in the past year as part of the definition for whether people needed substance use treatment. With the changes to the questionnaire in 2022, the term "specialty facility" was dropped from the 2022 and future NSDUH data products.

Consequently, the definition of the need for substance use treatment was revised beginning with the 2022 NSDUH. Respondents were classified as needing substance use treatment if they had an SUD in the past year, as defined in Section 3.4.7, or they received substance use treatment in the past year, as defined in Section 3.4.8.1. As noted previously, the "other services" described in Section 3.4.8.2 were not counted as substance use treatment. Because the variables for SUD and receipt of substance use treatment in the past year were statistically imputed, the variable for the need for substance use treatment does not have missing data.

People who received substance use treatment in the past year were classified as needing substance use treatment. The changes to the questions for inpatient and outpatient treatment in June 2024 that were described previously also affected the need for substance use treatment measure. Consequently, estimates from 2024 for the need for substance use treatment are considered not to be comparable with estimates from 2022 and 2023.

3.4.8.5 Perceived Unmet Need for Substance Use Treatment and Reasons for Not Receiving Treatment

Questions about whether respondents perceived an unmet need for substance use treatment and the reasons for not receiving treatment were also revised for the 2022 NSDUH. The questions did not change for 2024.

- These questions were asked only if respondents did not report any substance use treatment in the past year (i.e., inpatient or outpatient treatment, MAUD or MOUD, telehealth treatment, or treatment in a prison, jail, or juvenile detention center).
- Respondents were asked whether they sought treatment rather than being asked whether they made an effort to get treatment.
- Respondents who did not report seeking treatment were asked whether they thought they should get treatment rather than being asked whether they thought they needed treatment.

NSDUH respondents were classified as having a perceived unmet need for substance use treatment if they did not receive substance use treatment in the past year and reported either of the following:

- They did not receive services from a support group, from a peer support specialist or recovery coach, in an emergency room, or from withdrawal management services, but they sought or thought they should get treatment in the past 12 months for their use of alcohol or drugs.
- They received the other services described in the preceding bullet (see Section 3.4.8.2), but they sought or thought they should get additional professional counseling, medication, or other treatment in the past 12 months for their use of alcohol or drugs.[55]

Questions about reasons for people not receiving substance use treatment (i.e., barriers to treatment) were asked of respondents who reported a perceived unmet need for substance use treatment. The questionnaire changes for 2022 also included an expanded list of possible barriers to the receipt of treatment. For each reason for not receiving treatment, respondents were asked whether that reason was "one of the reasons" or "not one of the reasons" they did not seek or get professional counseling, medication, or other treatment for their use of alcohol or drugs.

Although questions for the perceived unmet need for substance use treatment and reasons for people not receiving treatment did not change for 2024, the questions for inpatient and outpatient treatment changed in June 2024, as described previously. These changes affected who was eligible to receive questions about unmet need. Consequently, estimates from 2024 for the perceived unmet need for substance use treatment and reasons for not receiving treatment are considered not to be comparable with estimates from 2022 and 2023.

Variables for the perceived unmet need for substance use treatment and reasons for not receiving substance use treatment were not statistically imputed and had missing data. Respondents with missing data were excluded from the analyses used to produce published estimates for these measures. See Section 3.3.2 for a discussion of the potential bias in estimates because of missing data.

[55] The question about the receipt of overdose reversal medicine added for 2024 was not included in the criteria asking whether respondents sought or thought they should get professional counseling, medication, or other treatment or *additional* professional counseling, medication, or other treatment. Beginning in 2024, respondents who reported receiving overdose reversal medication but no other treatment or services were still asked whether they sought or thought they should get professional counseling, medication, or other treatment. Likewise, respondents who did not receive substance use treatment but they received overdose reversal medicine and services from a support group, from a peer support specialist or recovery coach, in an emergency room, or from withdrawal management services were still asked whether they sought or thought they should get additional professional counseling, medication, or other treatment.

3.4.9 Estimation of Serious and Other Levels of Mental Illness

3.4.9.1 Background

The 1992 Alcohol, Drug Abuse, and Mental Health Administration Reorganization Act created SAMHSA and required the new organization to develop a definition and methodology for estimating SMI among adults for use by states in developing their plans for use of block grant funds distributed by SAMHSA. A technical advisory group convened by SAMHSA was tasked with developing a definition of SMI, which was published in the *Federal Register* in 1993 (SAMHSA, 1993):

> Pursuant to Section 1912(c) of the Public Health Service Act, as amended by Public Law 102-321, "adults with serious mental illness" are defined as the following:
>
> - Individuals aged 18 and over, who currently or at any time during the past year, have had diagnosable mental, behavioral, or emotional disorder of sufficient duration to meet diagnostic criteria specified within DSM-III-R that has resulted in functional impairment, which substantially interferes with or limits one or more major life activities.
> - These disorders include any mental disorder (including those of biological etiology) listed in DSM-III-R or their ICD-9-CM equivalent (and subsequent revisions), with the exception of DSM-III-R "V" codes, substance use disorders, and developmental disorders, which are excluded unless they co-occur with other diagnosable serious mental illness.
> - All of these disorders have episodic, recurrent, or persistent features; however, they vary in terms of severity or disabling effects. Functional impairment is defined as difficulties that substantially interfere with or limit role functioning in one or more major life activities including basic daily living skills (e.g., eating, bathing, dressing); instrumental living skills (e.g., maintaining a household, managing money, getting around the community, taking prescribed medication); and functioning in social, family, and vocational/educational contexts.
> - Adults who would have met functional impairment criteria during the referenced year without benefit of treatment or other support services are considered to have serious mental illness.

In December 2006, a new technical advisory group was convened within SAMHSA to solicit recommendations for data collection strategies to address SAMHSA's legislative requirements. Although the technical advisory group recognized the ideal way to estimate SMI in NSDUH would be to administer a clinical diagnostic interview annually to all adult respondents, this approach was not feasible because of constraints on the interview time and the need for trained mental health clinicians to conduct the interviews. Therefore, the approach recommended by the technical advisory group and adopted by SAMHSA for NSDUH was to utilize short scales in the NSDUH interview to separately measure psychological distress and functional impairment. NSDUH used the resulting data collected from these short scales in a statistical model to predict whether a respondent had mental illness.

To create the model, a Mental Health Surveillance Study (MHSS) was initiated in 2007 as part of NSDUH to develop and implement methods to estimate SMI. Models using the short scales for psychological distress and impairment to predict mental illness status were developed from a subsample of adult respondents who had completed the NSDUH interview and were administered a clinical psychological diagnostic interview soon afterward. For the clinical interview data, people were classified as having SMI if they had a diagnosable mental, behavioral, or emotional disorder in the past 12 months according to the criteria in the *Diagnostic and Statistical Manual of Mental Disorders*, 4th edition (DSM-IV; American Psychiatric Association, 1994), other than a developmental disorder or SUD, that resulted in substantial functional impairment. This estimation methodology was implemented in the 2008 NSDUH.

To recalibrate estimate(s) of mental illness using DSM-5 criteria in the future, the Mental Illness Calibration Study (MICS) is being conducted as part of the 2023 and 2024 NSDUHs. At the end of the NSDUH interview, adult respondents are randomly selected and asked to participate in a follow-up clinical interview conducted using DSM-5 diagnostic criteria. The goal of MICS is to fit a prediction model for SMI among adults aged 18 or older that can be used to create updated model-based estimates of SMI and other mental illness categories at the national and subgroup levels, such as by age group and race/ethnicity. Instrumentation from the clinical interviews will provide the "gold standard" for the presence of a diagnosable mental disorder and level of functional impairment for use in an updated prediction model to estimate mental illness in NSDUH based on the DSM-5 criteria. For more information on the MICS sample design, see the *2023 National Survey on Drug Use and Health (NSDUH) Methodological Resource Book, Section 2: Sample Design Report* (CBHSQ, 2024a) or the *2024 National Survey on Drug Use and Health (NSDUH) Methodological Resource Book, Section 2: Sample Design and Experience Report* (CBHSQ, forthcoming b).

3.4.9.2 Historical Summary of the 2012 Model

The model used for 2024 to predict mental illness was developed for the 2012 NSDUH (subsequently referred to as the "2012 model"). This 2012 model was previously updated from a model developed in 2008 (subsequently referred to as the "2008 model").

The 2008 model was created from a randomly selected subsample of approximately 1,500 adults in 2008 who had completed the NSDUH interview. Respondents were recruited for a follow-up clinical interview consisting of a diagnostic assessment for mental disorders.[56] In order to determine the optimal scale for measuring functional impairment in NSDUH, roughly half of the adult respondents were assigned to receive an abbreviated eight-item version of the World Health Organization Disability Assessment Schedule (WHODAS; Novak et al., 2010), and the other half were assigned to receive the Sheehan Disability Scale (SDS; Leon et al., 1997). The WHODAS was chosen as the impairment scale to be administered in the 2009 and subsequent NSDUHs (Office of Applied Studies, 2009), although the SDS is included in NSDUH for measurement of impairment during the most severe depressive episode in the past 12 months.

[56] *Structured Clinical Interview for the DSM-IV-TR Axis I Disorders, Research Version, Non-patient Edition (SCID-I/NP)*; clinical interviews would require the use of a DSM-5 diagnostic assessment to identify mental disorders according to DSM-5 criteria (First et al., 2002).

For more information on the 2008 MHSS design and analysis, see Colpe et al. (2009) and Office of Applied Studies (2009). Information about the 2008 model is available in Appendix B of the *Results from the 2012 National Survey on Drug Use and Health: Mental Health Findings* (CBHSQ, 2013b) and in Section 3.4.7 of the 2020 Methodological Summary and Definitions report (CBHSQ, 2021b).

Based on the accumulated MHSS clinical data collected from 2008 to 2012, however, SAMHSA determined the 2008 model had some important shortcomings not detected in the original model fitting because of the small sample of clinical interview respondents in 2008. Specifically, estimates of SMI and AMI among young adults based on the 2008 model were higher than the estimates for this age group based on the clinical interview data. In addition, improvements were needed in the weighting procedures for the MHSS clinical data to account better for undercoverage and nonresponse. Only NSDUH respondents in 2008 who answered their surveys in English were eligible for the clinical follow-up, and people with mental illness appeared to be more likely to participate in the follow-up.

Because of these concerns, SAMHSA fit a more accurate model using combined 2008-2012 clinical data for the 2012 model. To reduce bias and improve prediction, additional mental health-related variables and an age variable were included in the model. In addition, to protect against potential coverage and nonresponse error, alternatives for the weights were applied to the clinical sample data for the model development.

The next sections describe the instruments and items used to measure the variables employed in the 2012 model. Specifically, sections include descriptions of the instrument used to measure mental illness in the clinical interviews, followed by the scales and items in the main NSDUH interviews used as predictor variables in the model (i.e., the Kessler-6 [K6] and WHODAS total scores, age, MDE, and suicidal thoughts); see Sections 3.4.12 and 3.4.13 for more information on the measurement of MDE and serious thoughts of suicide among adults, respectively. Next, procedures for the MHSS clinical interview sampling and weighting and for developing the 2012 model are described. Section 3.4.9.9 discusses SEs for the mental illness estimates based on the 2012 model. Remaining sections discuss miscellaneous issues for the mental illness variables.

3.4.9.3 Clinical Measurement of Mental Illness

Mental illness was measured in the MHSS clinical interviews using an adapted version of the *Structured Clinical Interview for the DSM-IV-TR Axis I Disorders, Research Version, Non-patient Edition (SCID-I/NP)* (First et al., 2002) and was differentiated by the level of functional impairment based on the Global Assessment of Functioning (GAF) scale (Endicott et al., 1976).[57]

[57] The GAF is a numeric scale used by mental health clinicians to quantify the severity of mental disorders and the extent to which mental disorders negatively affected a person's daily functioning. In the MHSS, GAF scores were assigned by clinical interviewers at the end of each SCID interview based on information gathered throughout the interview about symptoms of mental disorders and related impairment. This procedure differs from use of the WHODAS in NSDUH, which relies on respondents' (rather than clinicians') perceptions of the extent to which their symptoms of psychological distress affected their day-to-day functioning.

Past year disorders assessed through the SCID included mood disorders (e.g., MDE, manic episode), anxiety disorders (e.g., panic disorder, generalized anxiety disorder, posttraumatic stress disorder), eating disorders (e.g., anorexia nervosa), intermittent explosive disorder, and adjustment disorder. In addition, the presence of psychotic symptoms was assessed. SUDs were also assessed, although these disorders were not used to produce estimates of mental illness.

- Respondents were classified as having *any mental illness* (AMI) if they were determined to have any of the mental disorders assessed in the SCID (not including SUDs), regardless of the level of functional impairment.
- Respondents were classified as having *low (mild) mental illness* if they had any of the mental disorders assessed in the SCID (not including SUDs), but these disorders resulted in no more than mild impairment, based on GAF scores of greater than 59.
- Respondents were classified as having *moderate mental illness* if they had any of the mental disorders assessed in the SCID (not including SUDs), and these disorders resulted in moderate impairment, based on GAF scores of 51 to 59.
- Respondents were classified as having *serious mental illness* (SMI) if they had any of the mental disorders assessed in the SCID (not including SUDs), and these disorders resulted in substantial impairment in carrying out major life activities, based on GAF scores of 50 or below. The SMI diagnosis was used as the response variable in both the 2008 and 2012 prediction models.

The SCID and the GAF in combination from the MHSS were considered to be the "gold standard" for measuring mental illness.

3.4.9.4 K6

The K6 in the main NSDUH interview consists of two sets of six questions in the mental health section for adult respondents. These questions ask adult respondents how frequently they experienced symptoms of psychological distress during two different time periods: (1) during the past 30 days, and (2) if applicable, the 1 month in the past year when they were at their worst emotionally. Respondents were asked about the second time period only if they indicated there was a month in the past 12 months when they felt more depressed, anxious, or emotionally stressed than they felt during the past 30 days. All questions had the same response categories:

1 All of the time

2 Most of the time

3 Some of the time

4 A little of the time

5 None of the time

Don't know/Refused

The six questions in the K6 scale for the past month are as follows:

NERVE30	During the past 30 days, how often did you feel nervous?
HOPE30	During the past 30 days, how often did you feel hopeless?
FIDG30	During the past 30 days, how often did you feel restless or fidgety?
NOCHR30	During the past 30 days, how often did you feel so sad or depressed that nothing could cheer you up?
EFFORT30	During the past 30 days, how often did you feel that everything was an effort?
DOWN30	During the past 30 days, how often did you feel down on yourself, no good or worthless?

In the 2024 NSDUH, all adult respondents with item nonresponse for these psychological distress items had their scores statistically imputed (see Section 2.3.3). Thus, there were no missing values in the 2024 survey for measures of psychological distress (based on the K6 distress scale) used in the mental illness prediction model. Imputation also mitigated potential effects of nonresponse from adult respondents who broke off the interview before or during the mental health section of the interview.

To create the score corresponding to the past month, the imputation-revised values for the six items for the past 30 days (NERVE30, HOPE30, FIDG30, NOCHR30, EFFORT30, and DOWN30) on the K6 scale were recoded from 0 to 4 so that "all of the time" was coded as 4, "most of the time" as 3, "some of the time" as 2, "a little of the time" as 1, and "none of the time" as 0. Summing across the transformed values for responses in these six items resulted in a score with a range from 0 to 24.

If respondents were asked about a month in the past 12 months when they felt more depressed, anxious, or emotionally stressed than they felt during the past 30 days, they were asked comparable K6 items for that particular month in the past 12 months. The imputation and scoring procedures for these K6 items corresponding to the worst month in the past 12 months were the same as those described previously for the past 30 days.

The maximum of the two K6 total scores for the past 30 days or past 12 months was used for MHSS analysis purposes and in the adult respondents' final data. If respondents were asked K6 items for both the past 30 days and past 12 months, two K6 total scores were calculated for these periods, and the maximum of the two scores was identified. For respondents who were asked the K6 questions only for the past 30 days, the maximum score was the score for the past 30 days.

An alternative K6 total score was also created from the maximum K6 score. Maximum K6 scores less than 8 were recoded as 0. A maximum score of 8 was recoded as 1, a maximum score of 9 was recoded as 2, and so on, until a score of 24 was recoded as 17 in the alternative score variable. This alternative K6 score was used in the 2012 SMI prediction model because SMI prevalence typically was extremely low for respondents with past year K6 scores of less than 8. The prevalence started increasing only when scores were 8 or greater.

3.4.9.5 WHODAS

The WHODAS was modified for use in a general population survey such as NSDUH by making minor changes to question wording and reducing its length (Novak, 2007). A subset of eight items was found to capture the information represented in the full 16-item scale with no significant loss of information.

Respondents were asked the WHODAS questions if they reported having at least some symptoms of psychological distress in the past 30 days or in their worst period in the past 12 months at least a little of the time (i.e., their answers yielded a K6 score greater than zero). Similar to the K6 variables, statistical imputation was used to replace missing data in the WHODAS variables (see Section 2.3.3). Thus, there were no missing values in the 2024 survey for measures of adult SMI and other mental illness measures created from a model using the WHODAS scores. Imputation also mitigated potential effects of nonresponse from adult respondents who broke off the interview before or during the mental health section of the interview.

Respondents were not asked the WHODAS questions if their maximum reported K6 score was zero. Respondents had a maximum unimputed K6 score of zero if their response was "none of the time," "don't know," or "refused" in the past 30 days and in their worst period in the past 12 months (if applicable) for all six symptoms of psychological distress. The K6 score was zero if all K6 questions had missing data because respondents answered all questions as "don't know" or "refused" or if they broke off the interview before being asked the K6 questions. If respondents were not asked the WHODAS questions because of missing data in the K6 questions, but the K6 scores were imputed to a value greater than zero, then missing data in the skipped WHODAS variables also were statistically imputed.

The imputation-revised values for the eight WHODAS items included in the main NSDUH mental health section of the interview were coded on a 0 to 3 scale, with imputed responses of "no difficulty" coded as 0, "mild difficulty" coded as 1, "moderate difficulty" coded as 2, and "severe difficulty" coded as 3. Some items had an additional category for respondents who did not engage in a particular activity (e.g., they did not leave the house on their own). Respondents who reported they did not engage in an activity were asked a follow-up question to determine whether they did not do so because of emotions, nerves, or mental health. Respondents with an imputation-revised value of "yes" to these follow-up questions were subsequently assigned to the "severe difficulty" category. Respondents with an imputation-revised value of "no" to these follow-up questions were assigned to the "no difficulty" category. Summing across these codes for the eight responses resulted in a total score with a range from 0 to 24.

An alternative WHODAS total score was used in the 2012 SMI prediction model starting from the imputation-revised WHODAS items. Individual item scores of less than 2 were recoded as 0 and item scores of 2 to 3 were recoded as 1. The individual alternative item scores then were summed to yield a total alternative score ranging from 0 to 8. Creation of an alternative version of the WHODAS score assumed a dichotomous measure dividing respondents into two groups (i.e., severely impaired vs. less severely impaired) would fit better than a linear continuous measure in models predicting SMI.

3.4.9.6 Suicidal Thoughts, MDE, and Age

In addition to the K6 and WHODAS scales, the 2012 model included the following measures as predictors of SMI: (1) serious thoughts of suicide in the past year, (2) having a past year MDE, and (3) adjusted age, as defined later in this section. The first two variables were added to the model to decrease the error rate in the predictions (i.e., the sum of the false-negative and false-positive rates relative to the clinical interview results). A recoded age variable reduced the bias in estimates for particular age groups, especially for 18- to 25-year-olds.

All adult respondents in NSDUH were asked the following question in the mental health section about serious thoughts of suicide: "At any time in the past 12 months, that is from [DATEFILL] up to and including today, did you seriously think about trying to kill yourself?"[58] Definitions for MDE in the lifetime and past year periods were based on questions in the adult depression section and are discussed in Section 3.4.12. For the modeling, beginning in 2021, missing data from adult respondents for whether they had serious thoughts of suicide or for having a past year MDE were statistically imputed. See the 2023 Editing and Imputation Report (CBHSQ, 2025c) for more information on how the adult suicidality and MDE measures were imputed for the 2023 NSDUH; these procedures did not change for the 2024 NSDUH.

For respondents aged 18 to 30, an adjusted age was created by subtracting 18 from the respondent's current age, resulting in values ranging from 0 to 12. For a respondent aged 18, for example, the adjusted age was 0 (i.e., 18 minus 18), and for a respondent aged 30, the adjusted age was 12 (i.e., 30 minus 18). For respondents aged 31 or older, the adjusted age was assigned a value of 12.

3.4.9.7 Sampling and Weighting of the 2012 Model

The target annual respondent sample sizes for the MHSS clinical interviews were 1,500 in 2008 (750 of which received the WHODAS and were used in developing the 2008 model), 500 in 2009 and 2010, and 1,500 in 2011 and 2012. Respondent sample sizes were roughly equal across quarters.

A stratified Bernoulli selection process was used in which all eligible NSDUH respondents were given an independent probability of selection into the MHSS based on their strata. In 2008 and the first two quarters in 2009, stratification was based on K6 scores in an attempt to minimize the variance of the estimate for SMI prevalence. In the last two quarters in 2009, stratification attempted to minimize the variance of the AMI prevalence estimate rather than the variance of the SMI estimate. This change reduced the probability a respondent with an extremely large weight would be selected. Starting from 2010, stratification for the MHSS sample incorporated information on functional impairment levels (WHODAS scores) and age in addition to K6 scores. Younger age groups were undersampled for the MHSS clinical sample to reverse the impact of the oversampling of young adults aged 18 to 25 in the main survey (see Section A.1 in Appendix A in the 2012 NSDUH Mental Health Findings report [CBHSQ, 2013b]). This

[58] In the question about serious thoughts of suicide (SUI01), "[DATEFILL]" refers to the date at the start of a respondent's 12-month reference period. The interview program sets the start of the 12-month reference period as the same month and day as the interview date but in the previous calendar year.

undersampling of younger age groups resulted in a more equally allocated clinical sample by age. More details about the sample design for the MHSS clinical study can be found in the 2012 Sample Design Report (CBHSQ, 2013a).

Special clinical sample analysis weights were created. Each was the product of the following seven weight components: (1) the NSDUH analysis weight; (2) a coverage adjustment for Hispanic people who completed the main NSDUH interview in English to account for Hispanic people who completed it in Spanish and thus were not eligible for the English-language clinical follow-up interview; (3) the inverse of the selection probability for clinical follow-up; (4) a refusal adjustment to account for NSDUH respondents who were selected for the MHSS but declined to be contacted for the clinical interview; (5) another nonresponse adjustment to account for MHSS nonresponse among NSDUH respondents who had originally agreed to be recontacted for the clinical interview but did not complete the interview; (6) poststratification adjustments to reduce the variance of the resulting estimates by matching the weighted main NSDUH interview sample by age, sex, race/ethnicity, alternative K6 score, alternative WHODAS score, having had serious thoughts of suicide in the past year, and having had an MDE;[59] and (7) a yearly scaling factor. The first six weight components were created separately for each year.

The 2012 model was fit by assuming the relationship between SMI and the covariates of the model stayed the same from 2008 through 2012. Because the sample size, sampling allocation, and weight adjustments for the MHSS clinical samples differed across years, gains in statistical efficiency were realized by scaling the weights in each year using the following scaling factors: 12 percent for 2008, 4 percent for 2009, 14 percent for 2010, 35 percent for 2011, and 35 percent for 2012. The scaling factors were determined based on the relative sizes of the estimated variances for estimates of SMI, AMI, and past year MDE made directly from SCID diagnoses.[60]

3.4.9.8 2012 SMI Model

The 2012 SMI prediction model was fit with data from 4,912 WHODAS MHSS respondents from 2008 through 2012. The response variable Y equaled 1 when an SMI diagnosis was positive based on the clinical interview; otherwise, Y was 0. Letting $\mathbf{X}$ be a vector of the characteristics attached to a NSDUH respondent and letting the probability this respondent had SMI be $\pi = \Pr(Y = 1 \mid \mathbf{X})$, the 2012 SMI prediction model was as follows:

[59] Both the lifetime and past year measures of MDE in adults (see Section 3.4.12) were used in poststratification.

[60] Past year MDE was estimated based on responses to the SCID from the MHSS respondents and on responses from all adults to the main survey (see Section 3.4.12). These two measures were created independently. The reference here is to the SCID measure from the MHSS.

$$\text{Logit}(\hat{\pi})=\text{Log}(\hat{\pi}/1-\hat{\pi}) = B_0 + B_1 X_k + B_2 X_w + B_3 X_s + B_4 X_m + B_5 X_a$$

or (1)

$$\hat{\pi} = \frac{1}{1+\exp[-(-B_0 + B_1 X_k + B_2 X_w + B_3 X_s + B_4 X_m + B_5 X_a)]}$$

where $\hat{\pi}$ refers to the estimate of the SMI response probability π.

The beta values for equation (1) are displayed in Table 3.9. The following covariates in equation (1) came from the main NSDUH interview data and were statistically imputed:

- X_k = *Alternative Past Year K6 Score:* Past year maximum K6 score of less than 8 recoded as 0; past year maximum K6 score of 8 to 24 recoded as 1 to 17.
- X_w = *Alternative WHODAS Score:* WHODAS item score of less than 2 recoded as 0; WHODAS item score of 2 to 3 recoded as 1, then summed for a score ranging from 0 to 8.
- X_s = *Serious Thoughts of Suicide in the Past Year:* Coded as 1 if "yes"; coded as 0 otherwise.
- X_m = *Past Year MDE:* Coded as 1 if the criteria for past year MDE were met (see Section 3.4.12);[61] coded as 0 otherwise.
- X_a = *Adjusted Age:* Coded as age minus 18 if aged 18 to 30; coded as 12 otherwise.

Table 3.9 Final Serious Mental Illness Prediction Models in the 2008 to 2012 Mental Health Surveillance Studies

Variable	Beta	Beta SE	T Statistic	*P* Value	*DF*	Wald *P* Value[1]
WHODAS Sample (2008A-2012)						
Intercept	-5.9726640	0.3201	-18.6586	0.0000	-	-
Alt PY K6	0.0873416	0.0248	3.5247	0.0009	1	0.0009
Alt WHODAS	0.3385193	0.0349	9.7034	0.0000	1	0.0000
PY Suicidal Thoughts	1.9552664	0.2164	9.0342	0.0000	1	0.0000
PY MDE	1.1267330	0.2196	5.1308	0.0000	1	0.0000
Age1830	0.1059137	0.0244	4.3380	0.0001	1	0.0001

Age1830 = recoded age variable; Alt = alternative; *DF* = degrees of freedom; K6 = Kessler-6, a six-item psychological distress scale; MDE = major depressive episode; PY = past year; SE = standard error; WHODAS = eight-item World Health Organization Disability Assessment Schedule.
NOTE: Alt PY K6: past year K6 score of < 8 recoded as 0; past year K6 score of 8 to 24 recoded as 1 to 17.
NOTE: Alt WHODAS: WHODAS item score of < 2 recoded as 0; WHODAS item score of 2 to 3 recoded as 1, then summed for a score ranging from 0 to 8.
NOTE: PY suicidal thoughts: coded as 1 if respondent had serious thoughts of suicide in the past year; coded as 0 otherwise.
NOTE: PY MDE: coded as 1 if the criteria for past year MDE were met; coded as 0 otherwise.
NOTE: Age1830: coded as age minus 18 if aged 18 to 30; coded as 12 otherwise.
[1] The Wald *p* value is obtained from the overall model fitting.
Source: SAMHSA, Center for Behavioral Health Statistics and Quality, National Survey on Drug Use and Health, 2008-2012.

[61] In this situation, the past year MDE measure is from the main NSDUH interview (i.e., not from the SCID).

A cut point probability π_0 was determined so that if $\hat{\pi} \geq \pi_0$ for a particular respondent, then the respondent was predicted to be SMI positive; otherwise, the respondent was predicted to be SMI negative. The cut point (0.260573529) was chosen so that the weighted numbers of false positives and false negatives in the MHSS dataset were as close to equal as possible. The predicted SMI status for all adult NSDUH respondents was used to compute prevalence estimates of SMI. The model is directly related to the probability of being diagnosed with SMI in the MHSS.

A second cut point probability (0.0192519810) was determined so that any respondent with an SMI probability greater than or equal to the cut point was predicted to be positive for AMI, and the remainder were predicted to be negative for AMI. The second cut point was chosen so that the weighted numbers of AMI false positives and false negatives were as close to equal as possible. The model is not directly related to the probability of being diagnosed with AMI in the MHSS.

The 2024 national reports and tables show estimates for AMI and SMI. The 2024 Detailed Tables (CBHSQ, 2025h) also show estimates for "AMI excluding SMI." Adults with AMI excluding SMI currently or at any time in the past year have had a diagnosable mental, behavioral, or emotional disorder resulting in less than substantial impairment in carrying out major life activities.

3.4.9.9 Standard Errors for Mental Illness Estimates

For the national reports and tables, SEs for mental illness estimates (SMI, AMI, and AMI excluding SMI[62]) were computed using the NSDUH dichotomous variable values without taking into account any variance introduced through using a model based on the clinical subsample data. This ignores the added error resulting from fitting the 2012 SMI model, which can be very large (see the *2012 National Survey on Drug Use and Health: Methodological Resource Book [Section 16a, 2012 Mental Health Surveillance Study: Design and Estimation Report];* CBHSQ, 2014a). These *conditional* SEs (conditional on the model predictions being correct) are useful when making comparisons across years and across subpopulations (except those involved in modeling) within years because the errors due to model fitting are nearly the same across the estimates being compared.

3.4.9.10 Limitations in Using the Mental Illness Variables in Analyses

There are many advantages to using the cut point methodology described in this section to predict the SMI and AMI status for every adult responding to the NSDUH main survey interview. For some analyses, however, these predicted values should *not* be used. In particular, these predicted values should not be employed in analyses using the mental illness variables in conjunction with variables used or closely related to variables used in the prediction model. Because variables used in the prediction models would be expected to be correlated with the SMI or AMI probabilities, SMI or AMI should not be estimated for the following groups of adults:

[62] See the 2024 Detailed Tables (CBHSQ, 2025h).

among people with past year or lifetime MDE; among people with past year suicidal thoughts, suicide plans, or suicide attempts; or among people with particular K6 or WHODAS scores. For details, see the *Estimating Mental Illness among Adults in the United States: Revisions to the 2008 Estimation Procedures* report (CBHSQ, 2015).

3.4.10 Mental Health Treatment

The mental health treatment questions underwent considerable revisions for the 2022 NSDUH. Revisions for 2022 were intended to reflect contemporary changes in the delivery of mental health treatment services. The changes also made the content more similar between the alcohol and drug treatment and the mental health services utilization sections of the questionnaire. The following is a summary of key changes to these questions:

- Adolescents aged 12 to 17 and adults aged 18 or older received the same questions about mental health treatment. Before 2022, adolescents and adults received different sets of questions, such that measures differed for adolescents and adults.
- The instruction was dropped for respondents not to report mental health treatment associated with their use of alcohol or drugs.
- Terminology was updated, and questions were included about newer treatment options.
- Additional inpatient and outpatient locations were provided for adult respondents to report where they received treatment in the past 12 months. As noted previously, adolescents were asked about the receipt of treatment in these same inpatient and outpatient locations.
- The question about the use of medication that was prescribed to help with people's mental health applied to all respondents, regardless of age. Before 2022, only adults were asked whether they took prescribed medication to help with their mental health.
- The question about treatment received in a prison, jail, or juvenile detention center applied to all respondents, regardless of age. Before 2022, only adolescents were asked whether they received services in a prison, jail, or juvenile detention center.
- The question about the receipt of treatment over the phone or through video (i.e., telehealth treatment) was revised to emphasize that the treatment was from a therapist or other healthcare professional.

Starting in June 2024, additional changes were included in the questionnaire to improve respondent understanding of whether they received treatment for their mental health, emotions, or behavior at specific inpatient or outpatient locations (see Section 2.2.2). The following changes were made:

- Introductory text for questions about inpatient and outpatient treatment was revised to mention "any of the locations below" for response choices for specific locations.
- For each inpatient and outpatient location that was presented to respondents, text was added to indicate that the treatment was "for mental health, emotions, or behavior." For example, the response option for receiving treatment at a general medical clinic or doctor's office as an outpatient was revised to mention receiving treatment at a general

medical clinic or doctor's office as an outpatient "for mental health, emotions, or behavior."
- In the grid format for respondents to report the locations where they received inpatient treatment, the heading "Stayed overnight or longer for treatment in…" was replaced with "LOCATION."
- In the grid format for respondents to report the locations where they received outpatient treatment, the heading "Received outpatient treatment at…" was replaced with "LOCATION."

Because of the changes in 2022, the definition for the receipt of mental health treatment changed. Therefore, estimates from 2022 based on these outcomes should not be compared with estimates from 2021.

Analysis of the data from June to December 2024 indicated that the changes to the mental health treatment questions in June 2024 affected estimates of the receipt of any mental health treatment, inpatient mental health treatment, and outpatient mental health treatment in the past year. As a consequence of these changes, mental health treatment estimates overall and for inpatient or outpatient locations from 2024 are not comparable with those from 2022 and 2023. Estimates from 2024 for mental health treatment via telehealth or in a prison, jail, or juvenile detention center and for the receipt of prescription medication to treat mental health, emotions, or behavior (see Section 3.4.10.1) are still comparable with estimates from 2022 and 2023. Estimates for measures of the receipt of specific other services described in Section 3.4.10.2 also remain comparable with estimates from 2022 and 2023.

In addition, questions for whether respondents perceived an unmet need for mental health treatment and the reasons for respondents not receiving treatment were revised for the 2022 NSDUH. Due to the additional changes for 2024 that were described previously, estimates from 2024 for the perceived unmet need for mental health treatment and reasons for not receiving treatment also are not considered comparable with corresponding estimates from prior years. Specific changes are discussed in Section 3.4.10.3 in the context of measurement of these outcomes.

3.4.10.1 Locations and Types of Services for Mental Health Treatment

Receipt of mental health treatment includes the receipt of treatment in the past year to help people with their mental health, emotions, or behavior that was received in an inpatient location, in an outpatient location, via telehealth, or in a prison, jail, or juvenile detention center. The definition also includes the receipt of prescription medication to help with mental health, emotions, or behavior.

Inpatient treatment was defined as treatment in locations where people stayed overnight or longer and included the following locations:

- a hospital,
- a residential mental health treatment center,

- a residential drug or alcohol rehab or treatment center, or
- some other place the respondent stayed overnight or longer.

Outpatient treatment was defined as treatment people received in locations that did not require them to stay overnight or longer. These measures included the following locations:

- a mental health treatment center as an outpatient;
- a drug or alcohol treatment or rehab center as an outpatient;
- the office of a therapist, psychologist, psychiatrist, or mental health professional;
- a general medical clinic or doctor's office;
- a hospital as an outpatient;
- a school health or counseling center; or
- some other place as an outpatient.

As mentioned previously, however, estimates from 2024 based on variables for any receipt of mental health treatment in the past year and receipt of treatment in inpatient and outpatient locations are not comparable with estimates from 2022 and 2023. Estimates from 2024 for mental health treatment via telehealth or in a prison, jail, or juvenile detention center and for the receipt of prescription medication to treat mental health, emotions, or behavior remain comparable with estimates from 2022 and 2023.

Variables for the receipt of mental health treatment in the past year in inpatient locations, in outpatient locations, via telehealth, or in a prison, jail, or juvenile detention center were statistically imputed. The variable for the receipt of prescription medication to treat mental health, emotions, or behavior in the past year was also statistically imputed. Consequently, variables since 2022 for the receipt of mental health treatment in the past year do not have missing data.

3.4.10.2 Receipt of Other Services to Help with Mental Health

The 2024 NSDUH also collected information on the receipt of other services in the past year to help people with their mental health, emotions, or behavior. These other services include the following:

- services from a support group,
- services from a peer support specialist or recovery coach, or
- services in an emergency room or emergency department.

NSDUH did not classify these other services as "mental health treatment." For example, the subject matter experts who worked with SAMHSA to develop the revised mental health treatment questions for 2022 thought that services in an emergency room or emergency department were primarily intended to treat acute medical conditions, such as from a suicide attempt, rather than to engage patients in a longer term process of change to promote recovery from mental disorders. Experts also did not consider services from a support group or recovery coach to be mental health treatment because of issues related to licensure or reimbursement for services.

However, these other services were included in a separate aggregate measure presented in the 2024 Detailed Tables (CBHSQ, 2025h) that was created to cover the receipt of mental health treatment or other services. Because of the changes to the inpatient and outpatient questions in June 2024, the estimates from 2024 for the aggregate measure of the receipt of mental health treatment or other services are not comparable with those from prior years. However, estimates from 2024 for specific services from a support group through services in an emergency room mentioned previously remain comparable with estimates from 2022 and 2023. Variables for the receipt of other services were statistically imputed, so these variables do not have missing data.

3.4.10.3 Perceived Unmet Need for Mental Health Treatment and Reasons for Not Receiving Treatment

Questions about whether respondents perceived an unmet need for mental health treatment and the reasons for not receiving treatment were also revised for the 2022 NSDUH. The questions did not change for 2024.

- These questions were asked of respondents of all ages, including those aged 12 to 17, not just adults aged 18 or older.
- These questions were asked only if respondents did not report mental health treatment (i.e., inpatient or outpatient treatment, prescription medication, telehealth treatment, or treatment in a prison, jail, or juvenile detention center).
- Respondents were asked whether they sought treatment.
- Respondents who did not report seeking treatment were asked whether they thought they should get treatment.

NSDUH respondents were classified as having a perceived unmet need for mental health treatment if they did not receive mental health treatment in the past year and reported either of the following:

- They did not receive other services (i.e., from a support group, from a peer support specialist or recovery coach, or in an emergency room), but they sought or thought they should get treatment in the past 12 months for their mental health, emotions, or behavior.
- They received other services, described in the preceding bullet (see Section 3.4.10.2), but they sought or thought they should get additional professional counseling, medication, or other treatment in the past 12 months for their mental health, emotions, or behavior.

Questions about reasons for people not receiving mental health treatment (i.e., barriers to treatment) were asked of respondents who reported a perceived unmet need for mental health treatment. The questionnaire changes for 2022 also included an expanded list of possible barriers to treatment for adults aged 18 or older. Beginning in 2022, adolescents aged 12 to 17 were asked about these same barriers. For each reason for not receiving mental health treatment, respondents were asked whether that reason was "one of the reasons" or "not one of the reasons" they did not seek or get professional counseling, medication, or other treatment for their mental health, emotions, or behavior.

Although questions for the perceived unmet need for mental health treatment and reasons for people not receiving treatment did not change for 2024, the questions for inpatient and outpatient treatment changed in June 2024, as described previously. These changes affected who was eligible to receive questions about unmet need. Consequently, estimates from 2024 for the perceived unmet need for mental health treatment and reasons for not receiving treatment are considered not to be comparable with estimates from 2022 and 2023.

Variables for the perceived unmet need for mental health treatment and reasons for not receiving mental health treatment were not statistically imputed and had missing data. Respondents with missing data were excluded from the analyses to produce published estimates for these measures. See Section 3.3.2 for a discussion of the potential bias in estimates because of missing data.

3.4.11 Symptoms of Generalized Anxiety Disorder

The seven-item generalized anxiety disorder (GAD-7) screener was added to the 2024 NSDUH questionnaire in the youth experiences section for adolescents aged 12 to 17 and to the mental health section for adults aged 18 or older. The GAD-7 is a validated self-report measure to screen for GAD and assess symptoms of GAD in the past 2 weeks in adolescents and adults (Mossman et al., 2017; Spitzer et al., 2006). Symptoms of GAD include feeling nervous or on edge, excessively worrying about different things, having difficulty controlling thoughts of worry, having trouble relaxing, being restless, feeling irritable, and feeling that something awful might happen.

The GAD-7 questions had the same response categories for adolescents and adults:

1 Not at all
2 Several days
3 More than half the days
4 Nearly every day

Don't know/Refused

The GAD-7 questions are as follows:[63]

YGAD1 Over the last **2 weeks**, how often have you been bothered by **feeling nervous, anxious, or on edge**?

YGAD2 Over the last **2 weeks**, how often have you been bothered by **not being able to stop or control worrying**?

YGAD3 Over the last **2 weeks**, how often have you been bothered by **worrying too much about different things**?

[63] Names of questions are shown for adolescents, but wordings of the GAD-7 questions were the same for adolescents and adults.

YGAD4	Over the last **2 weeks**, how often have you been bothered by **having trouble relaxing**?
YGAD5	Over the last **2 weeks**, how often have you been bothered by **being so restless that it is hard to sit still**?
YGAD6	Over the last **2 weeks**, how often have you been bothered by **becoming easily annoyed or irritable**?
YGAD7	Over the last **2 weeks**, how often have you been bothered by **feeling afraid as if something awful might happen**?

In the 2024 NSDUH, all respondents with item nonresponse for these anxiety symptom questions had their responses statistically imputed (see Section 2.3.3). Thus, there were no missing values for the variables associated with the GAD-7 questions that were used in analyses. Imputation also mitigated potential effects of nonresponse from respondents who broke off the interview before or during the sections of the interview where the GAD-7 questions appeared.

To create respondents' GAD-7 scores, the imputation-revised values for the seven individual items were recoded from 0 to 3 so that "nearly every day" was coded as 3, "more than half the days" as 2, "several days" as 1, and "not at all" as 0. Summing across the transformed values for responses in these seven items resulted in a score with a range from 0 to 21.

GAD-7 scores were classified as follows, with higher scores indicating greater symptom severity (Spitzer et al., 2006; Terlizzi & Villaroel, 2020):

- 0 to 4 = no or minimal symptoms of GAD,
- 5 to 9 = mild symptoms,
- 10 to 14 = moderate symptoms, and
- 15 to 21 = severe symptoms.

3.4.12 Major Depressive Episode (Depression)

Two sections related to MDE were included in the 2024 questionnaire: an adult depression section and an adolescent depression section. These sections were originally derived from DSM-IV criteria for MDE and remained applicable to the more recent DSM-5 criteria. Consistent with the DSM-5 criteria, NSDUH does not exclude MDEs occurring exclusively in the context of bereavement. In addition, no exclusions were made for MDEs caused by medication, alcohol, illicit drugs, or any medical illness.

Questions on depression permit estimates to be calculated for the occurrence of an MDE in the population and receipt of treatment for an MDE. Separate sections were administered to adults aged 18 or older and adolescents aged 12 to 17. The adult questions were adapted from the depression section of the National Comorbidity Survey Replication (NCS-R), and the questions for adolescents were adapted from the depression section of the National Comorbidity Survey Replication Adolescent Supplement (NCS-A). To make the sections developmentally appropriate for adolescents, there are minor wording differences in a few questions between the adult and adolescent sections. Revisions to the questions in both sections were made

primarily to reduce their length and to modify the interviewer-administered NCS questions for self-administration in NSDUH.

According to DSM-5, people are classified as having had an MDE[64] in their lifetime if they had at least five or more of nine symptoms nearly every day (except where noted) in the same 2-week period, where at least one of the symptoms is a depressed mood or loss of interest or pleasure in daily activities. These symptoms are as follows:

1. depressed mood most of the day;
2. markedly diminished interest or pleasure in all or almost all activities most of the day;
3. significant weight loss when not sick or dieting, or weight gain when not pregnant or growing,[65] or decrease or increase in appetite;
4. insomnia or hypersomnia;
5. psychomotor agitation or retardation at a level observable by others;
6. fatigue or loss of energy;
7. feelings of worthlessness or excessive or inappropriate guilt;
8. diminished ability to think or concentrate or indecisiveness; and
9. recurrent thoughts of death or suicidality (i.e., recurrent suicidal ideation without a specific plan, making a specific plan, or making an attempt).

Unlike the other symptoms, recurrent thoughts of death or suicidality did not need to have occurred nearly every day.

Respondents who had an MDE in their lifetime are asked whether they had a period of depression during the past 12 months lasting 2 weeks or longer while also having some of the other symptoms mentioned previously for the lifetime period. Respondents reporting experiences consistent with their having an MDE in the past year are asked questions from the SDS to measure the level of functional impairment in major life activities that were caused by their MDE in the past 12 months (Leon et al., 1997).

NSDUH measures the nine symptoms associated with MDE as defined in DSM-5 with the following questions. The questions shown are taken from the adult depression section of the 2024 NSDUH questionnaire. A few of the questions in the adolescent section were modified slightly to use wording more appropriate for adolescents aged 12 to 17.

[64] "An MDE" refers to the occurrence of at least one MDE, rather than only one MDE. Similarly, reference to "the MDE" in a given period (e.g., the past 12 months) does not mean an individual had only one MDE in that period.

[65] Adolescent respondents aged 12 to 17 who reported that they gained weight were asked whether they gained weight because they were growing. Adult respondents aged 18 or older who reported that they gained weight were not asked this follow-up question.

1. **Depressed mood most of the day**

   The following questions refer to the worst or most recent period of time when the respondent experienced any or all of the following: sadness, discouragement, or lack of interest in most things.

   During that [worst/most recent] period of time …

   a. … did you feel sad, empty, or depressed **most of the day nearly every day**?
   b. … did you feel discouraged about how things were going in your life **most of the day nearly every day**?

2. **Markedly diminished interest or pleasure in all or almost all activities most of the day**

   a. … did you lose interest in almost all things like work and hobbies and things you like to do for fun?
   b. … did you lose the ability to take pleasure in having good things happen to you, like winning something or being praised or complimented?

3. **Weight**

   In answering the next questions, think about the [worst/most recent] period of time.

   a. Did you have a much smaller appetite than usual nearly every day during that time?
   b. Did you have a much **larger** appetite than usual nearly every day?
   c. Did you gain weight without trying to during that [worst/most recent] period of time?
      i. … because you were growing?
      ii. … because you were pregnant?
      iii. How many pounds did you gain?
   d. Did you **lose** weight without trying to?
      i. … because you were sick or on a diet?
      ii. How many pounds did you lose?

4. **Insomnia or hypersomnia**

   a. Did you have a lot more trouble than usual falling asleep, staying asleep, or waking too early nearly every night during that [worst/most recent] period of time?
   b. During that [worst/most recent] period of time, did you sleep a lot more than usual nearly every night?

5. **Psychomotor agitation or retardation**
    a. Did you talk or move more slowly than is normal for you nearly every day?
        i. Did anyone else notice that you were talking or moving slowly?
    b. Were you so restless or jittery nearly every day that you paced up and down or couldn't sit still?
        i. Did anyone else notice that you were restless?
6. **Fatigue or loss of energy**
    a. During that [worst/most recent] period of time, did you feel tired or low in energy nearly every day, even when you had not been working very hard?
7. **Feelings of worthlessness**
    a. Did you feel that you were not as good as other people nearly every day?
    b. Did you feel totally worthless nearly every day?
8. **Diminished ability to think or concentrate or indecisiveness**
    a. During that [worst/most recent] time period, did your thoughts come much more slowly than usual or seem confused nearly every day?
    b. Did you have a lot more trouble concentrating than usual nearly every day?
    c. Were you unable to make decisions about things you ordinarily have no trouble deciding about?
9. **Recurrent thoughts of death or recurrent suicidal ideation**
    a. Did you often think about death, either your own, someone else's, or death in general?
    b. During that period, did you ever think it would be better if you were dead?
    c. Did you think about committing suicide?

Beginning in 2021, missing data were statistically imputed in variables for whether adult respondents had an MDE in their lifetime and whether adult respondents had an MDE in the past 12 months. MDE variables were not statistically imputed for adolescents aged 12 to 17. Respondents aged 12 to 17 who had missing data for whether they had an MDE in the past 12 months were excluded from the analyses to produce published estimates for the 2024 NSDUH. See Section 3.3.2 for a discussion of the potential bias in estimates because of missing data.

NSDUH also collects data on functional impairment using the SDS, which is a measure of impairment due to mental health conditions in four major life activities or role domains. These four domains are defined separately for adults aged 18 or older and adolescents aged 12 to 17 to reflect the different roles associated with the two age groups. See Section 3.4.12.1 for details about the questions for adults and Section 3.4.12.2 for details about the questions for adolescents. There is one question for each of the four domains, and each question uses an 11-point scale ranging from 0 (no interference for adults and no problems for adolescents) to

10 (very severe interference for adults and very severe problems for adolescents). The impairment score is defined as the single highest severity level of role impairment across the four SDS role domains. Ratings greater than or equal to 7 on the scale were considered severe impairment.

In addition to past year MDE, NSDUH shows estimates for past year MDE with severe impairment. Estimates for severe impairment are calculated separately for adolescents and adults because the four domains are slightly different for the two groups. Missing data for MDE with severe impairment were statistically imputed for adults but not for adolescents aged 12 to 17. Respondents aged 12 to 17 who had missing data for impairment were excluded from the analyses to produce published estimates for MDE with severe impairment in the 2024 NSDUH. See Section 3.3.2 for a discussion of the potential bias in estimates because of missing data.

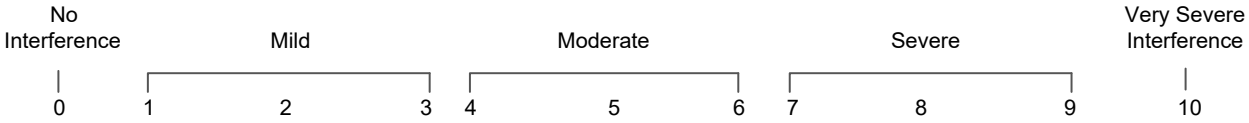
Because variables for lifetime and past year MDE among adults and MDE with severe impairment in the past year were statistically imputed, the main analysis weight described in Section 2.3.4 was used to produce estimates in national reports and tables for the 2024 NSDUH. For adolescents aged 12 to 17, the main person-level analysis weight was used to produce estimates of MDE and MDE with severe impairment in the past year because the number of break-offs among adolescents was minimal.

3.4.12.1 Adult Depression Section: Functional Impairment

The questions pertaining to the four domains of functional impairment for adults aged 18 or older are listed below. The scale is shown below for the first domain but applies to all four domains.

ASDSHOME Think about the time in the past 12 months when these problems with your mood were **most severe.**

Using the 0 to 10 scale shown below, where 0 means **no** interference and 10 means very **severe** interference, select the number that describes how much these problems interfered with your **ability to do** each of the following activities during that period. You can use any number between 0 and 10 to answer.



How much did your [depression symptoms] interfere with your **ability to do home management tasks**, like cleaning, shopping, and working around the house, apartment, or yard?

ASDSWORK During that time in the past 12 months when your [depression symptoms] were most severe, how much did this interfere with your **ability to work**?

ASDSREL How much did your [depression symptoms] interfere with your **ability to form and maintain close relationships** with other people during that period of time?

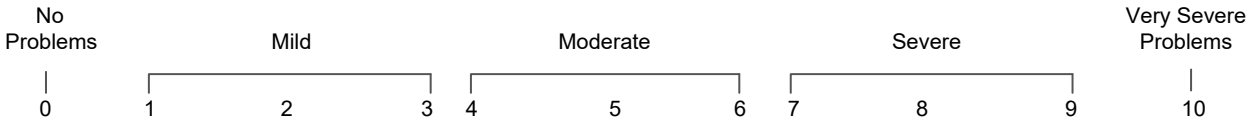
ASDSSOC How much did [depression symptoms] interfere with your **ability to have a social life** during that period of time?

3.4.12.2 Adolescent Depression Section: Functional Impairment

The questions pertaining to the four domains of functional impairment for adolescents aged 12 to 17 are listed below. The scale is also shown below for the first domain but applies to all four domains.

YSDSHOME Think about the time in the past 12 months when these problems with your mood were the **worst.**

Using the 0 to 10 scale shown below, where 0 means **no** problems and 10 means very **severe** problems, select the number that describes how much your [depression symptoms] caused problems with your **ability to do** each of the following activities during that time. You can use any number between 0 and 10 to answer.



How much did your [depression symptoms] cause problems with your **chores at home**?

YSDSWORK During that time in the past 12 months when your [depression symptoms] were worst, how much did this cause problems with your **ability to do well at school or work**?

YSDSREL How much did your [depression symptoms] cause problems with your **ability to get along with your family** during that time?

YSDSSOC How much did your [depression symptoms] cause problems with your **ability to have a social life** during that time?

3.4.13 Suicidal Thoughts and Behavior

The 2024 NSDUH included separate sets of questions asking adults aged 18 or older and adolescents aged 12 to 17 whether they had serious thoughts of suicide, made a suicide plan, or attempted suicide in the past 12 months. All adult and adolescent respondents were asked whether they made a suicide plan or attempted suicide regardless of whether they reported that they had serious thoughts of suicide in the past 12 months. Respondents who reported

that they made a suicide attempt were asked whether they received medical attention or stayed overnight in the hospital because of their suicide attempt.

3.4.13.1 Suicidal Thoughts and Behavior among Adults

The mental health section of the NSDUH questionnaire included questions about suicidal thoughts and behavior among adults. Beginning in 2021, the variables for suicidal thoughts and behavior among adults were statistically imputed, so these variables had no missing data for 2024. Estimates for suicidal thoughts and behavior among adults in the past year were created for 2024 using the standard analysis weight described in Section 2.3.4.

3.4.13.2 Suicidal Thoughts and Behavior among Adolescents

Questions about adolescents' suicidal thoughts and behaviors in the past 12 months have been included in the youth experiences section of the questionnaire since the 2022 NSDUH. Unlike the questions for adults, the questions about suicidal thoughts and behavior among adolescents included response choices for "I'm not sure" and "I don't want to answer," in addition to standard response choices for "yes" and "no." Adolescent respondents also could choose these response choices for "I'm not sure" and "I don't want to answer" instead of using function keys (as is the practice elsewhere in the interview) for answers of "don't know" or "refused," respectively.

Estimates for suicidal thoughts and behavior among adolescents in national reports and tables for 2024 included estimates for "I'm not sure," and "I don't want to answer," in addition to estimates for "yes" and "no." Responses of "don't know" were grouped with "I'm not sure," and refusals were grouped with "I don't want to answer." Thus, measures for suicidal thoughts and behavior among adolescents were not statistically imputed for 2024, but adolescent respondents who did not know or refused to report whether they had suicidal thoughts or behavior in the past year were not excluded from analyses. However, adolescents who broke off the interview before reaching these questions were excluded from the analyses.

The 2024 estimates for suicidal thoughts and behavior among adolescents were created using the main analysis weights, with no adjustment because of break-offs. As discussed in Section 2.3.4, review of the 2024 NSDUH data indicated that a small number of adolescents aged 12 to 17 broke off the interview before they reached the youth experiences section where the questions were located for suicidal thoughts and behavior among adolescents.

3.4.14 Modes of Marijuana Use

The marijuana section of the 2024 NSDUH questionnaire included questions to assess the ways that people used marijuana in the past year or past month. Respondents who reported using marijuana in these periods were asked to report whether they used marijuana in any of the following ways:

- smoking;
- vaping;
- dabbing waxes, shatter, or concentrates;
- eating or drinking;

- putting drops, strips, lozenges, or sprays in their mouth or under their tongue;
- applying lotion, cream, or patches to their skin;
- taking pills; or
- using it in some other way.

Respondents could report that they used marijuana in more than one way in the past year or past month. For example, respondents could report that they smoked marijuana and vaped it in the past year.

In 2024, the question on lifetime vaping of marijuana was moved from the emerging issues section to the earlier marijuana section of the interview. Respondents were asked whether they ever vaped marijuana if they used marijuana in their lifetime and they did not report that marijuana vaping was one of the ways they used marijuana in the past year or past month. The focus of the remainder of this section is on the modes of marijuana use in the past year or past month because lifetime estimates for marijuana vaping were not presented in reports and tables for the 2024 NSDUH.

3.4.14.1 Measures for Modes of Marijuana Use

The 2024 Detailed Tables (CBHSQ, 2025h) and the 2024 Key Substance Use and Mental Health Indicators report (CBHSQ, 2025g) presented past year estimates for all eight specific modes of marijuana use that were described at the beginning of this section, including use in some other way. The 2024 Detailed Tables also present an aggregate "other" category that includes modes for applying lotion, cream, or patches to the skin; putting drops, strips, lozenges, or sprays in the mouth or under the tongue; taking pills; and using marijuana in some other way.

Since the 2023 NSDUH, all variables for modes of marijuana use have been statistically imputed and do not have missing data. See the 2023 Editing and Imputation Report (CBHSQ, 2025c) for further details on the imputation processes that were implemented beginning with the 2023 survey; these procedures did not change for 2024. These revised methods were also applied to produce imputed variables for all modes of marijuana use for 2022, including revised marijuana vaping variables.

3.4.14.2 Data Inconsistencies

The following patterns of inconsistent data could occur for modes of marijuana use:

- Respondents did not report a particular mode of use (e.g., vaping) in the past year or past month but reported "some other way" and specified that mode for using marijuana in that period.
- Respondents did not report a particular mode of use in the past year but reported it as a mode of use for the past month.
- Respondents had missing data for the mode of marijuana use in the past month (i.e., they did not know or refused to report how they used it). However, they reported using marijuana in only one way in the past year.

These inconsistencies were resolved through logical editing procedures. See the 2023 Editing and Imputation Report (CBHSQ, 2025c) for further details; these editing procedures did not change for the 2024 NSDUH.

As noted previously, the question on lifetime vaping of marijuana was moved from the emerging issues section to the marijuana section of the interview for 2024. Because respondents were asked about lifetime vaping of marijuana only if they did not report that they vaped marijuana in the past year or past month, inconsistencies did not exist between lifetime vaping of marijuana and vaping in the past year or past month.

Before the 2024 NSDUH, data for marijuana vaping from the emerging issues section were not used in creating the published 2023 estimates for marijuana use in the lifetime, past year, or past month periods or for marijuana vaping in the past year or past month. Consequently, although inconsistencies could have remained in the 2023 data for marijuana use and marijuana vaping between the marijuana and emerging issues sections, they were not reflected in published estimates for 2023.

3.4.15 Recovery

Adult respondents aged 18 or older in 2024 were asked up to four questions about their recovery from substance use problems or mental health issues. These questions were in the emerging issues section of the questionnaire. Adult respondents first were asked whether they thought that they ever have had a problem with their own drug or alcohol use. If respondents answered "yes," they were asked whether they considered themselves to be in recovery or to have recovered from their own problem with drug or alcohol use. These first two questions on recovery from a substance use problem were followed by a set of two similar questions asking respondents whether they have ever had a problem with their own mental health and, if so, whether they considered themselves to be in recovery or to have recovered from their own mental health problem.

Adult recovery estimates were included in national reports and tables for the 2024 NSDUH among adults who reported ever having a substance use problem or mental health issue. The 2024 Detailed Tables (CBHSQ, 2025h) also presented recovery estimates among all adults, regardless of whether they perceived themselves ever to have had a problem. To generate estimates among the total adult population, adults who reported not having a problem were classified as not being in recovery or having recovered from a problem. Respondents were excluded from the respective analyses if they had unknown information for whether they ever had a substance use problem or mental health issues. Respondents were also excluded from analyses if they had unknown information for whether they perceived themselves to be in recovery or to have recovered from their respective problem (e.g., if respondents reported ever having had a substance use problem but did not know or refused to report whether they perceived themselves to be in recovery or to have recovered from their substance use problem). For a discussion on how procedures for handling missing data may bias estimates, see Section 3.3.2.

Consistent with the discussion in Section 3.3.3, data users are reminded that these estimates are based on self-reports. Specifically, these estimates reflect adults' *perceptions* of whether they had problems with their substance use or mental health issues, but not necessarily the clinical assessments by medical or mental health professionals or the internal consistency of respondents' answers. It is important to note that the terms "problem" and "recovery" were not defined for respondents. Therefore, how respondents subjectively defined these terms may have varied. In addition, data on adults' perceptions of whether they had a problem with their substance use or mental health and whether they perceived themselves to have recovered or to be in recovery from these problems were not edited relative to data in other sections of the interview for substance use, SUDs, substance use treatment, mental health conditions, or the receipt of mental health treatment (see Section 2.3.2). Therefore, perceptions may seem to be inconsistent with substance use and mental health data from earlier sections of the interview.

The emerging issues section that contained these questions on recovery occurred in the 2024 NSDUH interview after the mental health and adult depression sections. Also, the recovery variables were not statistically imputed for 2024. Therefore, 2024 estimates for recovery were created using the break-off analysis weight described in Section 2.3.4.

3.4.16 Definition of County Type

County type is based on the Rural-Urban Continuum Codes[66] developed by the U.S. Department of Agriculture. The county type measure used in 2024 was based on the 2013 Rural-Urban Continuum Codes. Because counties are defined for all NSDUH respondents, the county type measures did not have missing data.

To create the 2013 Rural-Urban Continuum Codes, all U.S. counties and county equivalents were first grouped according to their official metropolitan-nonmetropolitan status (i.e., statistical area definitions), as determined by the OMB in February 2013. This grouping distinguished metropolitan counties by the population size of their metropolitan area and nonmetropolitan counties by their degree of urbanization and adjacency to a metropolitan area. The OMB determined current metropolitan status by applying population and worker commuting criteria to the results of the 2010 census and the 2006-2010 ACS. No major changes were made in either the metropolitan-nonmetropolitan or urban-rural criteria between 2000 and 2010. However, the decennial census long form was eliminated in 2010, and the OMB used 5-year average commuting flow data from the 2006-2010 ACS rather than a point-in-time estimate to delineate metropolitan and micropolitan areas.

Nonmetropolitan counties in the three urban-sized categories were further subdivided by whether the county was adjacent to one or more metropolitan areas. A nonmetropolitan county was defined as adjacent if it physically adjoined one or more metropolitan areas and had at least 2 percent of its employed labor force commuting to central metropolitan counties. Nonmetropolitan counties not meeting these criteria were classed as nonadjacent. The

[66] These codes are also known as the Beale Codes.

2006-2010 ACS commuting flow data were also used to compute adjacency for the 2013 Rural-Urban Continuum Codes.

Metropolitan and nonmetropolitan categories were subdivided into three metropolitan and six nonmetropolitan categories, resulting in a nine-part county codification.

- *Large metropolitan statistical areas (MSAs) (large metropolitan)* have a total population of 1 million or more.
- *Small MSAs (small metropolitan)* have a total population of fewer than 1 million. Small MSAs are further subdivided into two categories:
  - 250,000 to 1 million population; and
  - less than 250,000 population.
- *Nonmetropolitan counties* were classified according to the aggregate size of their urban population and whether they were adjacent to a metropolitan area. Nonmetropolitan areas include counties in micropolitan statistical areas and counties outside of both metropolitan and micropolitan statistical areas and are classified as follows:
  - urban population of 20,000 or more, adjacent to a metropolitan area ("urbanized adjacent");
  - urban population of 20,000 or more, not adjacent to a metropolitan area ("urbanized nonadjacent");
  - urban population of 2,500 to 19,999, adjacent to a metropolitan area ("less urbanized adjacent");
  - urban population of 2,500 to 19,999, not adjacent to a metropolitan area ("less urbanized nonadjacent");
  - completely rural or less than 2,500 urban population, adjacent to a metropolitan area ("completely rural adjacent"); and
  - completely rural or less than 2,500 urban population, not adjacent to a metropolitan area ("completely rural nonadjacent").

For NSDUH, nonmetropolitan counties were categorized as "urbanized," "less urbanized," and "completely rural." The terms "urbanized," "less urbanized," and "completely rural" for counties are not based on the relative proportion of the county population in urbanized areas but rather are based on the absolute size of the population in urbanized areas. For example, some counties classified as "less urbanized" had over 50 percent of the county population residing in urbanized areas, but this percentage represented fewer than 20,000 people in the county.

4. Special Topics for the Prescription Drug Questions

This chapter discusses issues related to the measurement in the National Survey on Drug Use and Health (NSDUH) of the use and misuse of pain relievers, tranquilizers, stimulants, and sedatives that require a prescription in the United States. The chapter focuses specifically on topics for prescription drug questions in NSDUH for the following reasons:

- As shown in the in-person and web versions of the 2024 NSDUH questionnaire, the structure of the prescription drug sections differed from the structure of sections for other substances in NSDUH.
- Because the prescription drugs included in the NSDUH questionnaire may be legally used in the United States under proper medical supervision, questions for prescription drugs differentiated between use of prescription drugs for any reason and prescription drug misuse.[67]
- For most psychotherapeutic drug categories in the questionnaire (e.g., pain relievers), medications could require a prescription, but other drugs (e.g., aspirin, Tylenol®) could be available in the United States as "over-the-counter" (OTC) drugs without a prescription. As noted in this chapter, NSDUH respondents were instructed not to include OTC drugs in their answers because the focus of the NSDUH questions was on prescription drugs.
- For prescription pain relievers, all questions in NSDUH about specific pain relievers are for prescription opioids. As noted in this chapter, NSDUH respondents also could report that they used or misused other prescription pain relievers that are not opioids.
- As noted in Section 3.4.7, substance use disorder (SUD) criteria in the *Diagnostic and Statistical Manual of Mental Disorders*, 5th edition (DSM-5; American Psychiatric Association, 2013), differ according to whether people misused prescription drugs or used but did not misuse prescription drugs in the past year.

Moreover, NSDUH data have consistently shown the misuse of prescription psychotherapeutic drugs (prescription pain relievers, tranquilizers, stimulants, and sedatives) to rank second only to marijuana in terms of the most commonly used or misused illicit drugs in the United States. Prescription pain relievers have consistently been the most commonly misused prescription drugs among those that are asked about in NSDUH.

Most of the prescription drugs covered in NSDUH have additional controls beyond requiring a healthcare professional to write a prescription (see Section 4.2.1). Although the U.S. Food and Drug Administration (FDA) also regulates OTC drugs to ensure that they meet quality, effectiveness, and safety standards, a key aim of regulations for OTC drugs is to allow consumers to use these medications safely without requiring the supervision of a healthcare provider (Clarke, 2016; FDA, 2022). See the definition for "Nonprescription Cough or Cold Medicine Use" in Appendix A of the *Results from the 2024 National Survey on Drug Use and*

[67] Misuse of prescription drugs was broadly defined in NSDUH as use in any way a doctor did not direct people to use prescription drugs.

Health: Detailed Tables (Center for Behavioral Health Statistics and Quality [CBHSQ], 2025h) for how NSDUH collected information on the misuse of OTC cough or cold medicines.

4.1 Measurement of the Use and Misuse of Prescription Psychotherapeutic Drugs

This section discusses routing logic for the NSDUH prescription drug questions and general issues for the creation of prescription drug estimates in NSDUH national reports and tables. Several of these procedures are described in further detail in the *2023 National Survey on Drug Use and Health (NSDUH) Methodological Resource Book, Section 10: Editing and Imputation Report* (CBHSQ, 2025c).

NSDUH national reports and tables do not refer to "prescription-type" psychotherapeutic drugs because questions about the use of methamphetamine and illegally made fentanyl (IMF) are asked separately from questions about the use and misuse of prescription psychotherapeutic drugs.[68] Instead, NSDUH refers to "prescription psychotherapeutic drugs" or "prescription drugs."

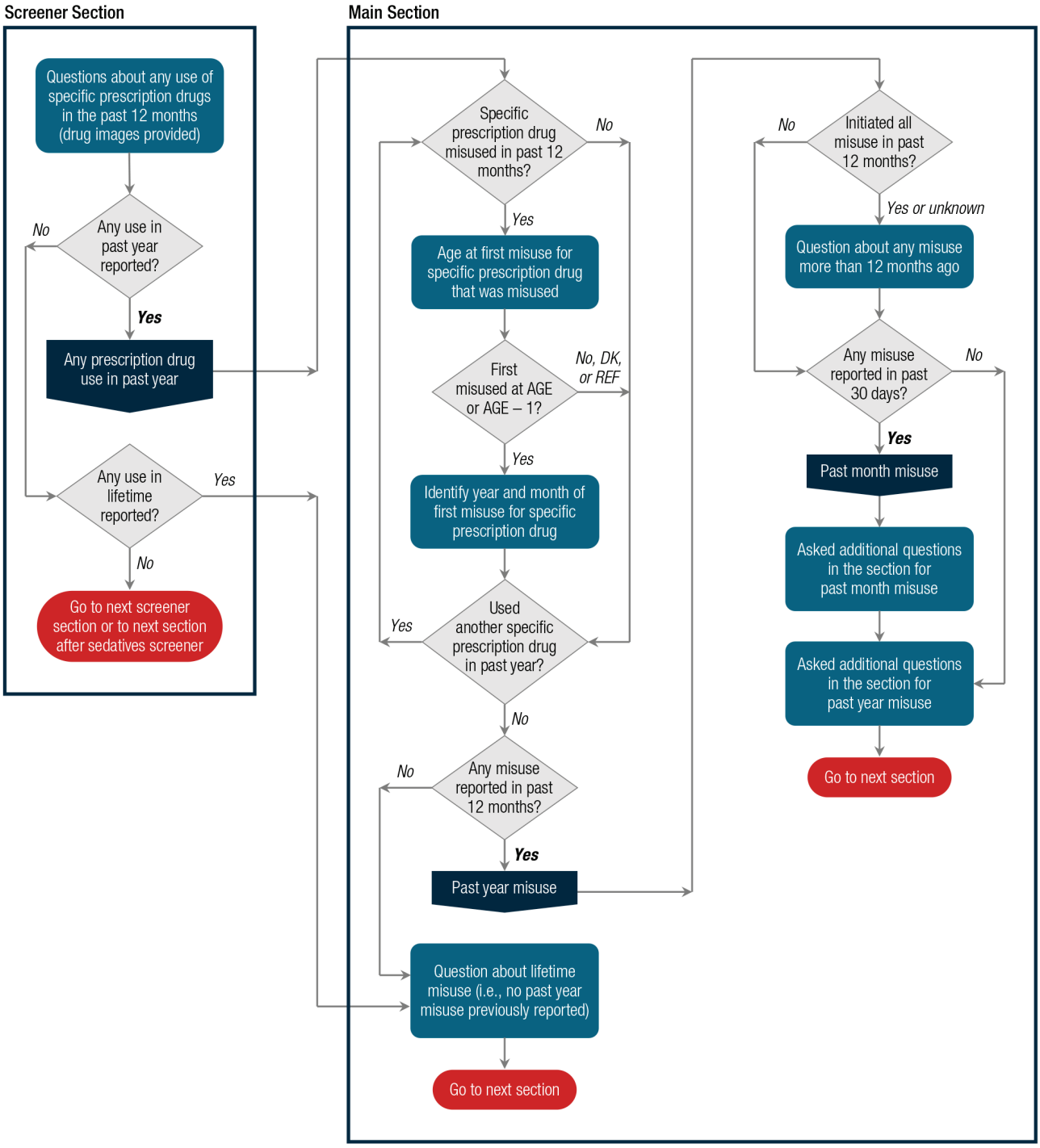
4.1.1 Routing Logic for Prescription Drug Questions

The 2024 NSDUH included questions about four categories of prescription psychotherapeutic drugs: pain relievers, tranquilizers, stimulants, and sedatives. Figure 4.1 provides an overview of the general routing logic for the prescription psychotherapeutic drug questions that were used to estimate use. These questions were used to estimate any use of prescription psychotherapeutic drugs in the past year and misuse in the past year and past month. This routing logic simplifies the cognitive task for respondents by separating whether they used a specific prescription drug for any reason and, if so, whether they used it in a way constituting misuse.

For each of the prescription psychotherapeutic drug categories, respondents were first asked whether they had used any drug from a series of specific prescription drugs in the past 12 months, as shown in the "Screener Section" of Figure 4.1. To help respondents answer these questions, most prescription drugs in the questionnaire included electronic images of pills or other forms of the drugs (where applicable) that were shown to respondents on the computer screen. These images can be found in the *2024 National Survey on Drug Use and Health (NSDUH): Prescription Drug Images for the 2024 Questionnaire* (CBHSQ, 2024c). Respondents also were asked whether they used "any other" prescription drug in that category in the past 12 months (e.g., any other prescription pain reliever). Beginning in 2024, respondents were asked to specify the names of the other prescription pain relievers that they used for any reason in the past 12 months. As in previous years, respondents were not asked to specify the names of the other prescription tranquilizers, stimulants, or sedatives that they used for any reason in that period.

[68] Although methamphetamine is available in the United States in prescription form (Desoxyn®), most methamphetamine used in the United States is produced in clandestine laboratories rather than by the pharmaceutical industry.

Figure 4.1 Routing Logic for the Past Year and Past Month Prescription Psychotherapeutic Drug Questions



DK = Don't Know; REF = Refused.

NOTE: For brevity, the figure refers to "specific" prescription drugs, that is, the detailed drugs that were presented to respondents. Although not shown in the figure, respondents were also asked in each screener section whether they used "any other" prescription drug in that category in the past 12 months. Respondents who reported in the screener section that they used other prescription drugs in the past 12 months were asked in the main section whether they misused any other prescription drug in that category in the past 12 months. Just as for the other prescription drugs in a given main section, respondents who reported that they misused any other prescription drug were asked when they first misused other prescription drugs.

Respondents who did not report use in the past 12 months of any specific prescription psychotherapeutic drug or any other prescription drug within a category (e.g., prescription pain relievers) were asked whether they ever, even once, used any prescription psychotherapeutic drug within that category (e.g., any prescription pain reliever). Respondents who reported use of prescription psychotherapeutics in any of these four psychotherapeutic drug categories in the past 12 months or the lifetime period were classified as users of any prescription psychotherapeutic drug.

To identify past year misusers of prescription psychotherapeutic drugs, respondents who reported that they used specific prescription psychotherapeutic drugs or any other prescription drug in the past 12 months were asked about the misuse of each specific drug they used in the past 12 months, as shown in the "Main Section" of Figure 4.1. Respondents were shown a list of the drugs they used in the past 12 months. For each drug that they used, respondents were asked whether they used it in the past 12 months "in any way not directed by a doctor" (i.e., misuse).[69] Respondents who previously reported that they used any other prescription drug in the past 12 months were asked whether they misused any other prescription drug in that period. If respondents reported that they misused other prescription drugs, they were asked to specify the names of the other prescription drugs that they misused. This request to respondents applied to all four prescription psychotherapeutic drug categories in the NSDUH questionnaire.

Respondents who reported misuse of one or more drugs within a psychotherapeutic drug category in the past 12 months were classified as having misused prescription drugs in the past year, as shown in Figure 4.1. Respondents who reported misuse in the past year were asked whether they misused any drug in that category (e.g., prescription pain relievers) in the past 30 days.[70] This question was used to estimate past month or "current" misuse, as shown in Figure 4.1. Respondents who reported (1) any use of prescription psychotherapeutics in a category in the past 12 months but no misuse in the past 12 months or (2) any use in their lifetime but not in the past 12 months were asked whether they ever, even once, misused any prescription psychotherapeutic drug within that category (e.g., any prescription pain reliever). Respondents who reported misuse in their lifetime were identified as having misused prescription psychotherapeutic drugs in their lifetime but not in the past 12 months. Respondents who reported misuse of prescription psychotherapeutics in any of these four psychotherapeutic drug categories in the past 30 days, past 12 months, or in the lifetime period were classified as having misused any prescription psychotherapeutic drug.

[69] Examples of ways that were presented to respondents included (1) use without a prescription of one's own; (2) use in greater amounts, more often, or longer than told to take a drug; and (3) use in any other way not directed by a doctor.

[70] The exception was that respondents were not asked the question about misuse in the past 30 days if prior answers for their age at first misuse or their year and month of first misuse of specific prescription drugs indicated that they initiated misuse of any prescription drug in that category in the past 30 days. Logically, these respondents had already reported misuse in the past 30 days.

4.1.2 Misuse of Other Prescription Drugs

As noted previously, if respondents reported that they misused other prescription drugs in a given prescription drug category (e.g., pain relievers), they were asked to specify the names of the other prescription drugs they misused. Figure 4.2 uses the example of prescription pain relievers to show how information from the other prescription drugs that respondents specified was used to assess whether respondents misused any other prescription pain reliever or whether they misused any prescription pain reliever in the past 12 months. Similar principles applied to the handling of data for the remaining prescription drug categories. These procedures were not applied to estimates for any use of prescription tranquilizers, stimulants, or sedatives in the past 12 months because respondents were not asked to specify the names of the other drugs in these categories that they used for any reason. See Section 4.1.5 for handling of data in 2024 for any use of other prescription pain relievers in the past 12 months.

For the example of the misuse of pain relievers from Figure 4.2, respondents beginning in 2024[71] were not counted as having misused any other prescription pain reliever in the past year if (1) the only other drugs they specified corresponded to the prescription pain relievers that they were asked about in the NSDUH questionnaire (e.g., Vicodin®),[72] (2) they specified only drugs that were OTC drugs (e.g., acetaminophen) or otherwise are not considered prescription drugs,[73] or (3) they specified only some combination of misuse of pain relievers in the questionnaire, OTC drugs, or other nonprescription drugs. Responses of "don't know," "refused," or nonspecific drugs (e.g., "painkillers") for the names of other prescription pain relievers that respondents specified were assumed to be potential indications of the misuse of prescription pain relievers. If respondents were not counted as having misused other pain relievers because the only drugs they specified were OTC or other nonprescription drugs, they also were not counted as having misused *any* prescription pain reliever in the past 12 months if they had not reported the misuse of any of the specific pain relievers in the questionnaire; if respondents reported that the only "prescription" pain relievers they misused in the past 12 months were OTC drugs or other nonprescription drugs, then they logically did not misuse prescription pain relievers in that period.

Because these respondents originally reported that they misused prescription pain relievers in the past year, they were not asked whether they ever misused pain relievers in their lifetime. Although these respondents were inferred not to have misused pain relievers in the past year, they could still be imputed to have misused pain relievers in their lifetime.

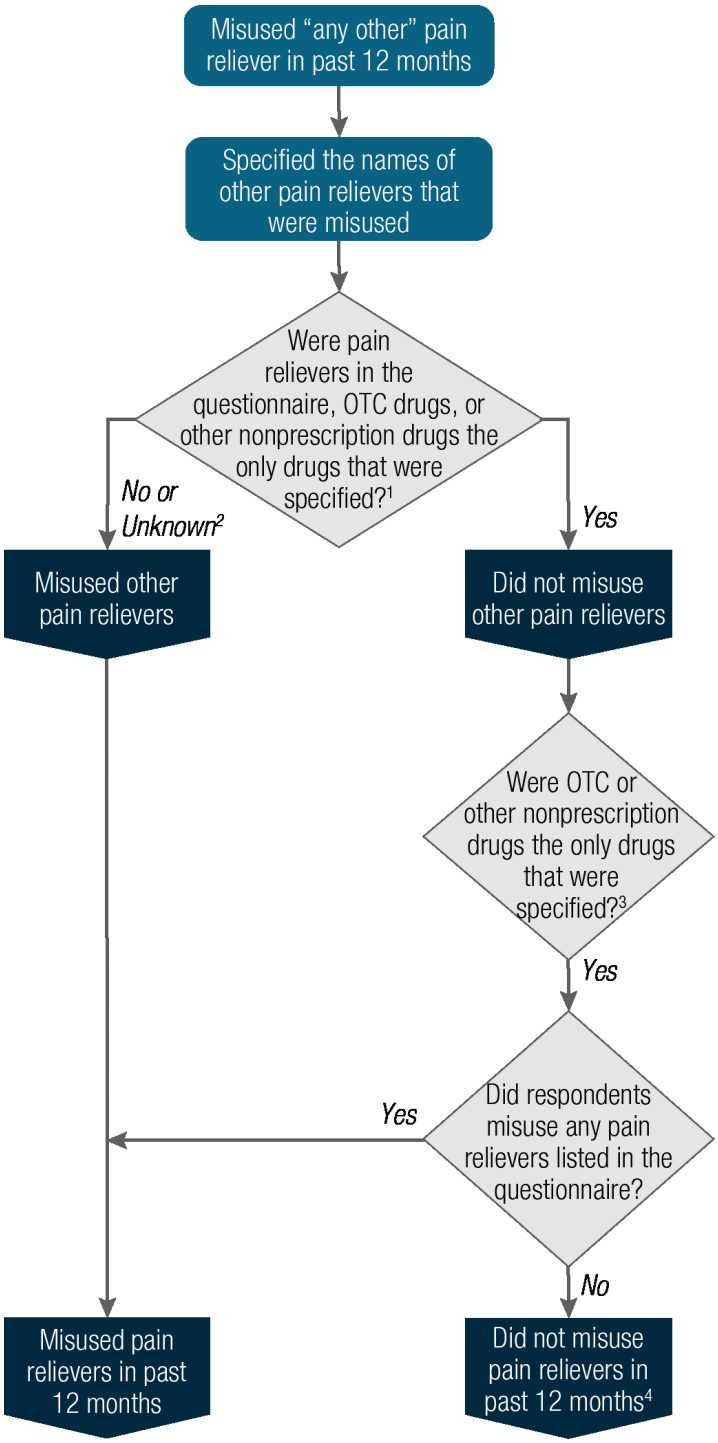
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[71] Beginning in 2024, reports of the misuse of other prescription drugs in a given category (e.g., pain relievers) were checked for reports of drugs that were not OTC drugs but were not considered to be prescription drugs. As in prior years, reports of the misuse of other prescription drugs in that category also were checked for reports of drugs that were OTC drugs or that were asked about in the questionnaire.

[72] For example, respondents who specified Vicodin® as the only other prescription pain reliever they misused in the past year were included in estimates for the past year misuse of hydrocodone products but were not included in estimates for the past year misuse of any other pain reliever.

[73] For example, the synthetic drug U-47700 is an opioid that is classified as having no currently accepted medical use and a high potential for abuse.

Figure 4.2 Handling of Data for the Past Year Misuse of Other Prescription Drugs: Pain Relievers as Example



OTC = over the counter.

[1] Respondents wrote in only the names of pain relievers that they were asked about in the questionnaire (e.g., Vicodin®), OTC drugs, or both.

[2] Unknown includes responses of "don't know," "refused," or a nonspecific drug (e.g., "painkillers").

[3] Respondents wrote in the names of only OTC drugs or nonprescription drugs and did not write in the names of pain relievers listed in the questionnaire.

[4] Lifetime misuse was unknown.

4.1.3 Measurement of Lifetime Misuse of Prescription Drugs

Consistent with the routing logic for the main sections for the misuse of prescription drugs described in Section 4.1.1 and shown in Figure 4.1, respondents who reported any use of specific prescription drugs in the past 12 months were asked whether they misused these specific drugs in that period. If respondents used prescription drugs for any reason in the past 12 months or in their lifetime, but they did not report the misuse of prescription drugs in the past 12 months, they were asked whether they ever, even once, misused any prescription drug in these categories. Respondents who reported that they never used prescription drugs for any reason in their lifetime logically did not misuse prescription drugs in their lifetime.

Although the NSDUH questions allow measurement of lifetime misuse of prescription psychotherapeutic drugs, the emphasis of the questions on misuse of specific prescription drugs in the past 12 months is assumed to result in underreporting of lifetime misuse of prescription drugs (see Section 3.3.3.2). Respondents who last misused prescription psychotherapeutic drugs more than 12 months ago would need to think about the prescription drugs that they misused at any point in their lifetime, including drugs that are no longer available by prescription in the United States. In contrast, a 12-month time frame is closer to the interview date and is assumed to allow better recall by respondents. Therefore, the 2024 Detailed Tables (CBHSQ, 2025h) do not show estimates for the lifetime misuse of prescription drugs.

4.1.4 Measurement of Any Lifetime Use of Prescription Tranquilizers, Stimulants, or Sedatives

Consistent with the routing logic for the screener sections for any use of prescription drugs described in Section 4.1.1 and shown in Figure 4.1, respondents who did not report the use of prescription tranquilizers, stimulants, or sedatives in the past year were asked in the respective prescription drug screener sections whether they ever, even once, used any prescription drug in these categories. Respondents who reported use in the past 12 months or the lifetime period were classified as users of these prescription drugs.

As for the lifetime misuse of prescription drugs described in Section 4.1.3, the emphasis of the questions on any use of specific prescription tranquilizers, stimulants, and sedatives in the past 12 months is assumed to result in underreporting of any lifetime use of prescription drugs in these categories (see Section 3.3.3.2). The lifetime reference period for any use of these prescription drugs involves a longer recall period than the past 12 months and could require respondents to think about the use of prescription drugs in these categories that are no longer available by prescription in the United States.

4.1.5 Measurement of Any Use of Prescription Pain Relievers

4.1.5.1 Any Past Year Use of Other Prescription Pain Relievers

Beginning in 2024, NSDUH respondents who reported any use of other pain relievers in the past 12 months were asked to specify the names of the other pain relievers they used. This additional information allowed respondents to be identified whose only use of prescription pain relievers in the past year involved the use of OTC drugs or other nonprescription drugs (see

Section 4.1.2). This information on the specific other pain relievers that respondents used in the past year also allowed respondents to be identified who used only pain relievers in the past year that are not prescription opioids. Section 4.4 discusses assumptions for any use of nonopioid pain relievers in the past year.

Responses for the names of other pain relievers that were specified for any use in the past 12 months were first reconciled with responses for the names of other pain relievers that respondents specified misusing in that period. If respondents specified the misuse of other drugs in the past year that they did not specify for any use of pain relievers in the past year, then respondents were logically inferred to have used these other drugs for any reason in the past year. In addition, if respondents specified that they did not know or refused to report the names of other pain relievers that they misused in the past year, but they did not have corresponding responses of "don't know" or "refused" for the other pain relievers they used for any reason, then these responses of "don't know" or "refused" were logically assigned for any use of other pain relievers in the past year. Responses of "don't know" or "refused" were assumed to indicate the potential use of prescription drugs.

After responses had been reconciled between any use of other prescription pain relievers and the misuse of other pain relievers, similar principles that were described in Section 4.1.2 and shown in Figure 4.2 applied for inferring that respondents did not use prescription pain relievers for any reason in the past year. These procedures were implemented when respondents' final data indicated any use of only prescription pain relievers that were asked about in the NSDUH questionnaire, OTC drugs, other nonprescription drugs, or some combination of these drugs.

4.1.5.2 Any Past Year Use of Prescription Pain Relievers Overall

If respondents in 2024 were logically inferred not to have used other pain relievers in the past year according to the procedures described in Section 4.1.5.1, and other pain relievers were the only pain relievers that respondents reported using, then these respondents were inferred not to have used prescription pain relievers for any reason in the past year. Consequently, estimates for any use of prescription pain relievers in the past year are not comparable with corresponding estimates in 2021-2023.

4.1.5.3 Any Lifetime Use of Prescription Pain Relievers Overall

Consistent with the routing logic for the screener sections for any use of prescription drugs described in Section 4.1.1 and shown in Figure 4.1, respondents who did not report the use of prescription pain relievers in the past year were asked whether they ever, even once, used any prescription pain reliever. Respondents who reported use of prescription pain relievers in the past 12 months or the lifetime period were classified as lifetime users of prescription pain relievers.

As noted in Section 4.1.5.2, respondents who were classified as having used OTC or other nonprescription drugs as the only prescription pain relievers they used in the past year were inferred not to have used prescription pain relievers in that period. Although these respondents

were inferred not to have used prescription pain relievers in the past year, they could still be imputed to have used pain relievers in their lifetime.

The emphasis of the questions in the pain relievers screener on the use of specific prescription pain relievers in the past 12 months is assumed to result in underreporting of any lifetime use of prescription pain relievers (see Section 3.3.3.2). Respondents who last used prescription pain relievers more than 12 months ago would need to think about the pain relievers that they used at any point in their lifetime, including those that are no longer available by prescription in the United States.

4.2 Prescription Drug Subtypes in NSDUH

4.2.1 Controlled Substances Act and Prescription Drug Subtypes

The Controlled Substances Act (CSA) of 1970 gives authority to the U.S. Drug Enforcement Administration within the U.S. Department of Justice to place controlled substances into "schedules" (CSA, n.d.). Schedules are defined according to factors such as (1) a substance's potential for abuse, (2) the state of current scientific knowledge regarding a drug, (3) risks to public health, or (4) the potential for physiological or psychological dependence. In principle, the classification of prescription drugs into these schedules could affect the availability of prescription drugs for misuse.

- *Schedule I substances* are deemed to have a high potential for abuse, have no currently accepted medical use in treatment in the United States, and have a lack of accepted safety for use under medical supervision. Drugs that are available by prescription in the United States are not in Schedule I, but some drugs that are chemically similar to prescription drugs may be, such as heroin (a nonprescription opioid).
- *Schedule II substances* have a high potential for abuse that can lead to severe psychological or physiological dependence. Unlike the drugs in Schedule I, however, the drugs in Schedule II have currently accepted medical uses in the United States under proper medical supervision. Several of the pain relievers and stimulants in NSDUH are in Schedule II.
- *Schedule III substances* have currently accepted medical uses. These substances have a lower potential for abuse than the substances in Schedule II. Abuse of these substances can lead to moderate or low physical dependence or a high degree of psychological dependence. Some of the stimulants in NSDUH prescribed for weight loss are in Schedule III.
- *Schedule IV substances* have currently accepted medical uses. These substances have a lower potential for abuse relative to the substances in Schedule III. Abuse of these substances can lead to limited physical or psychological dependence relative to the drugs in Schedule III. Several of the tranquilizers and sedatives in NSDUH are in Schedule IV.
- *Schedule V substances* have a lower potential for abuse relative to the substances in Schedule IV. The NSDUH questionnaire does not specifically ask about substances classified in Schedule V, such as cough medicines containing low dosages of codeine,

but respondents could specify these products as other prescription pain relievers they used (beginning in 2024) or misused.

Because of the greater risks associated with the drugs in Schedule II, the prescribing of these drugs is more tightly restricted and regulated than is the prescribing of drugs in Schedules III and IV (FDA, 2017).

4.2.2 Pain Reliever Subtypes and Their Status as Controlled Substances

Table 4.1 shows the CSA schedule numbers for the specific pain relievers included in the 2024 NSDUH questionnaire, grouped together into 11 subtypes of pain relievers for analysis. Figure 4.3 also shows these pain reliever subtypes and the specific pain relievers for each subtype, along with a 12th subtype for other prescription pain relievers. All of the pain reliever subtypes listed in Table 4.1 are prescription opioids, which are substances that act in the central nervous system (CNS) to reduce the perception of pain. As their name suggests, opioids include drugs found naturally in the opium poppy *Papaver somniferum*, such as morphine and codeine. Opioids also include drugs chemically similar to these naturally occurring substances, but they are manufactured in the laboratory (e.g., hydrocodone, fentanyl) (National Institute on Drug Abuse [NIDA], 2023; U.S. Drug Enforcement Administration, 2023).

Table 4.1 Specific Pain Relievers in the 2024 NSDUH, Their Status on the CSA Schedule, and Their Subtype Categorization

Subtype[1]	CSA Schedule[2]	Pain Relievers Included in a Subtype[3]
Hydrocodone Products	II	Examples in the questionnaire include Vicodin®, Lortab®, Norco®, Zohydro® ER, and generic hydrocodone. Subtype also includes any other pain reliever containing hydrocodone that respondents specified for any use or misuse in the past year.
Oxycodone Products	II	Examples in the questionnaire include OxyContin®, Percocet®, Percodan®, Roxicodone®, and generic oxycodone. Subtype also includes any other pain reliever containing oxycodone that respondents specified for any use or misuse in the past year.
Tramadol Products	IV	Examples in the questionnaire include Ultram®, Ultram® ER, Ultracet®, generic tramadol, and generic extended-release tramadol. Subtype also includes any other pain reliever containing tramadol that respondents specified for any use or misuse in the past year.
Codeine Products	II, III, or V[4]	Examples in the questionnaire include Tylenol® with codeine 3 or 4 and codeine pills. Codeine included in combination with pain relievers such as acetaminophen (e.g., Tylenol® with codeine 3 or 4) is classified as a Schedule III controlled substance. Codeine not included in combination with other pain relievers is classified as a Schedule II controlled substance.
Morphine Products	II[5]	Examples in the questionnaire include Avinza®, Kadian®, MS Contin®, generic morphine, and generic extended-release morphine. Subtype also includes any other pain reliever containing morphine that respondents specified for any use or misuse in the past year.

Table 4.1 Specific Pain Relievers in the 2024 NSDUH, Their Status on the CSA Schedule, and Their Subtype Categorization (continued)

Subtype[1]	CSA Schedule[2]	Pain Relievers Included in a Subtype[3]
Fentanyl Products	II[6]	Subtype is for prescription forms of fentanyl and includes Duragesic®, Fentora®, and generic fentanyl. Subtype also includes any other pain reliever containing fentanyl that respondents specified for any use or misuse in the past year.
Buprenorphine Products	III	Examples in the questionnaire include Suboxone®, generic buprenorphine, and generic buprenorphine plus naloxone. Subtype also includes any other pain reliever containing buprenorphine that respondents specified for any use or misuse in the past year.
Oxymorphone Products	II	Examples in the questionnaire include Opana®, Opana® ER, generic oxymorphone, and generic extended-release oxymorphone. Subtype also includes any other pain reliever containing oxymorphone that respondents specified for any use or misuse in the past year.
Demerol®	II	The active ingredient is meperidine. Subtype also includes any other pain reliever containing meperidine that respondents specified for any use or misuse in the past year.
Hydromorphone Products	II	Examples in the questionnaire include Dilaudid® or hydromorphone and Exalgo® or extended-release hydromorphone. Subtype also includes any other pain reliever containing hydromorphone that respondents specified for any use or misuse in the past year.
Methadone	II	Subtype also includes any other pain reliever containing methadone that respondents specified for any use or misuse in the past year.

CSA = Controlled Substances Act of 1970.

[1] Responses for specific pain relievers or any other pain reliever were categorized into subtypes for analysis.

[2] See the Controlled Substances Schedules.

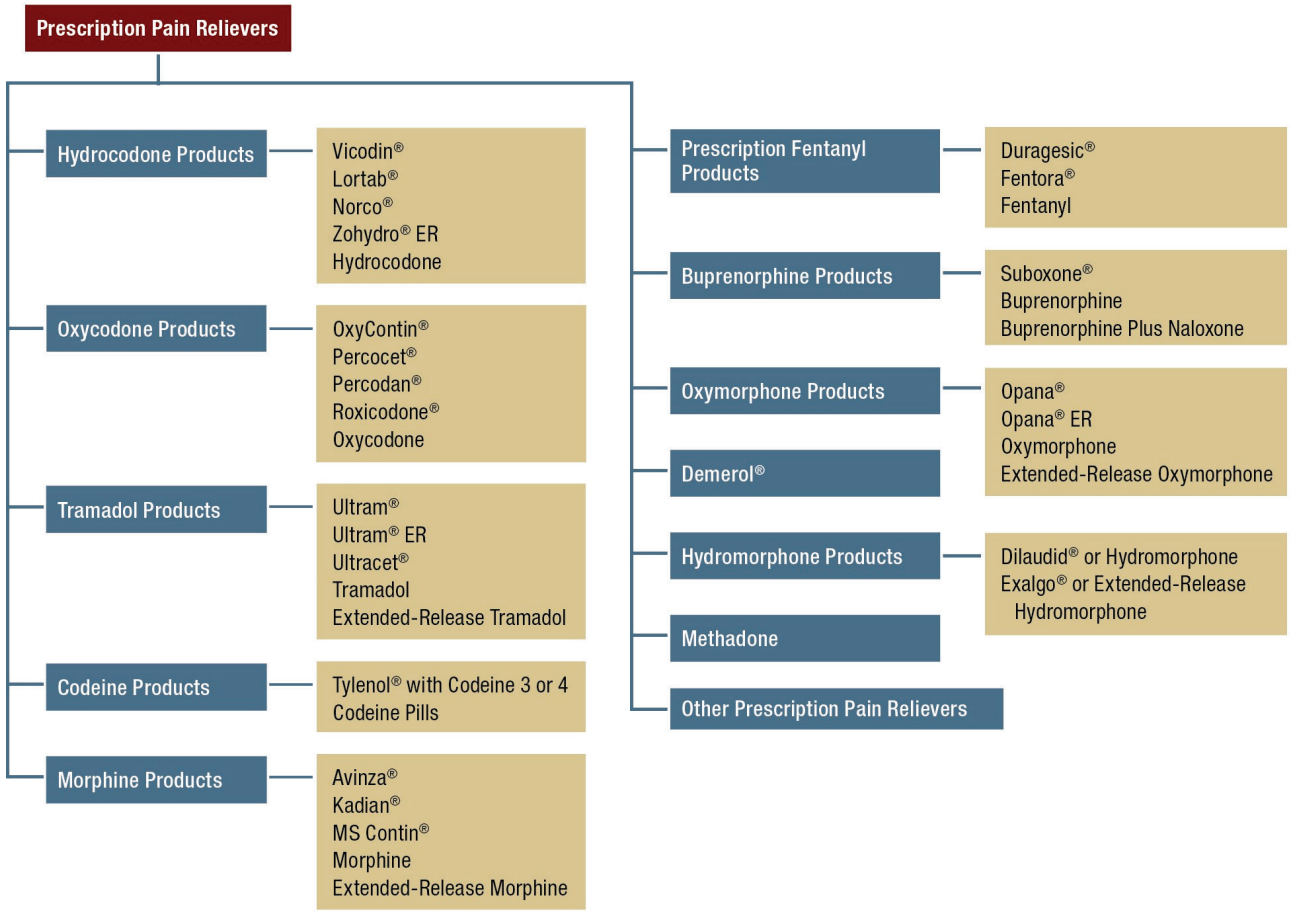
[3] Some brand names in the questionnaire may have been discontinued in the United States, but respondents may still recognize drugs in a subtype by these brand names.

[4] Beginning in 2024, cough medicines containing low dosages of codeine (which are classified as Schedule V controlled substances) that respondents specified as other pain relievers and that are available in the United States were counted as codeine products. However, codeine products that were available outside of the United States were counted as "other" prescription pain relievers rather than as codeine products because it was not known how people obtained or used these drugs.

[5] Source information on controlled substances from the U.S. Drug Enforcement Administration lists morphine products in combination with over-the-counter pain relievers in Schedule III. However, all examples of specific morphine products in the NSDUH questionnaire are in Schedule II.

[6] Prescription forms of fentanyl are Schedule II controlled substances. Schedule II does not include drug products containing illegally made fentanyl (IMF). See Sections 3.4.3 and 4.4 for more information on measures that include IMF.

Figure 4.3 Subtypes of Prescription Pain Relievers Based on Pain Reliever Questions in the 2024 NSDUH Questionnaire



NOTE: Prescription pain relievers as an aggregate category and categories for subtypes (e.g., hydrocodone products) represent measures for estimates shown in reports or tables for the 2024 NSDUH. The subtype categories were created from the specific drugs shown in the figure and were not presented directly to respondents, except where noted. If additional specific drugs are not shown for subtypes in the figure (e.g., Demerol®), then respondents were asked directly about those drugs.

NOTE: Prescription fentanyl products do not include illegally made fentanyl (IMF), which is produced in clandestine laboratories rather than by the pharmaceutical industry. These fentanyl products also do not include fentanyl analogues that are in Schedule II (e.g., carfentanil) or in Schedule I (e.g., ocfentanil). See Sections 3.4.3 and 4.4 for more information on IMF questions that have been included in the NSDUH questionnaire since 2022.

NOTE: The following drugs in this figure are generic drugs: hydrocodone, oxycodone, tramadol, extended-release tramadol, codeine pills, morphine, extended-release morphine, fentanyl, buprenorphine, buprenorphine plus naloxone, oxymorphone, extended-release oxymorphone, hydromorphone, extended-release hydromorphone, and methadone.

As noted previously, respondents who reported they used or misused other pain relievers in the past 12 months were asked to specify the names of the other pain relievers they used or misused. Although all of the pain reliever subtypes listed above are opioids, respondents could specify the use or misuse of other pain relievers that are not opioids, such as nonsteroidal anti-inflammatory drugs not classified as controlled substances (e.g., prescription-strength ibuprofen). Section 4.4 discusses implications of respondents' ability to specify that other pain relievers they used or misused in the past 12 months were not opioids.

As noted previously, most of the pain relievers in the NSDUH questionnaire are in the more stringently controlled Schedule II. Exceptions are products containing tramadol (Schedule IV); codeine plus acetaminophen (Schedule III), such as Tylenol® with codeine 3 or 4;[74] and buprenorphine (Schedule III). Respondents were reminded that Tylenol® with codeine 3 or 4 was not the same as OTC Tylenol®.

4.2.3 Stimulant Subtypes and Their Status as Controlled Substances

Table 4.2 shows the CSA schedule numbers for the specific stimulants included in the 2024 NSDUH questionnaire, grouped together into four subtypes of stimulants for analysis. Figure 4.4 also shows these stimulant subtypes and the specific stimulants for each subtype, along with a fifth subtype for other prescription stimulants.

Stimulants can be prescribed for multiple reasons, including treatment of attention-deficit/hyperactivity disorder (ADHD), weight reduction or control, or promoting wakefulness because of sleepiness associated with conditions such as narcolepsy or sleep apnea. Thus, unlike the other prescription drug categories, the intended purpose of prescribing stimulants is not always apparent from the name of the category. In contrast, the reason for prescribing pain relievers, tranquilizers, or sedatives is implied in the category name (i.e., pain relief, anxiety control, or sedation to relieve insomnia, respectively). For this reason, some of the subtypes of stimulants for 2024 shown in Table 4.2 and in Figure 4.4 refer to the condition for which the drugs are prescribed.

The amphetamines and stimulants containing methylphenidate that are primarily prescribed for the treatment of ADHD are in the more restrictive Schedule II. Stimulants in Table 4.2 that are prescribed for weight control are in Schedules III and IV.

As noted previously, methamphetamine is not included as a prescription stimulant in NSDUH unless the prescription form of methamphetamine (Desoxyn®) was specified as some other stimulant respondents had misused in the past year. However, no respondents in 2024 specified Desoxyn® as some other stimulant that they misused in the past year. When this drug was specified as another stimulant that respondents misused, it was classified as an amphetamine (Table 4.2) because it is chemically similar to other prescription amphetamines (e.g., Adderall®).[75]

[74] Although the brand name drugs Tylenol® with codeine 3 or 4 have been discontinued in the United States, respondents could still recognize generic codeine plus acetaminophen by these brand names.
[75] Because of the general principle of not using data from one section of the interview to edit variables in another section (see Section 2.3.2.1), reports of Desoxyn® outside of the stimulants section are not used to infer the use and misuse of amphetamines.

Table 4.2 Specific Stimulants in the 2024 NSDUH, Their Status on the CSA Schedule, and Their Subtype Categorization

Subtype[1]	CSA Schedule[2]	Stimulants Included in a Subtype[3]
Amphetamine Products[4]	II	Examples in the questionnaire include Adderall®, Adderall® XR, Dexedrine®, Vyvanse®, generic dextroamphetamine, generic amphetamine-dextroamphetamine combinations, and generic extended-release amphetamine-dextroamphetamine combinations. Subtype also includes any other amphetamine product that respondents specified for past year misuse of stimulants.[5] Vyvanse® is included because it is a Schedule II controlled substance and its active ingredient (lisdexamfetamine) is metabolized to dextroamphetamine.
Methylphenidate Products[4]	II	Examples in the questionnaire include Ritalin®, Ritalin® LA, Concerta®, Daytrana®, Metadate® CD, Metadate® ER, Focalin®, Focalin® XR, generic methylphenidate, generic extended-release methylphenidate, generic dexmethylphenidate, and generic extended-release dexmethylphenidate. Subtype also includes any other stimulant containing methylphenidate that respondents specified for past year misuse.
Anorectic (Weight-Loss) Stimulants	III or IV	Examples in the questionnaire include Didrex®, benzphetamine, Tenuate®, diethylpropion, phendimetrazine, and phentermine. Subtype also includes similar other products that respondents specified for past year misuse. Didrex®, benzphetamine, and phendimetrazine are Schedule III controlled substances. Tenuate®, diethylpropion, and phentermine are Schedule IV controlled substances.
Provigil®	IV	The active ingredient is modafinil. Subtype also includes any other stimulant containing modafinil that respondents specified for past year misuse. The drug is prescribed to improve wakefulness in adult patients with excessive sleepiness associated with narcolepsy, obstructive sleep apnea, or shift work disorder.

CSA = Controlled Substances Act of 1970.

[1] Responses for specific stimulants or any other stimulant were categorized into subtypes for analysis.

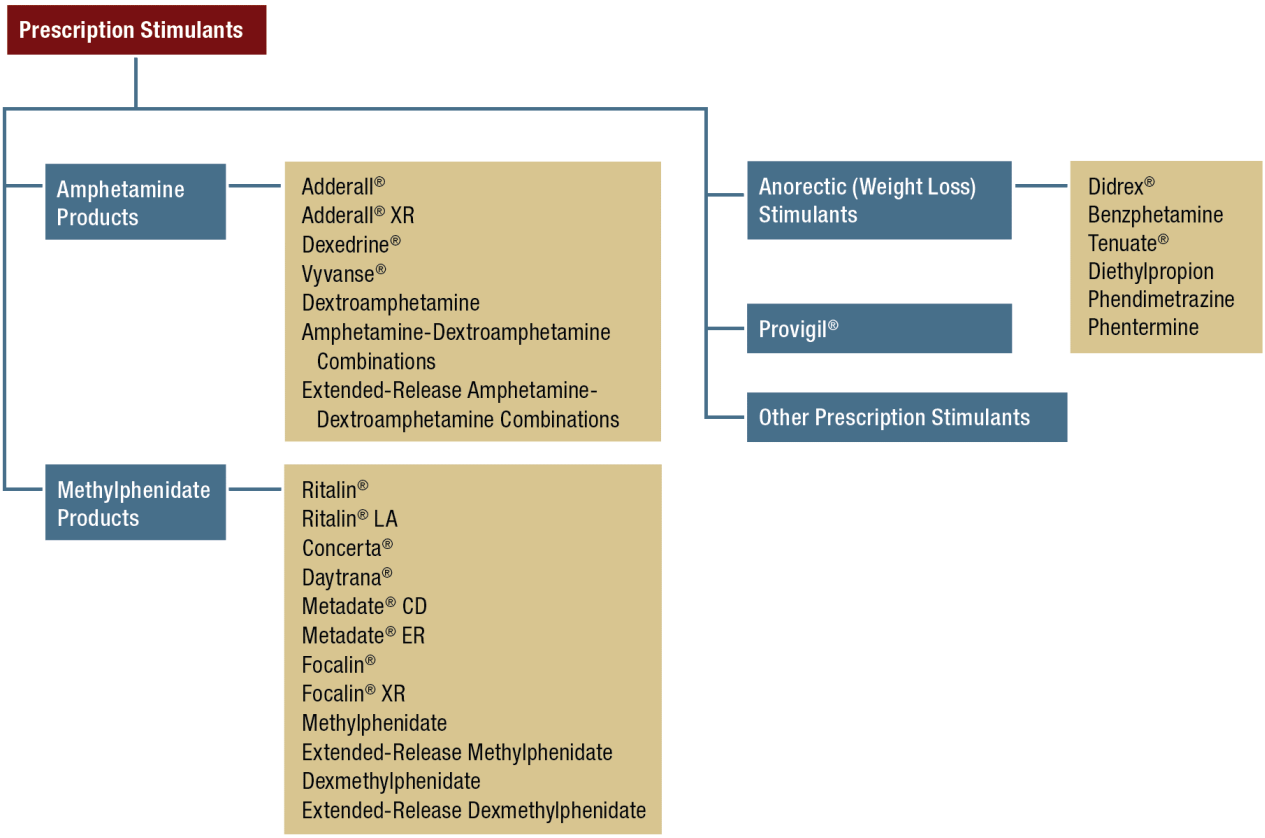
[2] See the Controlled Substances Schedules.

[3] Some brand names in the questionnaire may have been discontinued in the United States, but respondents may still recognize drugs in a subtype by these brand names.

[4] The amphetamine and methylphenidate products include stimulants primarily prescribed for the treatment of attention-deficit/hyperactivity disorder (ADHD).

[5] Desoxyn®, the prescription form of methamphetamine, was included as an amphetamine product. It has been specified only rarely as some other prescription stimulant but not in 2024.

Figure 4.4 Subtypes of Prescription Stimulants Based on Stimulant Questions in the 2024 NSDUH Questionnaire



NOTE: Prescription stimulants as an aggregate category and categories for subtypes (e.g., amphetamine products) represent measures for estimates shown in reports or tables for the 2024 NSDUH. The subtype categories were created from the specific drugs included in the figure and were not presented directly to respondents, except where noted. If additional specific drugs are not shown for subtypes in the figure (e.g., Provigil®), then respondents were asked directly about those drugs.
NOTE: Vyvanse® is included with amphetamine products because its active ingredient (lisdexamfetamine) is metabolized to dextroamphetamine.
NOTE: The following drugs in this figure are generic drugs: dextroamphetamine, amphetamine-dextroamphetamine combinations, extended-release amphetamine-dextroamphetamine combinations, methylphenidate, extended-release methylphenidate, dexmethylphenidate, extended-release dexmethylphenidate, benzphetamine, diethylpropion, phendimetrazine, and phentermine.

4.2.4 Tranquilizer Subtypes and Their Status as Controlled Substances

Table 4.3 shows the CSA schedule numbers for the specific tranquilizers included in the 2024 NSDUH questionnaire, grouped together into two subtypes of tranquilizers for analysis. Four additional subcategories of benzodiazepine tranquilizers and two additional subcategories of muscle relaxants were created. Figure 4.5 also shows these tranquilizer subtypes and subcategories, the specific tranquilizers for each subtype, and a third subtype for other prescription tranquilizers. Tranquilizers are usually prescribed to relax people, relieve anxiety, or relax muscle spasms.

Table 4.3 Specific Tranquilizers in the 2024 NSDUH, Their Status on the CSA Schedule, and Their Subtype and Subcategory Categorization

Subtype[1]	Subcategory	CSA Schedule[2]	Tranquilizers Included in a Subtype and Subcategory[3]
Benzodiazepine Tranquilizers	Alprazolam Products	IV	Subtype is for a benzodiazepine prescribed as a tranquilizer. Examples in the questionnaire for the subcategory include Xanax®, Xanax® XR, generic alprazolam, and generic extended-release alprazolam. Subcategory also includes any other tranquilizer containing alprazolam that respondents specified for past year misuse.
Benzodiazepine Tranquilizers	Lorazepam Products	IV	Subtype is for a benzodiazepine prescribed as a tranquilizer. Examples in the questionnaire for the subcategory include Ativan® and generic lorazepam. Subcategory also includes any other tranquilizer containing lorazepam that respondents specified for past year misuse.
Benzodiazepine Tranquilizers	Clonazepam Products	IV	Subtype is for a benzodiazepine prescribed as a tranquilizer. Examples in the questionnaire for the subcategory include Klonopin® and generic clonazepam. Subcategory also includes any other tranquilizer containing clonazepam that respondents specified for past year misuse.
Benzodiazepine Tranquilizers	Diazepam Products	IV	Subtype is for a benzodiazepine prescribed as a tranquilizer. Examples in the questionnaire for the subcategory include Valium® and generic diazepam. Subcategory also includes any other tranquilizer containing diazepam that respondents specified for past year misuse.
Muscle Relaxants	Cyclobenzaprine	None	The subcategory is for a muscle relaxant that is not a controlled substance. The drug also is known as Flexeril®, which is no longer available in the United States. Subcategory also includes any other tranquilizer containing cyclobenzaprine that respondents specified for past year misuse.
Muscle Relaxants	Soma®	IV	This is a muscle relaxant. The active ingredient is carisoprodol. Subcategory also includes any other tranquilizer containing carisoprodol that respondents specified for past year misuse.

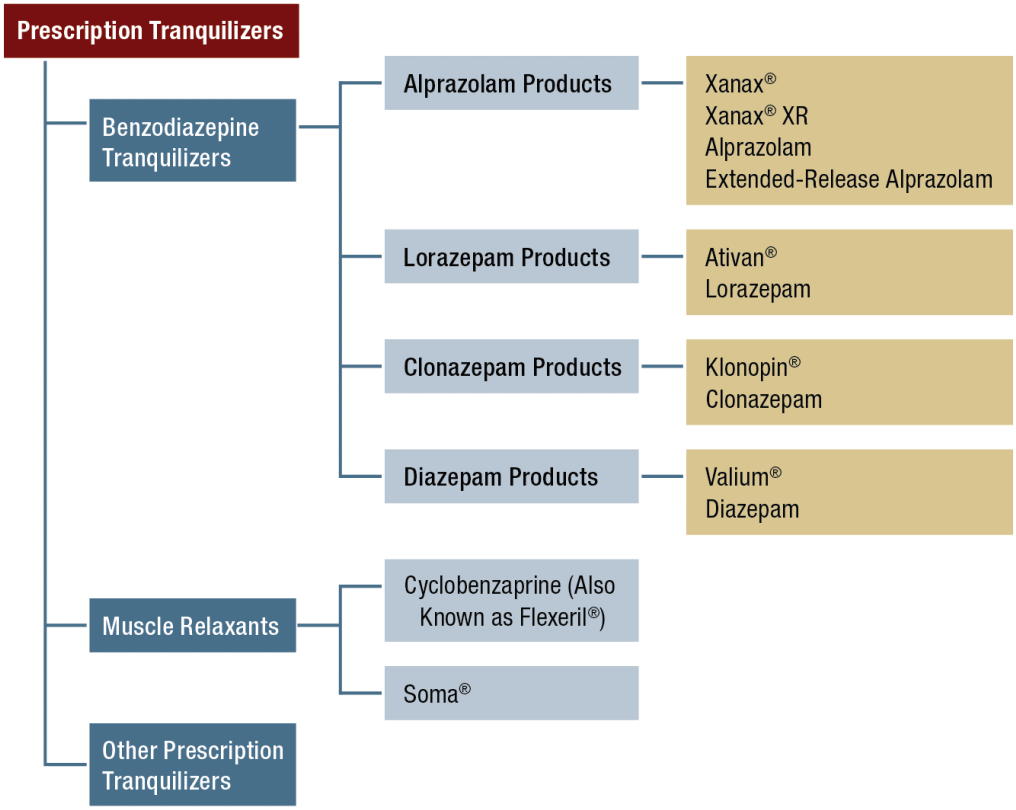
CSA = Controlled Substances Act of 1970.

[1] Responses for specific tranquilizers or any other tranquilizer were categorized into subtypes for analysis.

[2] See the Controlled Substances Schedules.

[3] Some brand names in the questionnaire may have been discontinued in the United States, but respondents may still recognize drugs in a subtype by these brand names.

Figure 4.5 Subtypes and Subcategories of Prescription Tranquilizers Based on Tranquilizer Questions in the 2024 NSDUH Questionnaire



NOTE: Prescription tranquilizers as an aggregate category, categories for subtypes (e.g., benzodiazepine tranquilizers), and additional subcategories (e.g., alprazolam products as a subcategory of benzodiazepine tranquilizers) represent measures for estimates shown in reports or tables for the 2024 NSDUH. The subtype categories and subcategories were created from the specific drugs included in the figure and were not presented directly to respondents, except where noted. If additional drugs are not shown for subtypes or subcategories in the figure (e.g., cyclobenzaprine [also known as Flexeril®]), then respondents were asked directly about those drugs.
NOTE: The following drugs in this figure are generic drugs: alprazolam, extended-release alprazolam, lorazepam, clonazepam, diazepam, and cyclobenzaprine.

Several of the tranquilizers in the 2024 NSDUH questionnaire are in the less restrictive Schedule IV. Cyclobenzaprine (also known as Flexeril®) is not classified by the U.S. Drug Enforcement Administration as a controlled substance but does require a prescription. Although cyclobenzaprine is not scheduled as a controlled substance, it is classified as a muscle relaxant. As shown in Table 4.3, another muscle relaxant in the questionnaire (Soma®) *is* a controlled substance. Despite cyclobenzaprine not being a controlled substance, the label for Flexeril® suggests that the drug can be misused.[76] Specifically, people can deliberately overdose on cyclobenzaprine. Because the drug may enhance the effects of alcohol and other CNS depressants, people would be misusing cyclobenzaprine if they take it with these other substances despite being directed by a doctor or other health professional not to do so.

[76] Product label information for Flexeril® is available on the FDA Center for Drug Evaluation and Research website. The product label for generic cyclobenzaprine is not available on the FDA website.

As discussed in Section 4.2.5, other benzodiazepines are prescribed as sedatives. Although both tranquilizers and sedatives cause drowsiness, including tranquilizers and sedatives that are benzodiazepines, a distinction between these drug categories is that tranquilizers are prescribed for anxiety relief or to relieve muscle spasms, whereas sedatives are prescribed specifically for the relief of insomnia. In particular, the types of benzodiazepine drugs prescribed as tranquilizers typically are metabolized more slowly than benzodiazepines prescribed as sedatives.[77,78] The rate of metabolism determines the duration and intensity of a drug's pharmacological effect on the body.

Because benzodiazepines are chemically similar regardless of whether they are prescribed as tranquilizers or sedatives, estimates for the use and misuse of any benzodiazepine in the past 12 months are included in 2024 NSDUH national reports and tables. Issues related to the measurement of any use and misuse of benzodiazepines are discussed further in Section 4.6.

4.2.5 Sedative Subtypes and Their Status as Controlled Substances

Table 4.4 shows the CSA schedule numbers for the specific sedatives included in the 2024 NSDUH questionnaire, grouped together into five subtypes of sedatives for analysis. Three additional subcategories of benzodiazepine sedatives were created. Figure 4.6 also shows these sedative subtypes and subcategories, the specific sedatives for each subtype, and a sixth subtype for other prescription sedatives. Sedatives are prescribed to relieve insomnia.

Most of the sedatives in the 2024 NSDUH questionnaire are in the less restrictive Schedule IV. However, some barbiturates are in Schedule II (Seconal®) or Schedule III (Butisol®). As noted in Section 4.2.4 on tranquilizers, the benzodiazepines prescribed as sedatives for the relief of insomnia (e.g., Halcion®) typically have a shorter duration of action compared with benzodiazepines prescribed for the treatment of anxiety (e.g., Xanax®).

[77] For example, the product label for Xanax®, which is prescribed as a tranquilizer, indicates the drug has an average half-life of 11.2 hours (i.e., the length of time for half of the dosage of the drug to be metabolized), with a range of 6.3 to 26.9 hours in healthy adults. In comparison, the product label for Halcion®, which is a benzodiazepine prescribed as a sedative, has a short half-life in the range of 1.5 to 5.5 hours. Product label information for these drugs is available on the FDA Center for Drug Evaluation and Research website.

[78] When a drug is metabolized, it is converted into *metabolites,* which are the substances that remain after the drug is broken down by the body. For more information, see the definition for "metabolite" by typing this word as a search term on the MedlinePlus web page.

Table 4.4 Specific Sedatives in the 2024 NSDUH, Their Status on the CSA Schedule, and Their Subtype and Subcategory Categorization

Subtype[1]	Subcategory	CSA Schedule[2]	Sedatives Included in a Subtype and Subcategory[3]
Zolpidem Products	N/A	IV	Examples in the questionnaire include Ambien®, Ambien® CR, generic zolpidem, and extended-release generic zolpidem. Subtype also includes any other sedative containing zolpidem that respondents specified for past year misuse.
Eszopiclone Products	N/A	IV	Examples in the questionnaire include Lunesta® and generic eszopiclone. Subtype also includes any other sedative containing eszopiclone that respondents specified for past year misuse.
Zaleplon Products	N/A	IV	Examples in the questionnaire include Sonata® and generic zaleplon. Subtype also includes any other sedative containing zaleplon that respondents specified for past year misuse.
Benzodiazepine Sedatives	Flurazepam	IV	Subtype is for a benzodiazepine prescribed as a sedative. The drug also is known as Dalmane®, which is no longer available in the United States. Subcategory also includes any other sedative containing flurazepam that respondents specified for past year misuse.
Benzodiazepine Sedatives	Temazepam Products	IV	Subtype is for a benzodiazepine prescribed as a sedative. Examples in the questionnaire for the subcategory include Restoril® and generic temazepam. Subcategory also includes any other sedative containing temazepam that respondents specified for past year misuse.
Benzodiazepine Sedatives	Triazolam Products	IV	Subtype is for a benzodiazepine prescribed as a sedative. Examples in the questionnaire for the subcategory include Halcion® and generic triazolam. Subcategory also includes any other sedative containing triazolam that respondents specified for past year misuse.
Barbiturates	N/A	II, III, or IV	Examples in the questionnaire include Butisol®, Seconal®, and phenobarbital. Subtype also includes any other barbiturate that respondents specified for past year misuse of sedatives. Seconal® (secobarbital) is a Schedule II controlled substance. Butisol® (butabarbital) is a Schedule III controlled substance. Phenobarbital is a Schedule IV controlled substance.

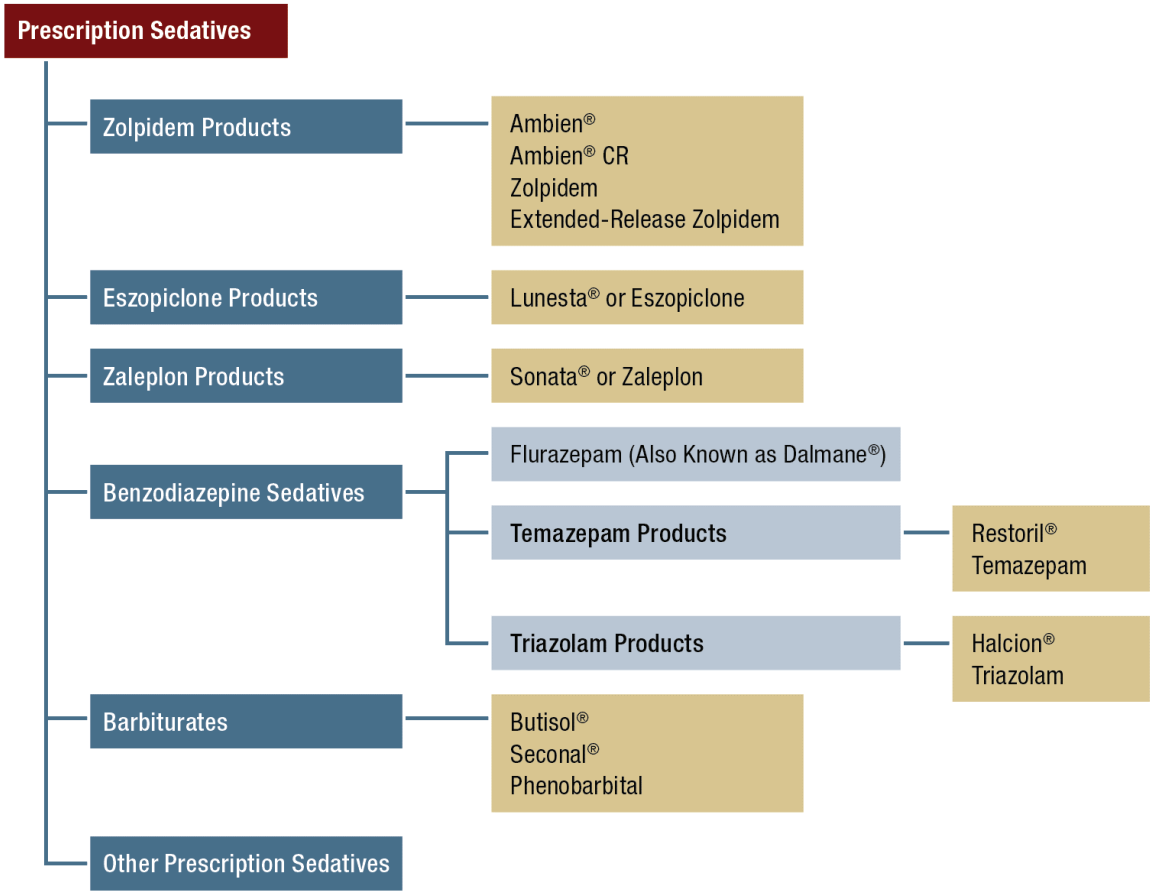
CSA = Controlled Substances Act of 1970; N/A = not applicable.

[1] Responses for specific sedatives or any other sedative were categorized into subtypes for analysis.

[2] See the Controlled Substances Schedules.

[3] Some brand names in the questionnaire may have been discontinued in the United States, but respondents may still recognize drugs in a subtype by these brand names.

Figure 4.6 Subtypes and Subcategories of Prescription Sedatives Based on Sedative Questions in the 2024 NSDUH Questionnaire



NOTE: Prescription sedatives as an aggregate category, categories for subtypes (e.g., zolpidem products), and additional subcategories (e.g., temazepam products as a subcategory of benzodiazepine sedatives) represent measures for estimates shown in reports or tables for the 2024 NSDUH. The subtype categories and subcategories were created from the specific drugs included in the figure and were not presented directly to respondents, except where noted. If additional drugs are not shown for subtypes or subcategories in the figure (e.g., flurazepam [also known as Dalmane®]), then respondents were asked directly about those drugs.

NOTE: The following drugs in this figure are generic drugs: zolpidem, extended-release zolpidem, eszopiclone, zaleplon, flurazepam, temazepam, triazolam, and phenobarbital.

4.3 Handling of Missing Data for Prescription Drugs

In 2024, a number of prescription drug variables underwent statistical imputation to account for item nonresponse and, therefore, had no missing data (Section 2.3.3). Imputed variables included all variables that were used to estimate any use and misuse in the past year for the overall categories of prescription pain relievers, tranquilizers, stimulants, and sedatives and the variables for prescription fentanyl products[79] and any benzodiazepine. Past year initiation variables for prescription drug misuse, corresponding to questions in Figure 4.1, and SUD

[79] Measures for the use of IMF also have been imputed since 2022. However, estimates in the *Results from the 2024 National Survey on Drug Use and Health: Detailed Tables* (CBHSQ, 2025h) for the use and misuse of pain reliever subtypes included prescription fentanyl products but not IMF.

variables for prescription drugs also were imputed (see Sections 2.3.3, 3.4.6, and 3.4.7). However, prescription drug variables for the following estimates in NSDUH national reports and tables did not undergo statistical imputation and, therefore, had missing data:

- reasons for the last misuse of prescription drugs within a given category (i.e., pain relievers, tranquilizers, stimulants, or sedatives);
- sources of prescription drugs for the last misuse of drugs in a given category in the past year; and
- most subtypes of prescription drugs (e.g., hydrocodone products) that were used or misused in the past year, except for prescription fentanyl products and any benzodiazepine.

When outcomes were not imputed, respondents with missing data were excluded from analyses. Bias may result when respondents with missing data are excluded from an analysis. For population totals (i.e., estimated numbers of people with a given characteristic), a negative bias will always occur if there are missing values in the domain variables, the outcome variable, or both. For the resulting outcomes (e.g., numbers of people who obtained the last prescription drug they misused from a particular source), this negative bias can yield estimates lower than the true population total.[80] When population proportions are estimated, there may or may not be bias, and the bias can be negative or positive. The direction and magnitude of the bias for proportions depend on how different the item respondents are from the item nonrespondents with respect to the outcome of interest.

Respondents could have missing data for whether they used or misused specific subtypes of prescription drugs in the past year. For example, respondents were presented with a list of prescription tranquilizers containing alprazolam and were asked to report which, if any, of these they had used in the past 12 months. Except in special situations, respondents who answered "don't know" or "refused" when presented with this list would have missing data for the past year use of alprazolam products. In turn, these respondents were not asked whether they misused specific alprazolam products in the past year.[81]

The availability of "OTHER, Specify" questions for any use of other pain relievers since 2024 could affect the occurrence of missing data for pain reliever subtypes. As for other prescription drug categories, respondents could answer "don't know" or "refused" to questions in the pain

[80] The estimated total will be lower than the true population total if the negative bias from excluding respondents with missing data outweighed other potential sources of random error (e.g., sampling error resulting from the selection of a sample) or nonrandom error (e.g., overreporting of the characteristic) that affected estimated totals in a positive direction.

[81] An exception to this general principle applied to respondents who specified they misused one or more prescription drugs for a given subtype as some "other" prescription drug they misused in the past year. For example, suppose respondents answered "don't know" when presented with the list of alprazolam products for any use in the past year. If these respondents reported the misuse of "other" tranquilizers in the past year and then specified an alprazolam product (e.g., Xanax®) was one of the other prescription tranquilizers they misused in the past year, then these respondents logically misused alprazolam products in the past year. These respondents also logically used alprazolam products in the past year for any reason.

relievers screener section that showed a specific list of pain relievers they used in the past year. However, respondents could specify use of these drugs as other pain relievers they used for any reason in the past year. For example, if respondents answered "don't know" or "refused" to the question about any use of hydrocodone products in the past year but they specified a hydrocodone product (e.g., Vicodin®) was one of the other prescription pain relievers they used in the past year, then these respondents logically used hydrocodone products in that period. Hence, these respondents no longer had missing data for the use of hydrocodone products in the past year. This special situation was in addition to the opportunity that respondents had to specify misuse of pain relievers in a given subtype despite having missing data for any use of pain relievers in that subtype.

Beginning in 2021, if variables for the use and misuse of prescription drug subtypes had missing data, respondents with missing values were excluded from analyses. Statistical imputation and excluding respondents with missing data for prescription drug subtype variables were adopted because missing data rates were low for most prescription drug variables. For example, see Table 2.4 in the *2022 National Survey on Drug Use and Health (NSDUH): Methodological Summary and Definitions* (CBHSQ, 2023). Remaining prescription drug variables that were not included in national reports and tables for the 2024 NSDUH were edited but not imputed. Potential biases associated with missing data discussed in this section and in Section 3.3.2 will apply to analyses using these edited variables.

4.4 Measures of Opioid Misuse and Use in NSDUH

4.4.1 Background

The subtypes of opioid pain relievers described in Section 4.2.2 are available in the United States by prescription as controlled substances. As opioids, however, they can produce the same kinds of adverse effects as heroin or other illegally manufactured opioids. People who misuse prescription opioids can develop an opioid use disorder or can overdose, sometimes fatally. According to the DSM-5 SUD criteria, people who are prescribed opioids for pain relief and take them under medical supervision also can develop opioid use disorder.[82] As noted in Section 4.2.2, most prescription opioids in the NSDUH questionnaire are in the more stringently controlled Schedule II category because of their high potential for abuse that can lead to severe psychological or physiological dependence.

NSDUH respondents were asked about their use and misuse of prescription *pain relievers* rather than being asked specifically about their use and misuse of prescription *opioids*. Respondents were assumed to be more likely to understand the term "pain relievers" rather than "opioids" because "pain relievers" indicates the purpose for which the drugs are likely to be taken. In

[82] As discussed in Section 3.4.7, NSDUH respondents beginning in 2021 who reported *any* use of prescription psychotherapeutic drugs (i.e., pain relievers, tranquilizers, stimulants, or sedatives) in the past year (i.e., not just misuse of prescription drugs) were asked the respective SUD questions for that category of prescription drugs. However, tolerance and withdrawal are normal physiological adaptations when people use these prescription drugs appropriately under medical supervision or abruptly discontinue use (Hasin et al., 2013).

contrast to "pain relievers," the term "opioids" could be too sophisticated for respondents at a 6th grade reading level. This term also would require respondents to know the chemical classification of a prescription drug. An online survey in 2020 of about 1,000 adult respondents in the United States indicated respondent difficulty in differentiating between medications that were opioids or nonopioids (Duff et al., 2021; DrFirst, 2020).

The 2024 NSDUH questionnaire included questions about 38 specific prescription pain relievers that fall into 11 opioid pain reliever subtypes (see Section 4.2.2 and Table 4.1). As noted in Section 4.1, however, respondents also were asked whether they used or misused any other prescription pain reliever in the past 12 months. Beginning in 2024, respondents who reported any use of other pain relievers in the past 12 months were asked to report the names of the other drugs they used, in addition to being asked to report the names of the other drugs they misused.

However, the general prescription pain reliever category includes prescription drugs that are not opioids, such as prescription strengths of nonsteroidal anti-inflammatory drugs (e.g., prescription-strength ibuprofen or naproxen). If NSDUH respondents reported using or misusing other prescription pain relievers in the past 12 months, these other pain relievers might not be opioids. Therefore, published NSDUH estimates of the use and misuse of prescription pain relievers are not completely synonymous with the use and misuse of prescription opioids.

The number of prescriptions for opioids has declined since the release of the Centers for Disease Control and Prevention (CDC) 2016 opioid prescribing guideline (Bohnert et al., 2018; Dowell et al., 2016), and there have been modest increases in the prescribing of nonopioids for chronic pain (Goldstick et al., 2022). The CDC updated clinical practice guideline in 2022 for prescribing opioids for chronic pain noted that nonopioid therapies are at least as effective as opioids for many types of acute pain, defined as pain lasting less than 1 month. The 2022 guideline also recommended nonopioid therapies as the preferred treatment for pain that is considered subacute (lasting 1 to 3 months) or chronic (lasting more than 3 months) (Dowell et al., 2022).

Because of these clinical practice guidelines and observed changes in medication prescribing for pain relief, some nonopioid pain relievers could emerge as drugs of abuse. For example, nonopioid drugs such as gabapentin (brand name Neurontin®) that are prescribed "off label"[83] for pain relief may have abuse potential, especially among people with a history of opioid misuse (Buttram, 2018; Buttram et al., 2017; Evoy et al., 2021; Havens, 2018; Smith et al., 2016). Detection of gabapentin increased in fatal drug overdoses in 2019-2020, but nearly 90 percent of the overdose deaths in which gabapentin was detected also involved opioids (Mattson et al., 2022). Use of gabapentin with opioids has been associated with an increased risk of overdose compared with the use of opioids alone (Chou et al., 2020). Other adverse

[83] "Off label" prescribing refers to the prescribing of a drug that has been approved for use in the United States, but the drug is being prescribed for a condition the drug is not approved to treat (FDA, 2018).

effects associated with gabapentin include blurred vision, cognitive effects, and sedation (McDonagh et al., 2020). In 2024, the first year in which data have been available for the specific other prescription pain relievers that respondents used for any reason in the past year, nearly 250 respondents reported that gabapentin was another pain reliever that they used in the past year. These reports are likely to underestimate the true prevalence of use of this drug (Kroutil et al., 2010).

4.4.2 Prescription Pain Reliever and Prescription Opioid Misuse

NSDUH respondents who reported the misuse of only any other prescription pain reliever are counted in NSDUH estimates of prescription pain reliever misuse, regardless of whether the other drugs they misused are opioids or nonopioids. Beginning with the 2024 NSDUH, published estimates are presented for the misuse of prescription opioids in the past 12 months. Prior analyses of NSDUH data have indicated that a small percentage of pain reliever misuse estimates can be attributed to respondents who explicitly reported the misuse of only nonopioids. For more information, see the 2023 Methodological Summary and Definitions report (CBHSQ, 2024b).

Figure 4.7, which refers to the "Pain Relievers Main Section" of the NSDUH questionnaire, describes the information that was used for creating variables to estimate the misuse of prescription opioids in the past year.

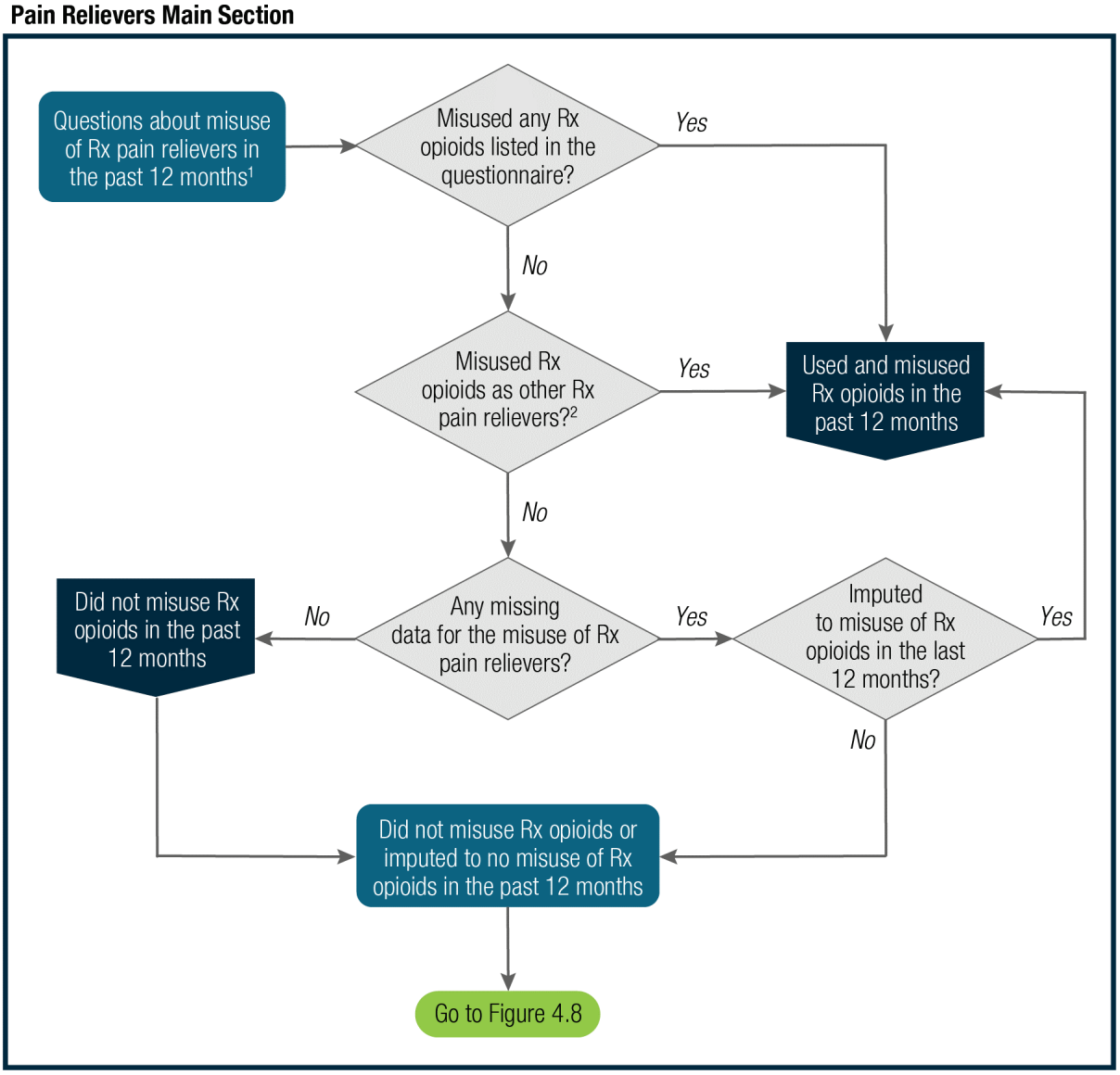
- Respondents who reported the past year misuse of any of the prescription opioids in the questionnaire (Table 4.1 and Figure 4.3) were classified as having misused prescription opioids in that period.[84]
- The following respondents who reported the past year misuse of only other pain relievers were classified as having misused prescription opioids:[84]
  - those who specified the misuse of prescription opioids as other pain relievers, including situations in which respondents specified the misuse of prescription opioids in the questionnaire despite not having reported their misuse;
  - those who did not know or refused to specify the names of other pain relievers they misused; and
  - those who provided a nonspecific answer, such as "painkillers," without indicating the specific pain reliever(s) they misused.

Consistent with the procedures described in Section 4.1.2, responses of "don't know," "refused," or a nonspecific other pain reliever were assumed to be potential indications of the misuse of prescription opioids. Respondents who reported that they misused only other prescription pain relievers in the past 12 months were classified as not having misused prescription opioids if the only pain relievers they specified misusing were nonopioid drugs. If respondents did not provide sufficient information to establish whether they misused

[84] Logically, these respondents also used prescription opioids for any reason in the past year.

prescription opioids in the past year, statistical imputation was used to classify respondents as having misused or not having misused prescription opioids in the past year.

Figure 4.7 Classification of the Misuse of Prescription Opioids in the Past Year



Rx = prescription.

NOTE: As per Figure 4.1, respondents were first asked about any use of specific Rx pain relievers in the past 12 months. To illustrate the logic for identifying the use and misuse of prescription opioids, the pain relievers main section is presented in this figure. Logically, if respondents misused Rx opioids in the past 12 months, then they also used them for any reason in that period.

[1] As per Figure 4.1, if respondents were asked about their misuse of specific Rx pain relievers in the past 12 months, then they previously reported using them for any reason in that period.

[2] Respondents were assumed to have misused Rx opioids if they did not know or refused to report the other pain relievers they misused or they reported a nonspecific drug (e.g., "painkillers").

The percentage of people aged 12 or older who misused prescription opioids in the past year decreased from 3.0 percent in 2021 to 2.6 percent in 2024. For more information, see the *Key Substance Use and Mental Health Indicators in the United States: Results from the 2024 National Survey on Drug Use and Health* report (CBHSQ, 2025g). About 95 percent of people in each year who misused any prescription pain reliever in the past year were classified as having misused prescription opioids.

4.4.3 Any Opioid Misuse, Including Heroin

In national reports and tables for the 2024 NSDUH, the categories for the misuse of prescription opioids and the use of heroin were combined into an overall category for opioid misuse. As shown in the 2024 Key Substance Use and Mental Health Indicators report, most misuse of opioids in the past year involves the misuse of prescription opioids rather than heroin use (CBHSQ, 2025g). Therefore, including data from respondents who used heroin in the past year does not appreciably affect estimates of any opioid misuse.

The percentage of people aged 12 or older who misused any opioid in the past year (i.e., prescription opioids or heroin) decreased from 3.2 percent in 2021 to 2.7 percent in 2024, as discussed in the 2024 Key Substance Use and Mental Health Indicators report (CBHSQ, 2025g). Including data from respondents who used heroin in the past year changed the estimates in 2021 from 3.0 percent for prescription opioid misuse to 3.2 percent for any opioid misuse, including heroin (Section 4.4.2). Likewise, estimates in 2024 changed from 2.6 percent for prescription opioid misuse to 2.7 percent for any opioid misuse, including heroin.

4.4.4 Any Opioid Misuse, Including Heroin and Illegally Made Fentanyl

Beginning with the 2022 NSDUH, respondents aged 12 or older were asked whether they ever used IMF and, if so, how long it had been since they last used it. The questionnaire explained that IMF is fentanyl that people cannot get from a doctor or pharmacy and that IMF can come in forms such as powder, pills, blotter paper, or mixed with heroin or other drugs. Therefore, the categories for the misuse of prescription opioids and the use of heroin or IMF can be used to create an overall category for opioid misuse that includes IMF; corresponding estimates are not available for 2021.

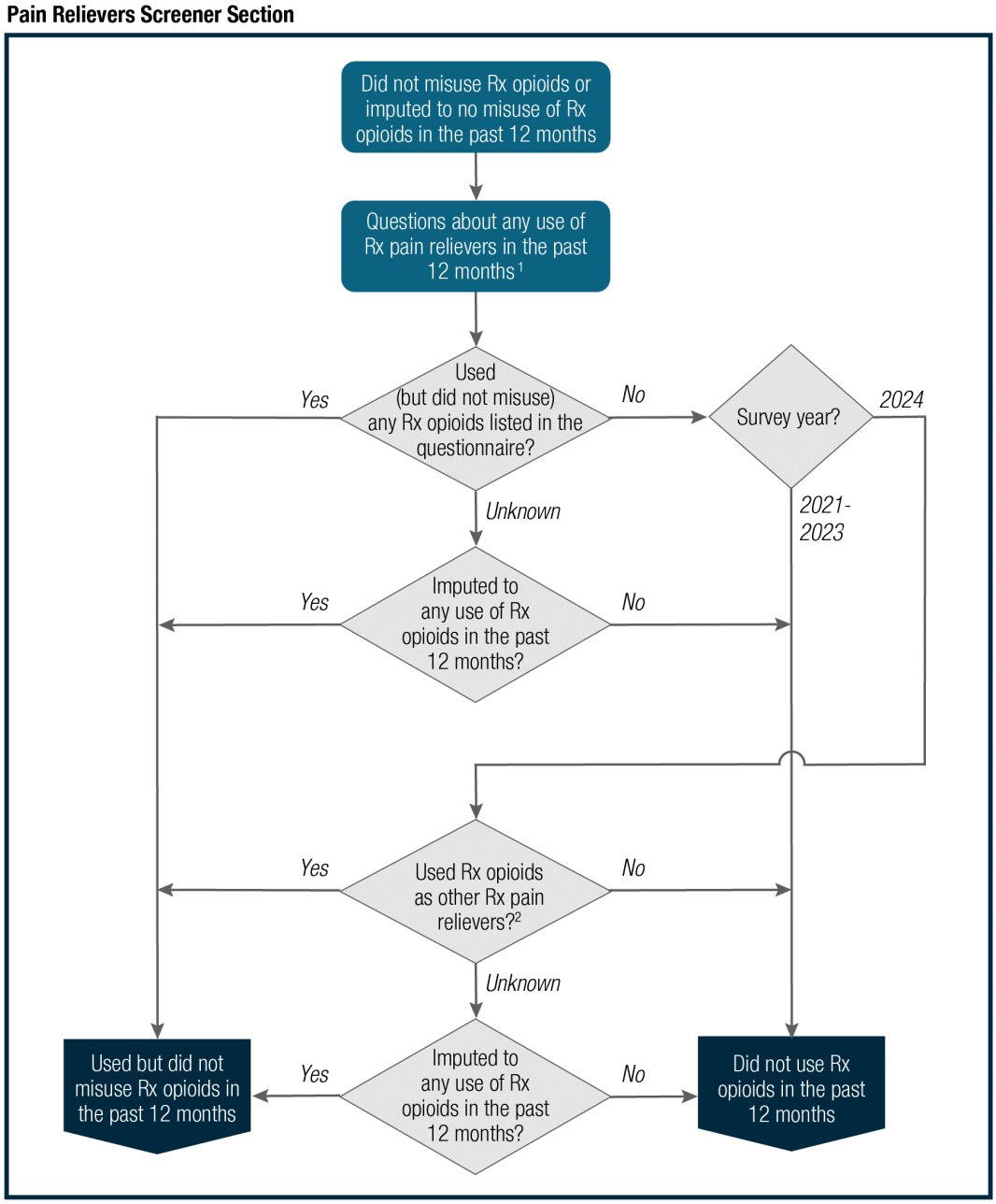
Similar to the opioid misuse measure discussed in Section 4.4.3, including data from respondents who used heroin or IMF in the past year did not appreciably affect estimates of any opioid misuse. In 2024, for example, estimates for the misuse of opioids among people aged 12 or older were 2.7 percent if data were not included from respondents who used IMF in the past year (Section 4.4.3) and 2.8 percent if data were included from respondents who used IMF.

4.4.5 Any Past Year Prescription Pain Reliever and Prescription Opioid Use

Figure 4.8, which refers to the "Pain Relievers Screener Section" of the NSDUH questionnaire, describes the information that was used for creating variables to estimate any use of prescription opioids in the past year. Logically, if respondents were classified as having misused prescription opioids in the past year, they also used prescription opioids for any reason in that

period. If respondents were classified as not having misused prescription opioids in the past year, they still could have used prescription opioids in the past year.

Figure 4.8 Classification of the Use of Prescription Opioids in the Past Year



Rx = prescription.

NOTE: As per Figure 4.1, respondents were first asked about any use of specific Rx pain relievers in the past 12 months. To illustrate the logic for identifying the use and misuse of prescription opioids, the pain relievers main section is presented in Figure 4.7. Logically, if respondents misused Rx opioids in the past 12 months, then they also used them for any reason in that period.

[1] Classification of respondents as having used but not misused Rx opioids applied if respondents did not report misuse in the past 12 months.

[2] Respondents in 2024 were assumed to have used Rx opioids if they did not know or refused to report the other pain relievers they used or they reported a nonspecific drug (e.g., "painkillers"). Respondents were classified as "unknown" for the use of Rx opioids as other pain relievers if they did not know or refused to report whether they used other pain relievers in the past 12 months.

If respondents were not classified as having misused prescription opioids in the past year, the following information was used to identify whether they had used prescription opioids for any reason in the past year:

- Respondents who reported the past year use of any of the prescription opioids in the questionnaire (Table 4.1 and Figure 4.3) were classified as having used prescription opioids in that period.
- Respondents in 2021-2023 who did not use any of the prescription opioids in the questionnaire were classified as not having used prescription opioids in the past year. Reports of the use of only other prescription pain relievers were assumed to involve the use of only nonopioids.
- For respondents in 2024, information on the other pain relievers that respondents specified for any use in the past year was available for classifying respondents as having used prescription opioids. Specifically, the following respondents were classified as having used prescription opioids in the past year:
  - those who specified the use of prescription opioids as other pain relievers, including situations in which respondents specified the use of prescription opioids in the questionnaire despite not having reported their misuse;
  - those who did not know or refused to specify the names of other pain relievers they used; and
  - those who provided a nonspecific answer, such as "painkillers," without indicating the specific pain reliever(s) they used.

If respondents in 2021-2023 did not report the misuse of prescription opioids, reported the use of only other prescription pain relievers, and definitely did not use pain relievers in any of the 11 prescription opioid subtypes,[85] then the most conservative approach was to assume that the other prescription pain relievers were all nonopioid drugs, except when respondents were assumed to have misused opioids in the past year, as noted previously (see Figure 4.7). Respondents in 2024 who did not report the misuse of prescription opioids, reported the use of only other prescription pain relievers, and definitely did not use pain relievers in any of the 11 prescription opioid subtypes were classified as not having used prescription opioids if the only pain relievers they specified using were nonopioid drugs.

If respondents in 2021-2024 did not provide sufficient information to establish whether they used prescription opioids in the past year, statistical imputation was used to classify respondents as having used or not having used prescription opioids in the past year. For example, opioid pain reliever use could be unknown if respondents did not report use in the past year of specific pain relievers in the 11 prescription opioid subtypes in the questionnaire, but they had missing data for any of these subtypes.

[85] That is, respondents had no missing data for the past year use of specific drugs in the 11 prescription opioid subtypes in the NSDUH questionnaire.

About 1 in 5 people aged 12 or older in 2021-2023 were assumed to have used prescription opioids in the past year because they reported using prescription opioids in the NSDUH questionnaire or were assumed to have misused prescription opioids. In 2024, about 1 in 5 people aged 12 or older (21.9 percent) also were estimated to have used prescription opioids in the past year.

Tests of the linear trend for prescription opioid use in 2021-2024 among people aged 12 or older indicated no change over this period. Testing was not conducted for a trend in the percentage of past year users of prescription pain relievers who used prescription opioids because the measure in 2024 of any use of prescription pain relievers in the past year was not comparable with the measure in 2021 to 2023 (see Section 4.1.5.2).

4.4.6 Any Past Year Opioid Use Including Heroin

As for opioid misuse, factoring in reports of heroin use in the past year did not appreciably change the estimates of any past year use of opioids compared with the estimates in the preceding section for any past year use of prescription opioids. In 2024, for example, the estimate for any past year use of opioids including heroin also was 21.9 percent.

4.4.7 Any Past Year Opioid Use Including Heroin and Illegally Made Fentanyl

With the addition of new questions for 2022 about the use of IMF, an overall category has been created for any opioid use that included any past year use of prescription opioids, heroin, or IMF, in addition to measures for any past year use of opioids that included prescription opioids or heroin. However, factoring in reports of IMF use in the past year did not appreciably change the estimates of any past year use of opioids compared with the estimates in preceding sections for any past year use of prescription opioids or any past year use of opioids including heroin. In 2024, for example, estimates for the use of opioids among people aged 12 or older were 21.9 percent if data were not included from respondents who used IMF in the past year (see Sections 4.4.5 and 4.4.6) and 22.0 percent if data were included from respondents who used IMF.

4.4.8 Opioid Use Disorder in the Past Year

Although reports of the use and misuse of prescription pain relievers could include drugs that are not prescription opioids, SUD questions for prescription pain relievers in the NSDUH interview apply to DSM-5 criteria for opioid use disorder. Therefore, beginning with the 2024 NSDUH, respondents were not counted as having an opioid use disorder in the past year if they used prescription pain relievers but not prescription opioids in the past year and they did not use heroin in the past year. See Section 3.4.7 for information on how opioid use disorder is defined. Because questions about the use of IMF in the past year are asked in a later section of the interview, after respondents have been asked SUD questions, estimates of opioid use disorder take into account disorder symptoms for heroin or prescription opioids but not IMF.

Excluding data from respondents who used only nonopioid pain relievers and did not use heroin in the past year did not affect conclusions about trends in opioid use disorder and had minimal effect on estimates. Percentages of people aged 12 or older who were classified as having a

heroin use disorder or a pain reliever use disorder[86] showed no change from 2021 to 2024. There also was no change from 2021 to 2024 for percentages of people aged 12 or older with an opioid use disorder after data had been excluded from respondents who used only nonopioid pain relievers and did not use heroin in the past year. Among people aged 12 or older in 2024, 1.8 percent were classified as having a heroin use disorder or pain reliever use disorder, and 1.7 percent were classified as having an opioid use disorder.

4.5 Measures of Tranquilizer or Sedative Use and Misuse in NSDUH

4.5.1 Background

Drugs defined in NSDUH as tranquilizers or sedatives have a number of important features in common.

- Both are CNS depressants, and, therefore, both cause drowsiness (NIDA, 2023).
- Most have a common effect on specific activity in the brain (NIDA, 2023).[87]
- Most are classified in Schedule IV under the CSA (Tables 4.3 and 4.4).
- Chemically similar drugs (e.g., benzodiazepines) are prescribed as either tranquilizers or sedatives.

The Multum Lexicon® database of drugs has a category for "anxiolytics, sedatives, and hypnotics" that includes drugs defined in NSDUH as tranquilizers or sedatives (National Center for Health Statistics, 2023). Because of these similarities, national reports and tables for the 2023 NSDUH include estimates for the misuse of any tranquilizer or sedative.

4.5.2 Creation of Measures for Tranquilizer or Sedative Misuse

The following measures for the misuse of tranquilizers or sedatives were included in reports or tables for the 2024 NSDUH:

- misuse of tranquilizers or sedatives in the past 12 months (i.e., past year),
- misuse of tranquilizers or sedatives in the past 30 days (i.e., past month), and
- tranquilizer or sedative use disorder in the past 12 months.

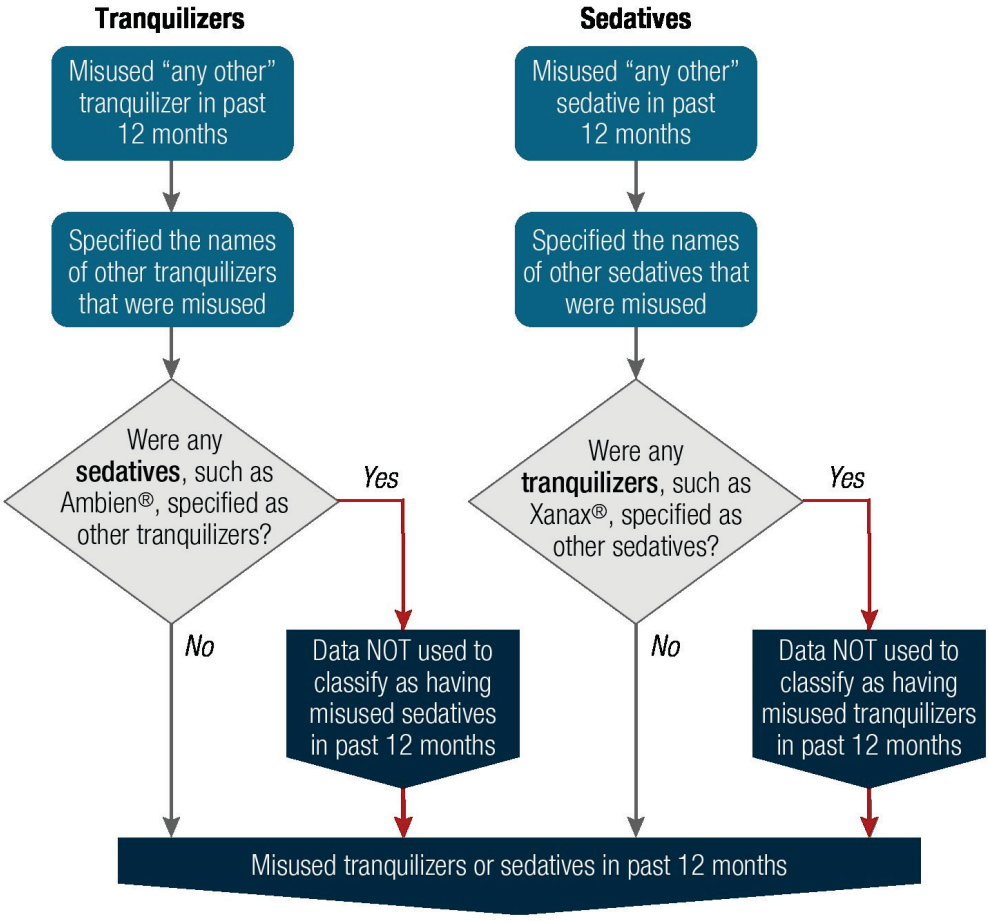
4.5.2.1 Past Year or Past Month Misuse of Tranquilizers or Sedatives

Respondents were classified as having misused prescription tranquilizers or sedatives in the past 12 months if they reported the misuse of prescription tranquilizers, prescription sedatives, or both in that period. A similar principle applied to the classification of respondents as having misused tranquilizers or sedatives in the past 30 days.

[86] Pain reliever use disorder estimates could include data from respondents who reported the use of only nonopioid pain relievers.
[87] Although prescription opioids also cause drowsiness, they do not act on the brain in the same way as tranquilizers or sedatives.

As previously noted, respondents who reported that they misused other tranquilizers in the past 12 months were asked to specify the names of the other tranquilizers they misused. Similarly, respondents who reported that they misused other sedatives in that period were asked to specify the names of the other sedatives they misused. Figure 4.9 describes how data were handled if respondents specified a prescription tranquilizer (e.g., Xanax®) as some "other sedative" they misused, or vice versa. Consistent with the principle discussed in Section 2.3.2.1, if respondents specified the misuse of a prescription tranquilizer as some other sedative they had misused, for example, these data for the misuse of other sedatives were not used to edit the data for the use and misuse of tranquilizers in that section of the interview. However, the reporting of prescription tranquilizers as other sedatives or the reporting of prescription sedatives as other tranquilizers did not affect the creation of the aggregate measures for the misuse of prescription tranquilizers or sedatives in the past year or past month.

Figure 4.9 Classification of Misuse of Tranquilizers or Sedatives in the Past 12 Months Based on Cross-Reporting of Tranquilizers or Sedatives as "Other" Drugs



4.5.2.2 Tranquilizer or Sedative Use Disorder

Beginning in 2021, respondents were classified as having a tranquilizer or sedative use disorder in the past 12 months if they had a tranquilizer use disorder related to any use of prescription tranquilizers in the past year, a sedative use disorder related to any use of prescription sedatives in the past year, or both disorders. The SUD criteria for these substances are described in Section 3.4.7.

4.5.3 Estimates Not Created for Tranquilizer or Sedative Use and Misuse

The following estimates for the use or misuse of any tranquilizer or sedative were not created for the 2024 national reports and tables; these estimates could be created for prescription pain relievers and prescription stimulants:

- initiation of the misuse of tranquilizers or sedatives in the past year, and
- frequency of misuse of any tranquilizer or sedative in the past month.

As discussed in Section 3.4.6.2, the potential underreporting of lifetime (but not past year) misuse of prescription drugs could result in some people being misclassified as having initiated the misuse of any prescription tranquilizer or sedative in the past year, when in fact they first misused any prescription tranquilizer or sedative more than 12 months prior to the interview date. Therefore, aggregate estimates were not created for the 2024 NSDUH for the initiation of misuse of any tranquilizer or sedative.

Aggregate estimates for the frequency of misuse of tranquilizers or sedatives in the past 30 days were not created because these data are not mutually exclusive for respondents who misused both tranquilizers and sedatives in that period. For these respondents, the number of days they misused prescription tranquilizers or sedatives in the past 30 days cannot be summed to estimate the total number of days respondents misused tranquilizers or sedatives because respondents could have misused both prescription drug types on the same day.

4.6 Measures of Benzodiazepine Use and Misuse in NSDUH

4.6.1 Background

As noted in Sections 4.2.4, 4.2.5, and 4.5, prescription drugs categorized as benzodiazepines can be prescribed as either tranquilizers or sedatives. The benzodiazepines listed in Tables 4.3 and 4.4 are classified as Schedule IV controlled substances. Therefore, these drugs have the potential to produce physical or psychological dependence. Because benzodiazepines are CNS depressants, they cause drowsiness and can impair motor skills important for tasks such as operating a motor vehicle or machinery. People also can overdose on benzodiazepines, especially when taken in combination with other CNS depressants such as opioids (NIDA, 2023, 2024b).

Estimates were included in 2024 NSDUH national reports and tables for the use and misuse of any benzodiazepine in the past 12 months, regardless of whether benzodiazepines were classified as tranquilizers or sedatives. The next section discusses the creation of measures from the NSDUH data for any benzodiazepine use and misuse.

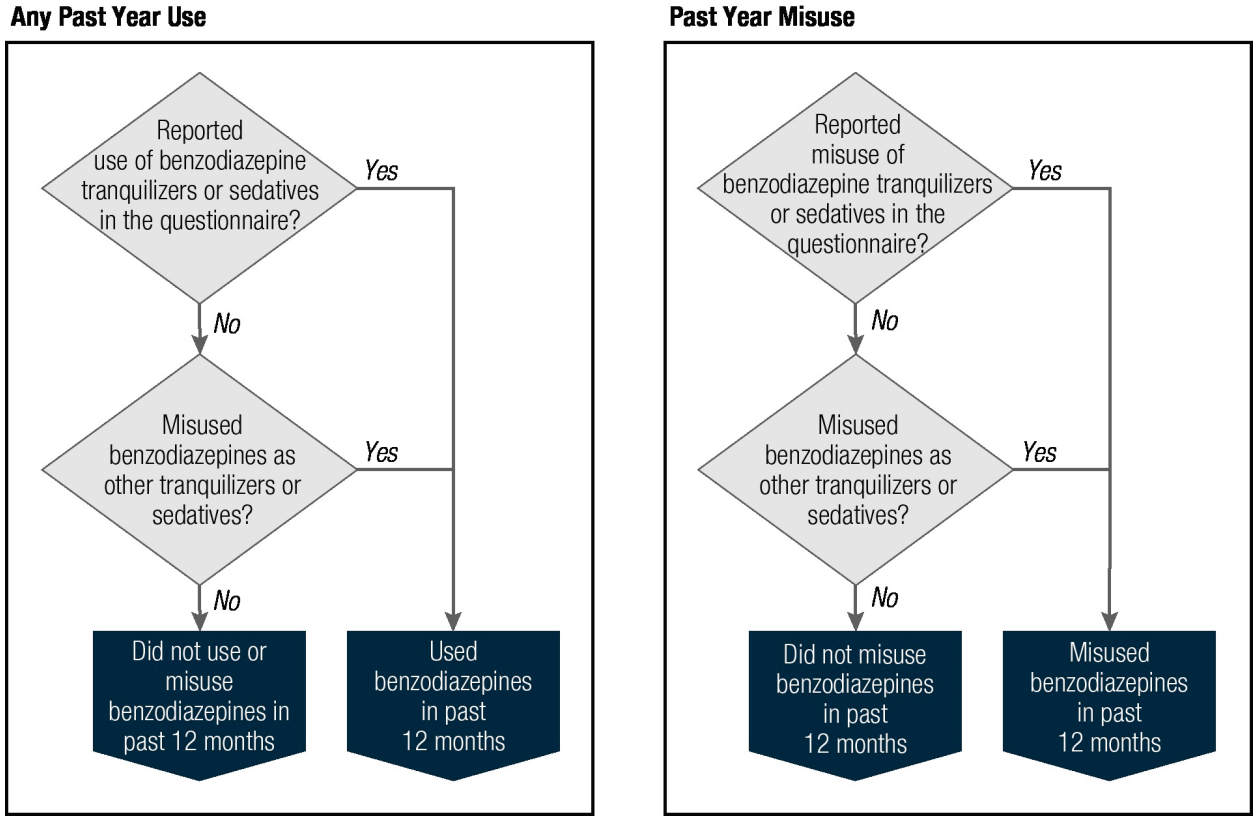
4.6.2 Creation of Measures for Benzodiazepine Use and Misuse

Respondents were classified as having used any benzodiazepine tranquilizer or sedative in the past 12 months if they reported the use of one or more of the benzodiazepines shown in Figures 4.5 and 4.6. As noted in Section 4.3, variables for the past year use or misuse of any benzodiazepine were statistically imputed for 2024 (see Section 2.3.3).

Figure 4.10 presents information on how measures of benzodiazepine use and misuse were created for NSDUH. Respondents who reported use of specific benzodiazepines included in the tranquilizers section or sedatives section in the past 12 months were classified as having used benzodiazepines for any reason in the past 12 months. Respondents also were classified as having used benzodiazepines for any reason in the past 12 months if they specified the misuse of benzodiazepines as other tranquilizers or sedatives they misused in that period. Logically, respondents who misused other benzodiazepines in the past 12 months used them for any reason. Similarly, respondents who reported they misused a benzodiazepine tranquilizer or sedative in the past 12 months—either from a response to a direct question (e.g., the direct question about misuse of Xanax® in the past 12 months) or as some other tranquilizer or sedative they misused in that period—were classified as having misused any benzodiazepine in the past 12 months. Consistent with the principle of not editing across sections of the interview (see Section 2.3.2.1), however, reports of benzodiazepines in sections other than tranquilizers or sedatives were not included in the measures of benzodiazepine use or misuse.

In addition, estimates in the 2024 Detailed Tables (CBHSQ, 2025h) for the use and misuse of any benzodiazepine tranquilizer and specific benzodiazepine tranquilizer subtypes in Figure 4.5 were based solely on reports from the tranquilizers section of the interview. Similarly, estimates for the use and misuse of any benzodiazepine sedative and specific benzodiazepine sedative subtypes shown in Figure 4.6 were based solely on reports from the sedatives section. For example, if respondents did not report the past year use or misuse of Xanax® in the tranquilizers section of the interview, they were not counted as having used or misused a benzodiazepine tranquilizer or an alprazolam product, even if they specified the past year misuse of Xanax® as some other sedative. However, respondents in this example were classified in a category for the past year misuse of any miscellaneous prescription benzodiazepine for the 2024 Detailed Tables. The miscellaneous prescription benzodiazepine estimates were intended to provide data users with an indication of the extent of reporting of benzodiazepines across the respective categories for tranquilizers and sedatives.

Figure 4.10 Identification of Past Year Use and Misuse of Benzodiazepines



4.6.3 Estimates Not Created for Benzodiazepine Use and Misuse

The following measures and associated estimates were not created for benzodiazepines:

- lifetime use or misuse of benzodiazepines,
- misuse of any benzodiazepine in the past month,
- initiation of benzodiazepine misuse in the past year, and
- an SUD attributable to the use of benzodiazepines in the past year.

These estimates were not created because the interview sections for tranquilizers and sedatives also included drugs that are not benzodiazepines.

Aside from the potential for respondents to underreport lifetime use or misuse of tranquilizers or sedatives (see Section 3.3.3.2), measures for the lifetime use or misuse of benzodiazepines could not be created because respondents who did not report use or misuse of tranquilizers or sedatives in the past 12 months were not asked whether they ever used or misused benzodiazepines. Similarly, if respondents reported that they misused any prescription tranquilizer or sedative in the past 30 days, they could have misused drugs that are not benzodiazepines (e.g., Ambien®).

As noted in Section 3.4.6, NSDUH respondents in 2024 were asked about the initiation of misuse of prescription psychotherapeutic drugs only for the individual prescription drugs they had misused in the past 12 months. If respondents reported past year initiation of misuse for all tranquilizers or sedatives they misused in that period, they were not asked whether they ever misused benzodiazepine tranquilizers or sedatives more than 12 months prior to the interview. Consequently, past year initiation of the misuse of benzodiazepines could not be determined.

Similar to the issue described previously for the misuse of benzodiazepines in the past 30 days, past year users of tranquilizers or sedatives since 2021 have been asked, respectively, about SUD symptoms attributable to their use of any tranquilizer or any sedative in the past 12 months.[88] Thus, if respondents reported the use of benzodiazepine sedatives and sedatives that were not benzodiazepines in the past 12 months, it could not be determined whether the SUD symptoms they reported applied to the benzodiazepine sedatives or the sedatives that were not benzodiazepines.

[88] As discussed in Section 3.4.7, NSDUH respondents beginning in 2021 who reported any use of prescription tranquilizers or sedatives in the past year (i.e., not just misuse) were asked the respective SUD questions for that category of prescription drugs.

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Substance Abuse and Mental Health Services Administration

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SAMHSA's mission is to lead public health and service delivery efforts that promote mental health, prevent substance misuse, and provide treatments and supports to foster recovery while ensuring access and better outcomes for all.

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